

State Manager is a capability of AWS Systems Manager. The article supporting (A) is entitled: Schedule Amazon RDS stop and start using AWS Systems Manager.

upvoted 2 times

satyaamm Most Recent 1 year, 4 months ago

Selected Answer: D

If A had AWS System Manager State Manager then it would be A. Else Lambda is suitable here.

upvoted 1 times

lofzee 2 years ago

Selected Answer: D

D.

You need to use AWS Systems Manager State Manager, not Systems Manager Session Manager.

upvoted 2 times

ManikRoy 2 years, 1 month ago

Selected Answer: D

Agree with study_aws1 comment.

Lambda and event bridge solution is the correct answer.

The option A was only possible if it had mentioned System manager state manager

upvoted 2 times

phonglktv 2 years, 1 month ago

Selected Answer: C

we still pay for RDS even when the instance stops. So for cost-optimize -> C

upvoted 2 times

NSA_Poker 2 years ago

While the instance is stopped, you're charged for storage and backups, but not for the DB instance hours.

<https://aws.amazon.com/blogs/database/schedule-amazon-rds-stop-and-start-using-aws-lambda/#:~:text=While%20the%20instance%20is%20stopped%2C%20you%E2%80%99re%20charged%20for%20storage%20and%20backups%2C%20but%20not%20for%20the%20DB%20instance%20hours.>

upvoted 2 times

LP0905 2 years, 3 months ago

Selected Answer: D

Although both A and D is a workable solution, the requirements is to minimum cost.

The benefits of automating the startup and shutdown of RDS DB instances using Lambda allows organizations to further reduce compute costs and simplify the administration of database environments that don't need to be running continuously.

<https://aws.amazon.com/blogs/database/schedule-amazon-rds-stop-and-start-using-aws-lambda/>

For using system manager to accomplish the task works however keep in mind that although we're stopping the databases, the storage costs for the databases still apply.

<https://aws.amazon.com/blogs/database/schedule-amazon-rds-stop-and-start-using-aws-systems-manager/>

Initially I also thought that A would be the correct answer however looking at the administration and cost I would go for D as a better solution instead.

upvoted 3 times

thewalker 2 years, 4 months ago

Selected Answer: A

To automatically shutdown an RDS instance during 09:00 PM to 09:00 AM and have it available between 09:00 AM to 09:00 PM, you can use AWS Systems Manager Maintenance Windows.

Create two Maintenance Windows - one to stop the RDS instance at 09:00 PM and another to start it at 09:00 AM.

For each Maintenance Window, select the "AWS-StopRDSInstance" and "AWS-StartRDSInstance" runbooks respectively and specify the cron expression for the schedule.

Tag the RDS instance with a name so it can be identified by the runbooks.

The runbooks will then automatically stop and start the RDS instance on the specified schedule without needing any manual intervention.

upvoted 1 times

thewalker 2 years, 4 months ago

This allows cost savings by shutting down the RDS instance during non-business hours while keeping it available during the day as per your requirements. Refer to the AWS documentation for more details on configuring Maintenance Windows and runbooks.

<https://docs.aws.amazon.com/prescriptive-guidance/latest/patterns/automatically-stop-and-start-an-amazon-rds-db-instance-using-aws-systems-manager-maintenance-windows.html>

https://repost.aws/questions/QUcVR5js8LSbOS_LE889Kdlg/automatically-stop-and-start-an-amazon-rds-db-instance-in-a-cdk-app
upvoted 1 times

theoChan 2 years, 4 months ago

Guys, we still have to pay for RDS instance even we stopped it, isn't it?

upvoted 3 times

NSA_Poker 2 years ago

While the instance is stopped, you're charged for storage and backups, but not for the DB instance hours.

<https://aws.amazon.com/blogs/database/schedule-amazon-rds-stop-and-start-using-aws-lambda/#:~:text=While%20the%20instance%20is%20stopped%2C%20you%E2%80%99re%20charged%20for%20storage%20and%20backups%2C%20but%20not%20for%20the%20DB%20instance%20hours.>
upvoted 2 times

Ruffyyit 2 years, 6 months ago

AWS Lambda functions can be used to start and stop RDS instances programmatically.

EventBridge scheduled rules can trigger the Lambda functions at specified times daily.

This allows fully automating the starting and stopping of RDS on a schedule to match usage patterns.

RDS billing is per hour when instance is running, so stopping when not in use significantly reduces costs.

Using Lambda and EventBridge is simpler and more robust than cron jobs on EC2.

ElastiCache and Systems Manager Session Manager are useful tools but do not directly address scheduled RDS start/stop.

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: D

You can use AWS Lambda and Amazon EventBridge to schedule a Lambda function to stop and start the idle databases with specific tags to save on compute costs.

<https://aws.amazon.com/blogs/database/schedule-amazon-rds-stop-and-start-using-aws-lambda/#:~:text=you%20to%20schedule%20a-,Lambda%20function,-to%20stop%20and%20start>
upvoted 3 times

lemur88 2 years, 9 months ago

Selected Answer: D

Here is the recommended solutions which describes choice D - <https://aws.amazon.com/blogs/database/save-costs-by-automating-the-start-and-stop-of-amazon-rds-instances-with-aws-lambda-and-amazon-eventbridge/>

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

AWS Lambda functions can be used to start and stop RDS instances programmatically.

EventBridge scheduled rules can trigger the Lambda functions at specified times daily.

This allows fully automating the starting and stopping of RDS on a schedule to match usage patterns.

RDS billing is per hour when instance is running, so stopping when not in use significantly reduces costs.

Using Lambda and EventBridge is simpler and more robust than cron jobs on EC2.

ElastiCache and Systems Manager Session Manager are useful tools but do not directly address scheduled RDS start/stop.

upvoted 4 times

cookieMr 2 years, 11 months ago

Selected Answer: D

By using AWS Lambda functions triggered by Amazon EventBridge scheduled rules, the company can automate the start and stop actions for the Amazon RDS for MySQL DB instance based on the 12-hour access period. This allows them to minimize costs by only running the DB instance when it is needed.

Option A is not the most suitable solution because it refers to IAM policies for AWS Systems Manager Session Manager, which is primarily used for interactive shell access to EC2 instances and does not directly address the requirement of starting and stopping the DB instance.

Option B is not the most suitable solution because it suggests using Amazon ElastiCache for Redis as a cache cluster, which may not provide the desired cost optimization for the DB instance.

Option C is not the most suitable solution because launching an EC2 instance and configuring cron jobs to start and stop it does not directly address the requirement of minimizing costs for the Amazon RDS DB instance.

upvoted 4 times

Siva007 3 years ago

Selected Answer: D

I got this question in real exam!

upvoted 5 times

srijrao 2 years, 11 months ago

why we need more than one lambda function to start and stop DB instance? btw how many questions came from this site?

upvoted 2 times

ccmc 3 years, 1 month ago

State Manager, a capability of AWS Systems Manager

upvoted 2 times

Ankit_EC_ran 3 years, 1 month ago

Selected Answer: D

Option D is correct

upvoted 2 times

A company sells ringtones created from clips of popular songs. The files containing the ringtones are stored in Amazon S3 Standard and are at least 128 KB in size. The company has millions of files, but downloads are infrequent for ringtones older than 90 days. The company needs to save money on storage while keeping the most accessed files readily available for its users. Which action should the company take to meet these requirements MOST cost-effectively?

- A. Configure S3 Standard-Infrequent Access (S3 Standard-IA) storage for the initial storage tier of the objects.
- B. Move the files to S3 Intelligent-Tiering and configure it to move objects to a less expensive storage tier after 90 days.
- C. Configure S3 inventory to manage objects and move them to S3 Standard-Infrequent Access (S3 Standard-1A) after 90 days.
- D. Implement an S3 Lifecycle policy that moves the objects from S3 Standard to S3 Standard-Infrequent Access (S3 Standard-1A) after 90 days. **Most Voted**

Correct Answer: D

Community vote distribution

D (66%)

B (34%)

Comments

rjam **Highly Voted** 3 years, 6 months ago

Selected Answer: D

Answer D

Why Optoin D ?

The Question talks about downloads are infrequent older than 90 days which means files less than 90 days are accessed frequently. Standard-Infrequent Access (S3 Standard-IA) needs a minimum 30 days if accessed before, it costs more.

So to access the files frequently you need a S3 Standard . After 90 days you can move it to Standard-Infrequent Access (S3 Standard-IA) as its going to be less frequently accessed

upvoted 42 times

MutiverseAgent 2 years, 10 months ago

I do not agree. The MOST cheaper option is B, because by choosing:

D) Files older than 90 days will live eternally in the S3 Infrequently access layer at \$0.0125/GB.

B) Using Intelligent-Tiering files older than 90 days can be moved DIRECTLY to the "Archive access tier" (Glacier instant retrieval) at \$0.004/GB, avoiding/skipping the "S3 Infrequently access layer". The question also seems to be according this assumption as says "and configure it to move objects to a less expensive storage tier after 90 days".

<https://aws.amazon.com/s3/pricing/?nc=sn&loc=4>

upvoted 4 times

MutiverseAgent 2 years, 10 months ago

I am taking back my answer, the right is D) as the "Archive access tier" check present in the "Intelligent-Tiering Archive configurations" is for "S3 Glacier flexible retrieval" which is not instant retrieval.

upvoted 8 times

Itsmetanmay 2 years, 1 month ago

Archive Instant Access tier (automatic)

If an object is not accessed for 90 consecutive days, the object moves to the Archive Instant Access tier. The Archive Instant Access tier provides low latency and high-throughput performance. --- This says it moves to instant retrieval class. Price for this is lesser than standard IA. Answer should be B

upvoted 2 times

zeronine75 **Highly Voted** 3 years, 6 months ago

Selected Answer: B

B/D seems possible answer. But, I'll go with "B".

In the following table, S3 Intelligent-Tiering seems not so expansive than S3 Standard.

https://aws.amazon.com/s3/pricing/?nc1=h_ls

And, in the question "128KB" size is talking about S3 Intelligent-Tiering stuff.

upvoted 16 times

FNJ1111 3 years, 5 months ago

also, there are probably several ringtones which aren't popular/used. Why keep them in S3 standard? The company would save money if s3 intelligent-tiering moves the unpopular ringtones to a more cost-effective tier than s3 standard.

upvoted 2 times

Wilson_S 3 years, 6 months ago

This link also has me going with "B." Specifying 128 KB in size is not a coincidence. <https://aws.amazon.com/s3/storage-classes/intelligent-tiering/>

upvoted 6 times

javitech83 3 years, 6 months ago

because of the link it is D.

There are no retrieval charges in S3 Intelligent-Tiering. S3 Intelligent-Tiering has no minimum eligible object size, but objects smaller than 128 KB are not eligible for auto tiering. These smaller objects may be stored, but they'll always be charged at the Frequent Access tier

upvoted 1 times

javitech83 3 years, 6 months ago

oh sorry it states objects are bigger than 128 KB. B is correct

upvoted 2 times

Wajif 3 years, 6 months ago

S3 Intelligent tiering is used when the access frequency is not known. I think 128KB is a deflector.

upvoted 7 times

pentium75 2 years, 5 months ago

Intelligent Tiering moves files how it thinks is right, not at a configurable interval such as "after 90 days".

upvoted 2 times

c12ab95 **Most Recent** 1 year ago

Selected Answer: B

Solution is B.

Key Analysis

Infrequent Access After 90 Days: S3 Intelligent-Tiering automatically transitions objects not accessed for 90 days to the Archive Instant Access tier (\$0.004/GB-month), which is 68% cheaper than S3 Standard-IA (\$0.0125/GB-month). This aligns perfectly with the requirement to save costs for older ringtones.

No Retrieval Fees: Unlike S3 Standard-IA (which charges retrieval fees), Intelligent-Tiering has no retrieval costs for data in Frequent, Infrequent, or Archive Instant Access tiers .

Automated Tiering: Intelligent-Tiering eliminates manual lifecycle policies. Objects older than 90 days with no access are moved automatically, reducing operational overhead . For new or frequently accessed ringtones, objects remain in the Frequent Access tier (same cost as S3 Standard), ensuring immediate availability .

S3 Intelligent-Tiering provides deeper cost savings than Standard-IA for infrequently accessed data, automates tier transitions, and avoids retrieval fees, making it the optimal choice.

upvoted 1 times

Skyskilo 1 year, 4 months ago

Selected Answer: D

Option D is the most cost-effective solution as it leverages an S3 Lifecycle policy to automatically transition infrequently accessed files to S3 Standard-IA after 90 days, saving costs while meeting the company's requirements.

upvoted 1 times

Rcosmos 1 year, 4 months ago

Selected Answer: D

B. Mova os arquivos para o S3 Intelligent-Tiering:

O S3 Intelligent-Tiering é uma solução flexível, mas tem uma pequena taxa mensal para monitorar os objetos. No caso descrito, uma política de ciclo de vida seria mais econômica para o comportamento de acesso conhecido (raros acessos após 90 dias).

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: D

Ans D - as well explained by rjam (1 year, 11 months ago)

upvoted 2 times

Duckydoo 1 year, 11 months ago

Selected Answer: D

I am going with D instead of B because Intelligent-Tiering may work against you in this case. The company knows that the files are rarely accessed after 90 days. So even if a file is accessed on the 89th day, they would still want to "archive" that file the next day (again because they fully know the typical access behavior). However, Intelligent Tiering will reset the 90-day inactive clock starting on the 89th day (i.e. it will not archive the file until the 189th day assuming no other access).

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: D

D for sure

upvoted 1 times

lofzee 2 years ago

Selected Answer: D

You can't configure Intelligent Tiering, that is the whole point of it. It's intelligent and moves things on its own so there is no 90 day configuration you can set.

Therefore D is the answer.

upvoted 1 times

lofzee 2 years ago

Actually scrap that. I'm changing my answer to B.

You can setup Intelligent Tiering to move data to an archive after 90 days.

https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage-class-intro.html?icmpid=docs_amazons3_console#sc-dynamic-data-access

<https://aws.amazon.com/about-aws/whats-new/2021/11/s3-intelligent-tiering-archive-instant-access-tier/>

Also the 128KB size is perfect for Intelligent Tiering.

Tricky question indeed, one of those pointless ones that doesn't really make you a better or worse AWS Engineer

upvoted 1 times

NSA_Poker 2 years ago

Selected Answer: B

Some songs make comebacks. Lifecycle policy isn't intelligent enough to deal with a resurgence in popularity of a ringtone. 128KB in size makes files eligible for Intelligent-Tiering.

upvoted 1 times

ManikRoy 2 years, 1 month ago

Selected Answer: D

Clearly option D is the correct. This is not a trick question.

upvoted 1 times

MomenAWS 2 years, 1 month ago

Selected Answer: B

I think this is a kind of tricky question (B VS D) However:

The Q stated "most accessed files readily available for its users."

From this and correct me if I am wrong, I think B is a better solution since Amazon S3 Intelligent-Tiering working on "moving data to the most cost-effective access tier based on access frequency" while the S3 Standard-IA doesn't.

Ref: <https://aws.amazon.com/s3/storage-classes/>

upvoted 1 times

richiexamaws 2 years, 2 months ago

Selected Answer: B

B. Move the files to S3 Intelligent-Tiering and configure it to move objects to a less expensive storage tier after 90 days.

Explanation:

S3 Intelligent-Tiering is a storage class that automatically moves objects between two access tiers: frequent access and infrequent access, based on access patterns.

By configuring S3 Intelligent-Tiering to move objects to a less expensive storage tier after 90 days of infrequent access, the company can save money on storage costs while ensuring that the most accessed files remain readily available.

This approach is more cost-effective than using S3 Standard-Infrequent Access (S3 Standard-IA) because it automatically adjusts storage tiers based on access patterns without the need for manual configuration or management.

upvoted 1 times

Kanagarajd 2 years, 3 months ago

Selected Answer: D

Right answer is D

upvoted 1 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

D is cheapest and managed by S3 Lifecycle policy

A: Not readily available

C: Wrong product

B: No choice of '90 days' so you'll be paying for Intelligent Tiering unnecessarily for files to drop out of frequent access after the first 90 days.

upvoted 2 times

Firdous586 2 years, 5 months ago

B is the correct answer Kindly follow the below link for more information as proof

<https://aws.amazon.com/s3/storage-classes/>

upvoted 1 times

awsgeek75 2 years, 4 months ago

So D is the correct answer as IA is cheaper than Intelligent tier.

upvoted 1 times

Ruffyt 2 years, 6 months ago

The key reasons:

S3 Lifecycle policies can automatically transition objects from S3 Standard to S3 Standard-IA after 90 days.

S3 Standard provides high performance for frequently accessed newer files.

S3 Standard-IA costs 20-30% less than S3 Standard for infrequently accessed files.

This matches access patterns - high performance for new files, cost savings for older files.

S3 Intelligent Tiering has higher request costs and complexity for this simple access pattern.

S3 Inventory lists objects and their properties but does not directly transition objects.

Lifecycle policies provide automated transitions without manual intervention.

upvoted 1 times

A company needs to save the results from a medical trial to an Amazon S3 repository. The repository must allow a few scientists to add new files and must restrict all other users to read-only access. No users can have the ability to modify or delete any files in the repository. The company must keep every file in the repository for a minimum of 1 year after its creation date.

Which solution will meet these requirements?

- A. Use S3 Object Lock in governance mode with a legal hold of 1 year.
- B. Use S3 Object Lock in compliance mode with a retention period of 365 days. **Most Voted**
- C. Use an IAM role to restrict all users from deleting or changing objects in the S3 bucket. Use an S3 bucket policy to only allow the IAM role.
- D. Configure the S3 bucket to invoke an AWS Lambda function every time an object is added. Configure the function to track the hash of the saved object so that modified objects can be marked accordingly.

Correct Answer: B

Community vote distribution

B (93%)

7%

Comments

elmogy **Highly Voted** 3 years ago

Selected Answer: B

B,

The key is "No users can have the ability to modify or delete any files" and compliance mode supports that.

I remember it this way: (governance is like government, they set the rules but they can allow some people to break it :D)

upvoted 62 times

Praewwara 2 years, 5 months ago

Amazon S3 Object Lock

1. Governance mode - Only users with special permissions can overwrite, delete, or alter object lock settings

2. Compliance mode - No user, including the root user in an AWS account, can overwrite, delete, or alter object lock settings

upvoted 13 times

Burrito69 2 years, 2 months ago

I liked that thought of yours.. can you do more of these please? Thank you

upvoted 5 times

Qjb8m9h **Highly Voted** 3 years, 6 months ago

Answer : B

Reason: Compliance Mode. The key difference between Compliance Mode and Governance Mode is that there are NO users that can override the retention periods set or delete an object, and that also includes your AWS root account which has the highest privileges.

upvoted 22 times

Zerotn3 3 years, 5 months ago

How about: The repository must allow a few scientists to add new files

upvoted 1 times

JayBee65 3 years, 5 months ago

Adding is not the same as changing :)

upvoted 9 times

abhishek2021 3 years ago

Compliance mode controls the object life span after creation.

how this option restricts all scientists from adding new file? please explain.

upvoted 4 times

satyaamm Most Recent 1 year, 4 months ago

Selected Answer: B

Compliance mode supports the feature of "No users can delete or modify the files".

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: B

Ans B - Compliance mode... but not sure that answers "allow only a few scientists to add new files"...

upvoted 3 times

toyaji 1 year, 10 months ago

Selected Answer: B

First of all, Regal hold has no expiration before you remove it. So A makes no sense.

After that Governance mode is breakable with permission, but Compliance mode is not even for root user cannot delete it.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/object-lock.html>

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: B

B for sure

upvoted 1 times

ManikRoy 2 years, 1 month ago

Selected Answer: B

I almost chose A for this deciving line lol but it would be compliance mode as no user should be able to change objects:-

The repository must allow a few scientists to add new files and must restrict all other users to read-only access.

upvoted 2 times

demigodnyi 2 years, 4 months ago

Can someone please explain why the answer is not A. It said that The repository must allow a few scientists to add new files. So, i think some user must have permission to change it.

upvoted 1 times

ManikRoy 2 years, 1 month ago

Write Once Read Many - in compliance mode there is no restriction in adding but only changing existing objects

upvoted 2 times

NSA_Poker 2 years ago

Legal hold – A legal hold provides the same protection as a retention period, but it HAS NO EXPIRATION DATE. Instead, a legal hold remains in place until you explicitly remove it. Legal holds are independent from retention periods and are PLACED ON INDIVIDUAL OBJECT VERSIONS.

upvoted 2 times

pentium75 2 years, 5 months ago

Unsure, B would meet the "must keep every file for a minimum of 1 year" requirement. (In theory C would too if you ignore the root user, but administrators could remove the policy.) But what about the 'a few scientists must be able to add new files'? None of the options mentions permissions for a special group.

upvoted 3 times

awsgeek75 2 years, 4 months ago

agree that something is missing for "some users".

ACD are not going to work flat out so B looks like right answer but with some language issues either in the question or the answer.

upvoted 1 times

LoXoL 2 years, 5 months ago

Agree. It looks like it's missing sth here.

upvoted 1 times

Ruffyt 2 years, 6 months ago

Both Compliance & Governance mode protect objects against being deleted or changed. But in Governance mode some people can have special permissions. In this question, no user can delete or modify files; so the answer is Compliance mode only. Neither of these modes restrict user from adding new files.

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

Compliance Mode best suits this scenario because once an object is locked in compliance mode, its retention mode can't be changed, and its retention period can't be shortened.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

B) seems to be the right option, because: Both option A) & B) allow to:

- Scientists add new files & other users read-only access.
- Keep files for a minimum of 1 year

Only option B allows to:

- Disable all users the ability to modify or delete any file.

If A) were the correct option some scientists will be able to modify files, as if they were in charge of putting an object lock same permission would allow them to remove the lock and consequently delete the file.

upvoted 2 times

MutiverseAgent 2 years, 10 months ago

Selected Answer: B

B) seems to be the right option, because: Both option A) & B) allow to:

- Scientists add new files & other users read-only access.
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If A) were the correct option some scientists will be able to modify files, as if they were in charge of putting an object lock same permission would allow them to remove the lock and consequently delete the file.

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: B

S3 Object Lock provides the necessary features to enforce immutability and retention of objects in an S3. Compliance mode ensures that the locked objects cannot be deleted or modified by any user, including those with write access. By setting a retention period of 365 days, the company can ensure that every file in the repository is kept for a minimum of 1 year after its creation date.

A does not provide the same level of protection as compliance mode. In governance mode, there is a possibility for authorized users to remove the legal hold, potentially allowing objects to be modified or deleted.

C can restrict users from deleting or changing objects, but it does not enforce the retention period requirement. It also does not provide the same level of immutability and protection against accidental or malicious modifications.

D does not address the requirement of preventing users from modifying or deleting files. It provides a mechanism for tracking changes but does not enforce the desired access restrictions or retention period.

upvoted 4 times

norris81 3 years ago

Am I the only one to worry about leap years ?

upvoted 1 times

cheese929 3 years, 1 month ago

Selected Answer: B

In compliance mode, a protected object version can't be overwritten or deleted by any user, including the root user in your AWS account. When an object is locked in compliance mode, its retention mode can't be changed, and its retention period can't be shortened. Compliance mode helps ensure that an object version can't be overwritten or deleted for the duration of the retention period.

In governance mode, users can't overwrite or delete an object version or alter its lock settings unless they have special permissions. With governance mode, you protect objects against being deleted by most users, but you can still grant some users permission to alter the retention settings or delete the object if necessary.

In Governance mode, Objects can be deleted by some users with special permissions, this is against the requirement.

upvoted 2 times

darn 3 years, 1 month ago

Selected Answer: B

its B, legal hold has no retention

upvoted 3 times

A large media company hosts a web application on AWS. The company wants to start caching confidential media files so that users around the world will have reliable access to the files. The content is stored in Amazon S3 buckets. The company must deliver the content quickly, regardless of where the requests originate geographically.

Which solution will meet these requirements?

- A. Use AWS DataSync to connect the S3 buckets to the web application.
- B. Deploy AWS Global Accelerator to connect the S3 buckets to the web application.
- C. Deploy Amazon CloudFront to connect the S3 buckets to CloudFront edge servers. **Most Voted**
- D. Use Amazon Simple Queue Service (Amazon SQS) to connect the S3 buckets to the web application.

Correct Answer: C

Community vote distribution

C (100%)

Comments

rjam **Highly Voted** 3 years, 7 months ago

key :caching

Option C

upvoted 15 times

Guru4Cloud **Highly Voted** 2 years, 9 months ago

Selected Answer: C

The reasons are:

Amazon CloudFront is a content delivery network (CDN) that caches content at edge locations around the world. Connecting the S3 buckets containing the media files to CloudFront will cache the content at global edge locations. This provides fast reliable access to users everywhere by serving content from the nearest edge location. CloudFront integrates tightly with S3 for secure, durable storage. Global Accelerator improves availability and performance for TCP/UDP traffic, not HTTP-based content delivery. DataSync and SQS are not technologies for a global CDN like CloudFront.

upvoted 9 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: C

CloudFront is most suitable here and is designed to do this.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: C

Ans C - Cloudfront is designed to do this...

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: C

Amazon CloudFront to the rescue

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: C

CloudFront is a content delivery network (CDN) service provided by AWS. It caches content at edge locations worldwide, allowing users to access the content quickly regardless of their geographic location. By connecting the S3 to CloudFront, the media files can be cached at edge locations, ensuring reliable and fast delivery to users.

A. is a data transfer service that is not designed for caching or content delivery. It is used for transferring data between on-premises storage systems and AWS services.

B. is a service that improves the performance and availability of applications for global users. While it can provide fast and reliable access, it is not specifically designed for caching media files or connecting directly to S3.

D. is a message queue service that is not suitable for caching or content delivery. It is used for decoupling and coordinating message-based communication between different components of an application.

Therefore, the correct solution is option C, deploying CloudFront to connect the S3 to CloudFront edge servers.

upvoted 5 times

jackky3123213 2 years, 12 months ago

Global Accelerator does not support Edge Caching

upvoted 1 times

Bmarodi 3 years ago

Selected Answer: C

Option C is correct answer.

upvoted 1 times

warioverde 3 years, 2 months ago

As far as I understand, Global Accelerator does not have caching features, so CloudFront would be the recommended service for that purpose

upvoted 2 times

Americo32 3 years, 3 months ago

Selected Answer: C

C correto

upvoted 1 times

ProfXsamson 3 years, 4 months ago

C, Caching == Edge location == CloudFront

upvoted 3 times

career360guru 3 years, 5 months ago

Selected Answer: C

C right answer

upvoted 2 times

k1kavi1 3 years, 5 months ago

Selected Answer: C

Agreed

upvoted 1 times

Wpcorgan 3 years, 6 months ago

C is correct

upvoted 1 times

MyNamelsJulien 3 years, 7 months ago

Selected Answer: C

Answer is C

upvoted 1 times

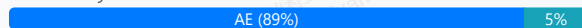
A company produces batch data that comes from different databases. The company also produces live stream data from network sensors and application APIs. The company needs to consolidate all the data into one place for business analytics. The company needs to process the incoming data and then stage the data in different Amazon S3 buckets. Teams will later run one-time queries and import the data into a business intelligence tool to show key performance indicators (KPIs).

Which combination of steps will meet these requirements with the LEAST operational overhead? (Choose two.)

- A. Use Amazon Athena for one-time queries. Use Amazon QuickSight to create dashboards for KPIs. **Most Voted**
- B. Use Amazon Kinesis Data Analytics for one-time queries. Use Amazon QuickSight to create dashboards for KPIs.
- C. Create custom AWS Lambda functions to move the individual records from the databases to an Amazon Redshift cluster.
- D. Use an AWS Glue extract, transform, and load (ETL) job to convert the data into JSON format. Load the data into multiple Amazon OpenSearch Service (Amazon Elasticsearch Service) clusters.
- E. Use blueprints in AWS Lake Formation to identify the data that can be ingested into a data lake. Use AWS Glue to crawl the source, extract the data, and load the data into Amazon S3 in Apache Parquet format. **Most Voted**

Correct Answer: AE

Community vote distribution



Comments

Wazhija **Highly Voted** 3 years, 7 months ago

Selected Answer: AE

I believe AE makes the most sense
upvoted 17 times

Six_Fingered_Jose **Highly Voted** 3 years, 7 months ago

Selected Answer: AE

yeah AE makes sense, only E is working with S3 here and questions wants them to be in S3
upvoted 12 times

Dharmarajan **Most Recent** 1 year, 4 months ago

Selected Answer: AE

A&E - least operational overhead - E over D
upvoted 1 times

sdelena 1 year, 4 months ago

Selected Answer: AE

A,E is ok
upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: AD

Ans A, D - A everyone seems to agree; I choose D over E because Parquet is aimed at columnar data - and that is not specified and may restrict query type access
upvoted 2 times

jaradat02 1 year, 10 months ago

Selected Answer: AE

AE satisfies the requirements that demand that the data should be stored in s3 and a one-time analytic will run on it.
upvoted 3 times

lofzee 2 years ago

Selected Answer: AE

C and D = too much overhead

B = incorrect because Athena is used for one time queries.

That leaves A and E

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: AE

A is a given due to Athena and QuickSight option.

Between C and E, the AWS Lake Formation is a more managed solution so it should have less operational overhead than writing Custom AWS Lambda.

AE should be preferred over AC.

upvoted 4 times

awsgeek75 2 years, 4 months ago

E is only confusing because of Apache Parquet format (like a grid?) what's the point of that in the context of this question?

upvoted 5 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AE

The reasons are:

AWS Lake Formation and Glue provide automated data lake creation with minimal coding. Glue crawlers identify sources and ETL jobs load to S3.

Athena allows ad-hoc queries directly on S3 data with no infrastructure to manage.

QuickSight provides easy cloud BI for dashboards.

Options C and D require significant custom coding for ETL and queries.

Redshift and OpenSearch would require additional setup and management overhead.

upvoted 9 times

Mia2009687 2 years, 11 months ago

Selected Answer: AE

It combines data from database and stream data, so data lake needs to be used.

And it wants to do one time query, so Athena is better.

upvoted 4 times

TTaws 2 years, 11 months ago

@Golcha once the data comes from different sources then you use GLUE

upvoted 3 times

Jeeva28 3 years ago

Selected Answer: AC

Less Overhead with option AC .No need to manage

upvoted 1 times

pentium75 2 years, 5 months ago

But C moves the data to Redshift while the question says you want it in S3 (and Athena from answer A also needs it in S3).

upvoted 4 times

Golcha 3 years, 1 month ago

Selected Answer: AC

No specific use case for GLUE

upvoted 1 times

TTaws 2 years, 11 months ago

once the data comes from different sources then you use GLUE

upvoted 3 times

pentium75 2 years, 5 months ago

C moves the data to Redshift while the question says you want it in S3 (and Athena from answer A also needs it in S3).

upvoted 2 times

TECHNOWARRIOR 3 years, 2 months ago

The Apache Parquet format is a performance-oriented, column-based data format designed for storage and retrieval. It is generally faster for reads than writes because of its columnar storage layout and a pre-computed schema that is written with the data into the files. AWS Glue's Parquet writer offers fast write performance and flexibility to handle evolving datasets. You can use AWS Glue to read Parquet files from Amazon S3 and from streaming sources as well as write Parquet files to Amazon S3. When using AWS Glue to build a data lake foundation, it automatically crawls your Amazon S3 data, identifies data formats, and then suggests schemas for use with other AWS analytic services[1][2][3][4].

upvoted 6 times

TECHNOWARRIOR 3 years, 2 months ago

ANSWER - AE: Amazon Athena is the best choice for running one-time queries on streaming data. Although Amazon Kinesis Data Analytics provides an easy and familiar standard SQL language to analyze streaming data in real-time, it is designed for continuous queries rather than one-time queries[1]. On the other hand, Amazon Athena is a serverless interactive query service that allows querying data in Amazon S3 using SQL. It is optimized for ad-hoc querying and is ideal for running one-time queries on streaming data[2]. AWS Lake Formation uses as a central place to have all your data for analytics purposes (E). Athena integrates perfectly with S3 and can make queries (A).

upvoted 5 times

jcramos 3 years, 2 months ago

Selected Answer: AE

AWS Lake Formation uses as a central place to have all your data for analytics purposes (E). Athena integrates perfectly with S3 and can make queries (A).

upvoted 5 times

jcramos 3 years, 2 months ago

Why S3 in Apache Parquet? <https://aws.amazon.com/about-aws/whats-new/2018/12/amazon-s3-announces-parquet-output-format-for-inventory/>

upvoted 2 times

JiyuKim 3 years, 4 months ago

Can anyone please explain me why B cannot be an answer?

upvoted 6 times

Shrestwt 3 years, 1 month ago

Kinesis Data Analytics is designed for continuous queries rather than one-time queries.

upvoted 8 times

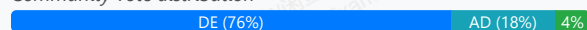
A company stores data in an Amazon Aurora PostgreSQL DB cluster. The company must store all the data for 5 years and must delete all the data after 5 years. The company also must indefinitely keep audit logs of actions that are performed within the database. Currently, the company has automated backups configured for Aurora.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

- A. Take a manual snapshot of the DB cluster.
- B. Create a lifecycle policy for the automated backups.
- C. Configure automated backup retention for 5 years.
- D. Configure an Amazon CloudWatch Logs export for the DB cluster. **Most Voted**
- E. Use AWS Backup to take the backups and to keep the backups for 5 years. **Most Voted**

Correct Answer: DE

Community vote distribution



Comments

JayBee65 **Highly Voted** 3 years, 5 months ago

I tend to agree D and E...

A - Manual task that can be automated, so why make life difficult?
 B - The maximum retention period is 35 days, so would not help
 C - The maximum retention period is 35 days, so would not help
 D - Only option that deals with logs, so makes sense
 E - Partially manual but only option that achieves the 5 year goal
 upvoted 54 times

aadityaravi8 2 years, 11 months ago

100% agree

upvoted 7 times

kmaneith **Highly Voted** 3 years, 6 months ago

Selected Answer: DE

dude trust me

upvoted 22 times

JayBee65 3 years, 5 months ago

No, please show your reasoning, you may be wrong. Remember, no one thinks they are wrong, but some always are :)

upvoted 16 times

Priyanshugpt486 2 years, 8 months ago

hehe... hehe

upvoted 2 times

bcd6d36 1 year, 1 month ago

I trust you

upvoted 1 times

Clouddon **Most Recent** 11 months, 1 week ago

Selected Answer: AE

A. Take a manual snapshot of the DB cluster.
 E. Use AWS Backup to take the backups and to keep the backups for 5 years.
 Explanation:

To meet retention and deletion requirements (5 years) and permanent audit log storage, here's what each selected option provides:

✓ A. Take a manual snapshot of the DB cluster

Manual snapshots do not expire automatically, unlike automated backups.

They can be retained as long as needed (e.g., up to 5 years) and deleted manually or via automation after that period.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: DE

Ans D, E

upvoted 1 times

lofzee 2 years ago

D + E I think.

D deals with the logs.

AWS Backup for backups...

Why make backups more difficult by not using the built in backup tool?

upvoted 3 times

rjjkc 2 years ago

Selected Answer: AD

I'd say "A" over "E" because in the option "E", it says use AWS Backup to take the "backups" not snapshot. "If you use AWS CLI, this is set using the parameter DeleteAfterDays. The retention period for snapshots can range between 1 day and 100 years (or indefinitely if you don't enter one), while the retention period for continuous backups can range from 1 day to 35 days. The creation date of a backup is the date the backup job started, not the date it completed. If your backup job doesn't complete on the same date it started, use the date on which it began to help calculate retention periods."

From here: <https://docs.aws.amazon.com/aws-backup/latest/devguide/creating-a-backup-plan.html>

upvoted 2 times

ManikRoy 2 years, 1 month ago

Selected Answer: DE

Agree with the reasoning of JayBee

upvoted 1 times

EMPERBACH 2 years, 1 month ago

Selected Answer: CD

C - C -

<https://www.bing.com/ck/a?!&&p=7e852c106834b9bdJmldtHM9MTcxMzM5ODQwMCZpZ3VpZD0xNzJiMThiOS1kM2RiLTZlZGEtMWNhZC0wYjRlZDlZDZmZDYmaW5zaWQ9NTQ0OQ&p=3&ver=2&hsh=3&fclid=172b18b9-d3db-6eda-1cad-0b4ed2bd6fd6&psq=does+Aurora+offer+auto+backup&u=a1aHR0cHM6Ly9kb2NzLmF3cy5hbWF6b24uY29tL0FtYXpvcjJEUy9sYXRlc3QvQXVyb3JhVXNlckd1aWRlOF1cm9yYS5NYW5hZ2luZy5CYWNrdXBzLmh0bWwJOn46dGV4dD1BdXJvcmlEIMjBiYWNRcyUyMHVwJTlweW91ciUyMGNsdXN0ZXIIMjB2b2x1bWUIMjBhdXRVbWFOaWNhbGx5JTlwYW5kLHRvJTlwYW55JTlwcG9pbmQIMjB3aXRoaW4IMjB0aGUIMjBiYWNRdXAIMjByZXRIbnRpb24IMjBwZXJpb2Qu&ntb=1>

D - Only answer for audit logs of activities on database

upvoted 2 times

jjcode 2 years, 3 months ago

My thoughts:

1. AWS backups is designed to make back ups
2. "configure backup retention for 5 years" with what? a script? maybe AWS backups???? are the back ups done with DD and stored in S3? i cannot trust this answer
3. "Take a manual snapshot of the DB cluster" this is not an amazon best practice they want us to use their tools AWS backups
4. "create a life cycle policy" assuming the back ups are stored in S3 (which is not a best practice) cannot trust this

that leaves D and E

upvoted 2 times

Ruffyt 2 years, 6 months ago

D AND E- makes more sense as we automate backups in Aurora DB

- Export data to CloudWatch to capture all log events and configure CloudWatch to retain logs indefinitely.

upvoted 2 times

awashenko 2 years, 8 months ago

Selected Answer: DE

D and E.

A would work as well, but D is the better option as its automated.

E is the only option that gets you to the 5 year retention.

upvoted 2 times

kambarami 2 years, 9 months ago

D AND E- makes more sense as we automate backups in Aurora DB

- Export data to CloudWatch to capture all log events and configure CloudWatch to retain logs indefinitely.

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: DE

DE makes more sense

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: CD

The reasons are:

Configuring the automated backups for the Aurora PostgreSQL DB cluster to retain backups for 5 years will meet the requirement to store all data for that duration.

Exporting the database logs to CloudWatch Logs will capture the audit logs of actions performed in the database. CloudWatch Logs retention can be configured to store logs indefinitely.

This meets the need to keep audit logs available beyond the 5 year data retention period.

Additional manual snapshots or using AWS Backup for backups is not necessary since automated backups are already enabled.

A lifecycle policy is useful for transitioning storage classes but does not apply here for a set 5 year retention.

upvoted 2 times

neverdie 3 years, 2 months ago

Selected Answer: AD

Automated backup is limited 35 days

upvoted 3 times

Training4aBetterLife 3 years, 4 months ago

Selected Answer: DE

Previously, you had to create custom scripts to automate backup scheduling, enforce retention policies, or consolidate backup activity for manual Aurora cluster snapshots, especially when coordinating backups across AWS services. With AWS Backup, you gain a fully managed, policy-based backup solution with snapshot scheduling and snapshot retention management. You can now create, manage, and restore Aurora backups directly from the AWS Backup console for both PostgreSQL-compatible and MySQL-compatible versions of Aurora.

To get started, select an Amazon Aurora cluster from the AWS Backup console and take an on-demand backup or simply assign the cluster to a backup plan.

upvoted 5 times

Training4aBetterLife 3 years, 4 months ago

https://aws.amazon.com/about-aws/whats-new/2020/06/amazon-aurora-snapshots-can-be-managed-via-aws-backup/?nc1=h_ls

upvoted 3 times

Zerotn3 3 years, 5 months ago

Selected Answer: DE

A is not a valid option for meeting the requirements. A manual snapshot of the DB cluster is a point-in-time copy of the data in the cluster. While taking manual snapshots can be useful for creating backups of the data, it is not a reliable or efficient way to meet the requirement of storing all the data for 5 years and deleting it after 5 years. It would be difficult to ensure that manual snapshots are taken regularly and retained for the required period of time. It is recommended to use a fully managed backup service like AWS Backup, which can automate and centralize the process of taking and retaining backups.

upvoted 4 times

Zerotn3 3 years, 5 months ago

Sorry, B and E that correct

B. Create a lifecycle policy for the automated backups.

This would ensure that the backups taken using AWS Backup are retained for the desired period of time.

upvoted 1 times

JayBee65 3 years, 5 months ago

I think a lifecycle policy would only keep backups for 35 days

upvoted 3 times

awashenko 2 years, 8 months ago

Thats not correct (i thought it was but I went and looked it up) Aurora only keeps backups from 1-35 days.

upvoted 2 times

A solutions architect is optimizing a website for an upcoming musical event. Videos of the performances will be streamed in real time and then will be available on demand. The event is expected to attract a global online audience.

Which service will improve the performance of both the real-time and on-demand streaming?

- A. Amazon CloudFront **Most Voted**
- B. AWS Global Accelerator
- C. Amazon Route 53
- D. Amazon S3 Transfer Acceleration

Correct Answer: A

Community vote distribution

A (75%)

B (25%)

Comments

Nigma **Highly Voted** 3 years, 6 months ago
A is right

You can use CloudFront to deliver video on demand (VOD) or live streaming video using any HTTP origin

Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses
upvoted 45 times

mr123dd **Highly Voted** 2 years, 5 months ago

Selected Answer: A

website = http = cloudfront, if it is UDP, then global accelerator
upvoted 12 times

bora4motion 1 year ago

I thought that a live video stream is UDP.
upvoted 2 times

Dharmarajan **Most Recent** 1 year, 4 months ago

Selected Answer: A

A because CloudFront can be used for video streaming. NOT GA because GA is good for services with lower level network protocols - and their speeds.
upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: A

Ans A - CloudFront delivers video on demand or live streaming video using any HTTP origin
upvoted 2 times

dragongoseki 1 year, 11 months ago

Selected Answer: A

A is right answer.
upvoted 1 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: A

A for sure
upvoted 1 times

trinh_le 1 year, 12 months ago

AD is correct. link here <https://docs.aws.amazon.com/aws-backup/latest/devguide/creating-a-backup-plan.html>

upvoted 1 times

lofzee 2 years ago

Selected Answer: A

A.

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/on-demand-streaming-video.html>

upvoted 2 times

ManikRoy 2 years, 1 month ago

Selected Answer: A

use CloudFront to deliver video on demand (VOD) or live streaming video using any HTTP origin

upvoted 2 times

lanceshen 2 years, 3 months ago

Selected Answer: B

Option A (Amazon CloudFront) is a content delivery network (CDN) service that can improve the performance of on-demand streaming by caching and delivering content from edge locations. While it can accelerate on-demand streaming, it may not provide the same level of optimization for real-time streaming as AWS Global Accelerator.

upvoted 3 times

MrPCarrot 2 years, 4 months ago

A is pperfect <https://d1.awsstatic.com/whitepapers/amazon-cloudfront-for-media.pdf>

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/on-demand-streaming-video.html>

CloudFront solves the problem of streaming over Amazon CDN on global scale.

"B" AWS Global Accelerator won't be suitable for streaming from web server as it does not provide edge caching like CDN. Global Accelerator only points the user to nearest functioning node which is helpful for real-time streaming but not best for on-demand.

upvoted 2 times

viru 2 years, 5 months ago

Selected Answer: A

<https://aws.amazon.com/cloudfront/streaming/>

upvoted 3 times

yayaayzo 2 years, 6 months ago

A IS THE RIGHT ANS

. Amazon CloudFront is a content delivery network (CDN) service offered by AWS. It is designed to deliver data, including videos and other media files, with low latency and high transfer speeds. This is a suitable option for optimizing website performance, especially for streaming content globally.

upvoted 2 times

MiniYang 2 years, 6 months ago

Selected Answer: B

Although CloudFront is a content delivery network (CDN) that can provide low-latency and high-performance content delivery, its performance for real-time streaming and on-demand streaming may not be as professional as AWS Global Accelerator

upvoted 1 times

awsgeek75 2 years, 4 months ago

<https://aws.amazon.com/global-accelerator/>

"AWS Global Accelerator is a networking service that helps you improve the availability, performance, and security of your public applications.

Global Accelerator provides two global static public IPs that act as a fixed entry point to your application endpoints, such as Application Load Balancers, Network Load Balancers, Amazon Elastic Compute Cloud (EC2) instances, and elastic IPs."

It is Anycast which connect user to closest resource on your server like ALB etc or ergional services.

CloudFront is CDN and pushes your content to edge locations near the user. This solves all issues of latency and performance

upvoted 1 times

Ruffyit 2 years, 6 months ago

You can use CloudFront to deliver video on demand (VOD) or live streaming video using any HTTP origin

Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses

upvoted 3 times

mhka1988 2 years, 7 months ago

Selected Answer: A

CloudFront offers several options for streaming your media to global viewers—both pre-recorded files and live events.

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/IntroductionUseCases.html#IntroductionUseCasesStreaming>

For video on demand (VOD) streaming, you can use CloudFront to stream in common formats such as MPEG DASH, Apple HLS, Microsoft Smooth Streaming, and CMAF, to any device.

For broadcasting a live stream, you can cache media fragments at the edge, so that multiple requests for the manifest file that delivers the fragments in the right order can be combined, to reduce the load on your origin server.

upvoted 4 times

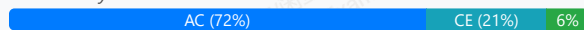
A company is running a publicly accessible serverless application that uses Amazon API Gateway and AWS Lambda. The application's traffic recently spiked due to fraudulent requests from botnets.

Which steps should a solutions architect take to block requests from unauthorized users? (Choose two.)

- A. Create a usage plan with an API key that is shared with genuine users only. **Most Voted**
- B. Integrate logic within the Lambda function to ignore the requests from fraudulent IP addresses.
- C. Implement an AWS WAF rule to target malicious requests and trigger actions to filter them out. **Most Voted**
- D. Convert the existing public API to a private API. Update the DNS records to redirect users to the new API endpoint.
- E. Create an IAM role for each user attempting to access the API. A user will assume the role when making the API call.

Correct Answer: AC

Community vote distribution



Comments

jdr75 **Highly Voted** 3 years, 2 months ago

Selected Answer: CE

C) WAF has bot identification and remedial tools, so it's CORRECT.

A) remember the question : "...block requests from unauthorized users?" -- an api key is involved in a authorization process. It's not the more secure process, but it's better than an totoally anonymous process. If you don't know the key, you can't authenticate. So the bots, at least the first days/weeks could not access the service (at the end they'll do, cos' the key will be spread informally). So it's CORRECT.

B) Implement a logic in the Lambda to detect fraudulent ip's is almost impossible, cos' it's a dynamic and changing pattern that you cannot handle easily.

D) creating a rol is not going to imply be more protected from unauth. request, because a rol is a "principal", it's not involved in the authorization process.

upvoted 10 times

debasishdtta 2 years, 5 months ago

Don't use API keys for authentication or authorization to control access to your APIs. If you have multiple APIs in a usage plan, a user with a valid API key for one API in that usage plan can access all APIs in that usage plan. Instead, to control access to your API, use an IAM role, a Lambda authorizer, or an Amazon Cognito user pool.

upvoted 2 times

pentium75 2 years, 5 months ago

E "An IAM role for EACH (!) user ATTEMPTING (!) to access the API"? Hello no.

upvoted 5 times

awashenko **Highly Voted** 2 years, 8 months ago

Selected Answer: AC

Agree A and C

I don't see how E is feasible as its a public API. How would you create an IAM role for each user?

upvoted 7 times

90c3163 10 months, 1 week ago

Cognito Identity Pools. But thats a lot of overhead.

upvoted 2 times

Mrigranj12 **Most Recent** 1 year, 4 months ago

Selected Answer: AC

Wrong choices: B-> Integrating logic in lambda is not feasible as you will have to feed data which will increase its size, execution time and therefore overall cost.

D-> If it happens then the public will not be able to access it as it is private only clients inside VPC will be able to access it.

E-> You are just giving them access

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: AC

Ans A, C -

A: using API keys and usage plans restricts access to your API to users who have the key, limiting fraudulent access.

C: designed to fight bots

upvoted 2 times

diddy99 1 year, 11 months ago

Selected Answer: AC

C) Everyone agrees on C

B) Almost impossible cos how do you detect fraudulent IP address from a publicly accessible application

D) It's a publicly accessible application, converting the API to a private one defeats the purpose

E) IAM role for each user trying to access a publicly accessible API is impossible. It like creating an IAM for each user that tries to use google AUTH for their website

A) By implementing API keys and usage plans, you can restrict access to your API to only those users who possess the key, helping to limit fraudulent access.

upvoted 2 times

jerry1 1 year, 2 months ago

but i think both option A and E imply that the api access actually have a targeted user
if A is possible, why cant E be an approach?

upvoted 1 times

dragongoseki 1 year, 11 months ago

Selected Answer: AC

AC is right answer.

upvoted 1 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: AC

AC for sure

upvoted 2 times

hb0011 2 years ago

Do the people voting E realize how insane that is? Creating a local IAM user in your account for every user that needs to access the API. No just... no.

upvoted 3 times

EMPERBACH 2 years, 1 month ago

Selected Answer: BC

B. Integrate logic within the Lambda function to ignore the requests from fraudulent IP addresses. -> you can think about CORS script write on Lambda to prevent fraudulent IP addresses.

C. Implement an AWS WAF rule to target malicious requests and trigger actions to filter them out. -> No comment here as it can use to filter traffic

upvoted 1 times

MrPCarrot 2 years, 4 months ago

C and D are the perfect answers

upvoted 2 times

debasishdtt 2 years, 5 months ago

Don't use API keys for authentication or authorization to control access to your APIs. If you have multiple APIs in a usage plan, a user with a valid API key for one API in that usage plan can access all APIs in that usage plan. Instead, to control access to your API, use an IAM role, a Lambda authorizer, or an Amazon Cognito user pool.

upvoted 1 times

awsgeek75 2 years, 5 months ago

Selected Answer: CD

I'll throw a curveball over here. "C" is a given as WAF rules can target malicious usage. For example:

<https://docs.aws.amazon.com/whitepapers/latest/aws-best-practices-ddos-resiliency/aws-waf-ip-reputation.html>

"D" Convert existing public API to a private API. This part is same as A. The additional bit over here is to change the DNS record to a new API endpoint which blocks the requests from unauthorised users also. The unauthorised users will not be redirected from public to private API endpoint. am assuming that the public API endpoint will be used for authorisation and only authorised users will be redirected to private endpoint. This is more robust as the actual API (private endpoint) never gets hit with requests from unauthorised bots and WAF redirects it back to public URL.

Happy to be corrected and challenged
upvoted 4 times

sidharthwader 2 years, 3 months ago

It's a globally published API if you make it private how do other people access it ? A would be the better solution than D
upvoted 2 times

ale_brd_111 2 years, 6 months ago

Selected Answer: AC

The combination of using an API key and implementing an AWS WAF rule provides the most comprehensive and effective way to block requests from unauthorized users and protect the company's serverless application from botnet attacks.

upvoted 4 times

MiniYang 2 years, 6 months ago

Selected Answer: CE

A. Create plans using API keys shared only with real users: While using API keys is a standard way to control access to APIs, using API keys alone may not completely prevent attacks from botnets. Malicious request.

B. Incorporate logic in the Lambda function to ignore requests from fraudulent IP addresses: This may be a solution, but filtering that relies more on IP addresses may not be as flexible as using AWS WAF.

D. Convert an existing public API to a private API. Update DNS records to redirect users to the new API endpoint: This approach makes the API private, but requires user redirects and may inconvenience existing users.

upvoted 1 times

Ruffyyit 2 years, 6 months ago

C) WAF has bot identification and remedial tools, so it's CORRECT.

A) remember the question : "...block requests from unauthorized users?" -- an api key is involved in a authorization process. It's not the more secure process, but it's better than an totoally anonymous process. If you don't know the key, you can't authenticate. So the bots, at least the first days/weeks could not access the service (at the end they'll do, cos' the key will be spread informally). So it's CORRECT.

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: AC

AWS WAF rule to target and filter out malicious requests and API key to authorize users.

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AC

The reasons are:

An API key with a usage plan limits access to only authorized apps and users. This prevents general public access.

WAF rules can identify and block malicious bot traffic through pattern matching and IP reputation lists.

Together, the API key and WAF provide preventative and detective controls against unauthorized requests.

The other options add complexity or are reactive. IAM roles per user is not feasible for a public API.

Ignoring requests in Lambda and changing DNS are response actions after an attack.

upvoted 3 times

An ecommerce company hosts its analytics application in the AWS Cloud. The application generates about 300 MB of data each month. The data is stored in JSON format. The company is evaluating a disaster recovery solution to back up the data. The data must be accessible in milliseconds if it is needed, and the data must be kept for 30 days.

Which solution meets these requirements MOST cost-effectively?

- A. Amazon OpenSearch Service (Amazon Elasticsearch Service)
- B. Amazon S3 Glacier
- C. Amazon S3 Standard **Most Voted**
- D. Amazon RDS for PostgreSQL

Correct Answer: C

Community vote distribution

C (90%)

10%

Comments

babaxoxo **Highly Voted** 3 years, 6 months ago

Selected Answer: C

Ans C:

Cost-effective solution with milliseconds of retrieval -> it should be s3 standard

upvoted 14 times

Guru4Cloud **Highly Voted** 2 years, 9 months ago

Selected Answer: C

The reasons are:

S3 Standard provides high durability and availability for storage

It allows millisecond access to retrieve objects

Objects can be stored for any duration, meeting the 30 day retention need

Storage costs are low, around \$0.023 per GB/month

OpenSearch and RDS require running and managing a cluster for DR storage

Glacier has lower cost but retrieval time is too high at 3-5 hours

S3 Standard's simplicity, high speed access, and low cost make it optimal for this small DR dataset that needs to be accessed quickly

upvoted 6 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: C

S3 Standard is the most cheapest.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: C

Ans C - S3 std: cost-effective, milliseconds of retrieval

upvoted 2 times

PaulGa 1 year, 7 months ago

Selected Answer: C

S3 Standard

upvoted 1 times

soufiyane 2 years, 2 months ago

Selected Answer: C

also s3 can store any form of data

upvoted 1 times

pentium75 2 years, 5 months ago

Selected Answer: C

Only Glacier class that would meet the requirement is Instant Retrieval, but it has 90 days minimum storage time which would kill the cost savings.

upvoted 5 times

xdkonorek2 2 years, 7 months ago

Selected Answer: B

300 MB / month storage without retrieval when file is single 300 MB file:

S3 Standard cost (Monthly): 0.01 USD

S3 Standard cost (Upfront): 0.00 USD

S3 Glacier Instant Retrieval cost (Monthly): 0.00 USD

if it was 3GB:

3 GB / month storage without retrieval when file is single 3GB file:

S3 Standard cost (Monthly): 0.07 USD

S3 Standard cost (Upfront): 0.00 USD

S3 Glacier Instant Retrieval cost (Monthly): 0.01 USD

When assumed no retrieval is required because it's DR solution, and it's a single file, Glacier Instant Retrieval wins, and when they mention S3 glacier we must choose one of the sub-category

upvoted 1 times

pentium75 2 years, 5 months ago

Only Glacier Instant Retrieval (which is not mentioned in B) would meet the access requirement, but "Objects that are archived to S3 Glacier Instant Retrieval and S3 Glacier Flexible Retrieval are charged for a minimum storage duration of 90 days."

upvoted 2 times

xdkonorek2 2 years, 7 months ago

but if 300 MB is divided into smaller files situation changes which is probably the case..

300 MB / month storage without retrieval when files are 600x0.5MB :

S3 Standard cost (Monthly): 0.01 USD

S3 Standard cost (Upfront): 0.00 USD

S3 Glacier Instant Retrieval cost (Monthly): 0.01 USD

S3 Glacier Instant Retrieval cost (Upfront): 0.02 USD

3 GB / month storage without retrieval when files are 6000x0.5 MB file:

S3 Standard cost (Monthly): 0.10 USD

S3 Standard cost (Upfront): 0.03 USD

S3 Glacier Instant Retrieval cost (Monthly): 0.13 USD

S3 Glacier Instant Retrieval cost (Upfront): 0.25 USD

upvoted 4 times

Its_SaKar 2 years, 9 months ago

Selected Answer: C

Answer is not B because S3 glacier and S3 glacier instant storage are two different types of storage class. So, answer here is C: S3 standard

upvoted 4 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: C

Data must be accessible in milliseconds and must be kept for 30 days = Amazon S3 Standard

upvoted 2 times

chanchal133 2 years, 9 months ago

Selected Answer: C

ANS - C

upvoted 1 times

Nazmul123 2 years, 10 months ago

Selected Answer: C

<https://aws.amazon.com/s3/storage-classes/glacier/instant-retrieval/>

upvoted 2 times

pentium75 2 years, 5 months ago

Not mentioned in B, and 90 days minimum storage time.

upvoted 1 times

cookieMr 2 years, 11 months ago

S3 Standard is a highly durable and scalable storage option suitable for backup and disaster recovery purposes. It offers millisecond access to data when needed and provides durability guarantees. It is also cost-effective compared to other storage options like OpenSearch Service, S3 Glacier, and RDS for PostgreSQL, which may have higher costs or longer access times for retrieving the data.

A. OpenSearch Service (Elasticsearch Service): While it offers fast data retrieval, it may incur higher costs compared to storing data directly in S3, especially considering the amount of data being generated.

B. S3 Glacier: While it provides long-term archival storage at a lower cost, it does not meet the requirement of immediate access in milliseconds. Retrieving data from Glacier typically takes several hours.

D. RDS for PostgreSQL: While it can be used for data storage, it may be overkill and more expensive for a backup and disaster recovery solution compared to S3 Standard, which is more suitable and cost-effective for storing and retrieving data.

upvoted 3 times

joehong 3 years ago

Selected Answer: B

<https://aws.amazon.com/s3/storage-classes/glacier/instant-retrieval/>

upvoted 4 times

pentium75 2 years, 5 months ago

"Objects that are archived to S3 Glacier Instant Retrieval and S3 Glacier Flexible Retrieval are charged for a minimum storage duration of 90 days." Also Instant Retrieval is not mentioned in B.

upvoted 2 times

KZM 3 years, 4 months ago

A. Incorrect

Amazon OpenSearch Service (Amazon Elasticsearch Service) is designed for full-text search and analytics, but it may not be the most cost-effective solution for this use case

B. Incorrect

S3 Glacier is a cold storage solution that is designed for long-term data retention and infrequent access.

C. Correct

S3 standard is cost-effective and meets the requirement. S3 Standard allows for data retention for a specific number of days.

D. PostgreSQL is a relational database service and may not be the most cost-effective solution.

upvoted 5 times

ngochieu276 3 years, 5 months ago

Selected Answer: B

S3 Glacier Instant Retrieval – Use for archiving data that is rarely accessed and requires milliseconds retrieval.

<https://docs.aws.amazon.com/amazonglacier/latest/dev/introduction.html>

upvoted 4 times

career360guru 3 years, 5 months ago

Selected Answer: C

Option C

upvoted 1 times

A company has a small Python application that processes JSON documents and outputs the results to an on-premises SQL database. The application runs thousands of times each day. The company wants to move the application to the AWS Cloud. The company needs a highly available solution that maximizes scalability and minimizes operational overhead.

Which solution will meet these requirements?

- A. Place the JSON documents in an Amazon S3 bucket. Run the Python code on multiple Amazon EC2 instances to process the documents. Store the results in an Amazon Aurora DB cluster.
- B. Place the JSON documents in an Amazon S3 bucket. Create an AWS Lambda function that runs the Python code to process the documents as they arrive in the S3 bucket. Store the results in an Amazon Aurora DB cluster. **Most Voted**
- C. Place the JSON documents in an Amazon Elastic Block Store (Amazon EBS) volume. Use the EBS Multi-Attach feature to attach the volume to multiple Amazon EC2 instances. Run the Python code on the EC2 instances to process the documents. Store the results on an Amazon RDS DB instance.
- D. Place the JSON documents in an Amazon Simple Queue Service (Amazon SQS) queue as messages. Deploy the Python code as a container on an Amazon Elastic Container Service (Amazon ECS) cluster that is configured with the Amazon EC2 launch type. Use the container to process the SQS messages. Store the results on an Amazon RDS DB instance.

Correct Answer: B

Community vote distribution

B (94%)

6%

Comments

babaxoxo **Highly Voted** 3 years, 6 months ago

Selected Answer: B

solution should remove operation overhead -> s3 -> lambda -> aurora
upvoted 19 times

markw92 2 years, 11 months ago

Aurora supports mysql and postgresql but question has database sql server. So, that eliminates B. So, the other logical answer is D. IMHO. Btw, i also thought the answer is B and started re-reading question carefully.

upvoted 6 times

JJJJIXI 2 years, 8 months ago

sql database, not sql server

upvoted 6 times

Zerotn3 **Highly Voted** 3 years, 5 months ago

Selected Answer: B

By placing the JSON documents in an S3 bucket, the documents will be stored in a highly durable and scalable object storage service. The use of AWS Lambda allows the company to run their Python code to process the documents as they arrive in the S3 bucket without having to worry about the underlying infrastructure. This also allows for horizontal scalability, as AWS Lambda will automatically scale the number of instances of the function based on the incoming rate of requests. The results can be stored in an Amazon Aurora DB cluster, which is a fully-managed, high-performance database service that is compatible with MySQL and PostgreSQL. This will provide the necessary durability and scalability for the results of the processing.

upvoted 13 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: B

Using Lambda, S3 and Aurora is the most suitable and provides least operational overhead for this problem.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: D

Ans D - because none of the other answers guarantee FIFO. What if a subsequent JSON operation is intended to update an existing database record (which is highly likely) - then the wrong change would be applied

upvoted 2 times

soufiyane 2 years, 2 months ago

Selected Answer: B

b is the right answer it very obvious

upvoted 1 times

Anantvir 2 years, 5 months ago

Guys I have a question.

We dont know how long the processing of JSON documents is going to take. What if that processing takes more than 15 min ? Lambda can run only for 15 correct ? Based on this the answer could be D

Please correct my understanding.

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

"D" has a lot of moving parts and more operational overhead even if each part is a managed service in itself. Also, if something can be done with Lambda, don't use an EC2 instance in any form as it always increases operational overhead (compared to Lambda).

upvoted 3 times

David_Ang 2 years, 7 months ago

Selected Answer: B

"D" is just like the most complex one, sometimes the admin make mistakes and don't realize. lambda is a service make for this

upvoted 2 times

Mandar15 2 years, 8 months ago

Selected Answer: B

B is correc

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

Main requirement is: 'scalability and minimized operational overhead' = serverless = Amazon S3 bucket, AWS Lambda function, Amazon Aurora DB cluster

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

- Using Lambda functions triggered by S3 events allows the Python code to automatically scale up and down based on the number of incoming JSON documents. This provides high availability and maximizes scalability.

- Storing the results in an Amazon Aurora DB cluster provides a managed, scalable, and highly available database.

- This serverless approach minimizes operational overhead since Lambda and Aurora handle provisioning infrastructure, deploying code, monitoring, patching, etc.

upvoted 3 times

aadityaravi8 2 years, 11 months ago

The answer is B. Place the JSON documents in an Amazon S3 bucket. Create an AWS Lambda function that runs the Python code to process the documents as they arrive in the S3 bucket. Store the results in an Amazon Aurora DB cluster.

This solution is highly available because Lambda functions are automatically scaled up or down based on the number of requests they receive. It is also scalable because you can easily add more Lambda functions to process more documents. Finally, it minimizes operational overhead because you do not need to manage any EC2 instances.

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: B

Using Lambda eliminates the need to manage and provision servers, ensuring scalability and minimizing operational overhead. S3 provides durable and highly available storage for the JSON documents. Lambda can be triggered automatically whenever new documents are added to the S3 bucket, allowing for real-time processing. Storing the results in an Aurora DB cluster ensures high availability and scalability for the processed data. This solution leverages serverless architecture, allowing for automatic scaling and high availability without the need for managing infrastructure, making it the most suitable choice.

A. This option requires manual management and scaling of EC2 instances, resulting in higher operational overhead and complexity.

C. This approach still involves manual management and scaling of EC2 instances, increasing operational complexity and overhead.

D. This solution requires managing and scaling an ECS cluster, adding operational overhead and complexity. Utilizing SQS adds complexity to the system, requiring custom handling of message consumption and processing in the Python code.

upvoted 5 times

Bmarodi 3 years ago

Selected Answer: B

Keywords here are : "maximizes scalability and minimizes operational overhead, hence option B is correct answer.

upvoted 2 times

channn 3 years, 2 months ago

Selected Answer: D

i vote for D as 'on-premises SQL database' is not mysql/postgre which can replace by aurora

upvoted 2 times

pentium75 2 years, 5 months ago

Why not? It's a "SQL database", NOT necessarily Microsoft SQL Server. But even if it would be SQL server, that could be migrated to Aurora.

upvoted 1 times

kerin 3 years, 3 months ago

B is the best option. <https://aws.amazon.com/rds/aurora/>

upvoted 1 times

mp165 3 years, 5 months ago

Selected Answer: B

agree...B is the best option S3, Lambda , Aurora.

upvoted 2 times

A company wants to use high performance computing (HPC) infrastructure on AWS for financial risk modeling. The company's HPC workloads run on Linux. Each HPC workflow runs on hundreds of Amazon EC2 Spot Instances, is short-lived, and generates thousands of output files that are ultimately stored in persistent storage for analytics and long-term future use.

The company seeks a cloud storage solution that permits the copying of on-premises data to long-term persistent storage to make data available for processing by all EC2 instances. The solution should also be a high performance file system that is integrated with persistent storage to read and write datasets and output files.

Which combination of AWS services meets these requirements?

- A. Amazon FSx for Lustre integrated with Amazon S3 **Most Voted**
- B. Amazon FSx for Windows File Server integrated with Amazon S3
- C. Amazon S3 Glacier integrated with Amazon Elastic Block Store (Amazon EBS)
- D. Amazon S3 bucket with a VPC endpoint integrated with an Amazon Elastic Block Store (Amazon EBS) General Purpose SSD (gp2) volume

Correct Answer: A

Community vote distribution

A (100%)

Comments

Marge_Simpson **Highly Voted** 3 years, 6 months ago

Selected Answer: A

If you see HPC and Linux both in the question.. Pick Amazon FSx for Lustre
upvoted 45 times

HayLLIHuK 3 years, 5 months ago

yeap, you're right!

upvoted 5 times

aba2s **Highly Voted** 3 years, 5 months ago

Selected Answer: A

Additional keywords: make data available for processing by all EC2 instances ==> FSx

In absence of EFS, it should be FSx. Amazon FSx For Lustre provides a high-performance, parallel file system for hot data

upvoted 11 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: A

Lustre is the most suitable when HPC is required.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: A

Ans A - HPC so its FSx and Lustre

upvoted 3 times

MrPCarrot 2 years, 4 months ago

A is the answer because Amazon FSx for Lustre provides a high-performance, scalable file system optimized for compute-intensive workloads like HPC.

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

Lustre is default when HPC is involved. <https://aws.amazon.com/fsx/lustre/>

B: mentions Windows and no-one asked for it.

C: S3 Glacier is too slow for HPC

D: I don't think this is possible, unless I'm mistake, how can you connect a VPC endpoint to EBS without an EC2 kind of instance?

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

HPC workloads running on Linux = Amazon FSx for Lustre

upvoted 3 times

Jeyaluxshan 2 years, 9 months ago

High performance - Lustre

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

The reasons are:

Amazon FSx for Lustre provides a high-performance, scalable file system optimized for compute-intensive workloads like HPC. It has native integration with Amazon S3.

Data can be copied from on-premises to an S3 bucket, acting as persistent long-term storage.

The FSx for Lustre file system can then access the S3 data for high speed processing of datasets and output files.

FSx for Lustre is designed for the Linux environments used in this HPC workload.

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: A

FSx for Lustre is a high-performance file system optimized for compute-intensive workloads. It provides scalable, parallel access to data and is suitable for HPC applications.

By integrating FSx for Lustre with S3, you can easily copy on-premises data to long-term persistent storage in S3, making it available for processing by EC2 instances.

S3 serves as the durable and highly scalable object storage for storing the output files, allowing for analytics and long-term future use.

Option B, FSx for Windows File Server, is not suitable because the workloads run on Linux, and this option is designed for Windows file sharing.

Option C, S3 Glacier integrated with EBS, is not the best choice as it is a low-cost archival storage service and not optimized for high-performance file system requirements.

Option D, using an S3 bucket with a VPC endpoint integrated with an Amazon EBS General Purpose SSD (gp2) volume, does not provide the required high-performance file system capabilities for HPC workloads.

upvoted 5 times

Bmarodi 3 years ago

Selected Answer: A

Option A is right answer.

upvoted 1 times

kerin 3 years, 3 months ago

FSx for Lustre makes it easy and cost-effective to launch and run the popular, high-performance Lustre file system. You use Lustre for workloads where speed matters, such as machine learning, high performance computing (HPC), video processing, and financial modeling.

Amazon FSx for Lustre is integrated with Amazon S3.

upvoted 3 times

SilentMilli 3 years, 5 months ago

Selected Answer: A

Amazon FSx for Lustre integrated with Amazon S3

upvoted 2 times

techhb 3 years, 5 months ago

Selected Answer: A

A is right choice here.

upvoted 1 times

career360guru 3 years, 5 months ago

Selected Answer: A

Option A is the best high performance storage with integration to S3

upvoted 2 times

wly_al 3 years, 5 months ago

Selected Answer: A

requirement is File System and workload running on linux. so S3 and FSx for windows is not an option

upvoted 2 times

Shasha1 3 years, 6 months ago

A

The Amazon FSx for Lustre service is a fully managed, high-performance file system that makes it easy to move and process large amounts of data quickly and cost-effectively. It provides a fully managed, cloud-native file system with low operational overhead, designed for massively parallel processing and high-performance workloads. The Lustre file system is a popular, open source parallel file system that is well-suited for a variety of applications such as HPC, image processing, AI/ML, media processing, data analytics, and financial modeling, among others. With Amazon FSx for Lustre, you can quickly create and configure new file systems in minutes, and easily scale the size of your file system up or down

upvoted 3 times

A company is building a containerized application on premises and decides to move the application to AWS. The application will have thousands of users soon after it is deployed. The company is unsure how to manage the deployment of containers at scale. The company needs to deploy the containerized application in a highly available architecture that minimizes operational overhead.

Which solution will meet these requirements?

A. Store container images in an Amazon Elastic Container Registry (Amazon ECR) repository. Use an Amazon Elastic Container Service (Amazon ECS) cluster with the AWS Fargate launch type to run the containers. Use target tracking to scale automatically based on demand. **Most Voted**

B. Store container images in an Amazon Elastic Container Registry (Amazon ECR) repository. Use an Amazon Elastic Container Service (Amazon ECS) cluster with the Amazon EC2 launch type to run the containers. Use target tracking to scale automatically based on demand.

C. Store container images in a repository that runs on an Amazon EC2 instance. Run the containers on EC2 instances that are spread across multiple Availability Zones. Monitor the average CPU utilization in Amazon CloudWatch. Launch new EC2 instances as needed.

D. Create an Amazon EC2 Amazon Machine Image (AMI) that contains the container image. Launch EC2 instances in an Auto Scaling group across multiple Availability Zones. Use an Amazon CloudWatch alarm to scale out EC2 instances when the average CPU utilization threshold is breached.

Correct Answer: A

Community vote distribution

A (100%)

Comments

goatbernard **Highly Voted** 3 years, 6 months ago

Selected Answer: A

AWS Fargate

upvoted 17 times

awsgeek75 **Highly Voted** 2 years, 5 months ago

Selected Answer: A

A is minimal overhead.

B has EC2 overhead

C EC2 instance overhead + container repository running on EC2 overhead

D AMI, CloudWatch alarm is overhead++

upvoted 8 times

awsgeek75 2 years, 4 months ago

Also, simply speaking, if the company is unsure how to manage deployed containers then Fargate is the only choice.

upvoted 2 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: A

AWS Fargate provides least operational overhead.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: A

Ans A - AWS Fargate - it does it all

upvoted 2 times

lofzee 2 years ago

Selected Answer: A

fargate is the compute managed for you.. A
upvoted 2 times

Ruffyit 2 years, 6 months ago

ECR+ECS+Fargate = Less overhead
upvoted 4 times

ACloud_Guru15 2 years, 7 months ago

Selected Answer: A

ECR+ECS+Fargate = Less overhead
upvoted 3 times

Sindokuhlep 2 years, 7 months ago

Selected Answer: A

Fargate
upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

Highly available architecture that minimizes operational overhead = Serverless = Elastic Container Registry, Amazon Elastic Container Service with AWS Fargate launch type
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

Using ECR provides a fully managed container image registry.
ECS with Fargate launch type allows running containers without managing servers or clusters. Fargate will handle scaling and optimization. Target tracking autoscaling will allow automatically adjusting capacity based on demand.
The serverless approach with Fargate minimizes operational overhead.
upvoted 3 times

MikeDu 2 years, 10 months ago

Selected Answer: A

AWF Fargate should be the best choice
upvoted 2 times

aadityaravi8 2 years, 11 months ago

A is the right answers undoubtedly.
upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: A

ECR provides a secure and scalable repository to store and manage container images. ECS with the Fargate launch type allows you to run containers without managing the underlying infrastructure, providing a serverless experience. Target tracking in ECS can automatically scale the number of tasks or services based on a target value such as CPU or memory utilization, ensuring that the application can handle increasing demand without manual intervention.

Option B is not the best choice because using the EC2 launch type requires managing and scaling EC2 instances, which increases operational overhead.

Option C is not the optimal solution as it involves managing the container repository on an EC2 instance and manually launching EC2 instances, which adds complexity and operational overhead.

Option D also requires managing EC2 instances, configuring ASGs, and setting up manual scaling rules based on CloudWatch alarms, which is not as efficient or scalable as using Fargate in combination with ECS.

upvoted 6 times

Bmarodi 2 years, 11 months ago

Nice explanations!
upvoted 1 times

Bmarodi 3 years ago

Selected Answer: A

ECS + Fargate satisfy requirements, hence option A is the best solution.
upvoted 2 times

studynoplay 3 years ago

Selected Answer: A

minimize operational overhead = Serverless
Fargate is Serverless
upvoted 2 times

NoinNothing 3 years, 1 month ago

Selected Answer: A

Correct is "A"
upvoted 1 times

jaswantn 3 years, 2 months ago

You can place Fargate launch type all in one AZ, or across multiple AZs. But Option A does not take care of High Availability requirement of question. With Option C we have multi AZ.
upvoted 2 times

wizcloudifa 2 years, 3 months ago

that was my doubt too, is Fargate by default highly available? I chose D as option C didn't have a scalability option in it, option D has an autoscaling group in it C doesn't
upvoted 1 times

A company has two applications: a sender application that sends messages with payloads to be processed and a processing application intended to receive the messages with payloads. The company wants to implement an AWS service to handle messages between the two applications. The sender application can send about 1,000 messages each hour. The messages may take up to 2 days to be processed. If the messages fail to process, they must be retained so that they do not impact the processing of any remaining messages.

Which solution meets these requirements and is the MOST operationally efficient?

- A. Set up an Amazon EC2 instance running a Redis database. Configure both applications to use the instance. Store, process, and delete the messages, respectively.
- B. Use an Amazon Kinesis data stream to receive the messages from the sender application. Integrate the processing application with the Kinesis Client Library (KCL).
- C. Integrate the sender and processor applications with an Amazon Simple Queue Service (Amazon SQS) queue. Configure a dead-letter queue to collect the messages that failed to process. **Most Voted**
- D. Subscribe the processing application to an Amazon Simple Notification Service (Amazon SNS) topic to receive notifications to process. Integrate the sender application to write to the SNS topic.

Correct Answer: C

Community vote distribution

C (90%)

10%

Comments

aba2s **Highly Voted** 3 years, 5 months ago

Selected Answer: C

Amazon SQS supports dead-letter queues (DLQ), which other queues (source queues) can target for messages that can't be processed (consumed) successfully.

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-dead-letter-queues.html>

upvoted 13 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: C

By integrating both the sender and processor applications with an SQS, messages can be reliably sent from the sender to the processor application for processing. SQS provides at-least-once delivery, ensuring that messages are not lost in transit. If a message fails to process, it can be retained in the queue and retried without impacting the processing of other messages. Configuring a DLQ allows for the collection of messages that repeatedly fail to process, providing visibility into failed messages for troubleshooting and analysis.

A is not the optimal choice as it involves managing and configuring an EC2 instance running a Redis, which adds operational overhead and maintenance requirements.

B is not the most operationally efficient solution as it introduces additional complexity by using Amazon Kinesis data streams and integrating with the Kinesis Client Library for message processing.

D, using SNS, is not the best fit for the scenario as it is more suitable for pub/sub messaging and broadcasting notifications rather than the specific requirement of message processing between two applications.

upvoted 7 times

Bmarodi 2 years, 11 months ago

Nice explanations always, thanks a lot!

upvoted 1 times

Bmarodi 2 years, 10 months ago

Nice explanations always, thanks a lot

upvoted 1 times

satyaamm Most Recent 1 year, 4 months ago

Selected Answer: C

Dead Letter queues are the most suitable here.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: B

Ans C - Use SQS and dead letter queue.

Not ideal because max visibility timeout for SQS is 12h and can't be extended by user.

I did consider B: Kinesis which is for real-time processing - altho' the question doesn't say it's not real-time. Anyway, SQS is basic and fits.

upvoted 2 times

dkw2342 2 years, 3 months ago

C) This option works. There is a 12h maximum visibility timeout, but:

"If you don't know how long it takes to process a message, create a heartbeat for your consumer process: Specify the initial visibility timeout (for example, 2 minutes) and then—as long as your consumer still works on the message—keep extending the visibility timeout by 2 minutes every minute."

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/working-with-messages.html>

upvoted 4 times

dkw2342 2 years, 3 months ago

None of the options fit.

A) Not operationally efficient

B) Kinesis is for real-time processing

D) SNS is not suitable for work queuing.

C) While this may be the "correct" answer, it also doesn't really fit the problem statement.

Maximum visibility timeout for SQS is 12h, also can't be extended by the consumer.

"If your consumer needs longer than 12 hours, consider using Step Functions."

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-visibility-timeout.html>

upvoted 2 times

MatAlves 1 year, 9 months ago

It doesn't seem to be the case though:

"For example, assume that a message spends 1 day in the original queue before it's moved to a dead-letter queue. If the dead-letter queue's retention period is 4 days, the message is deleted from the dead-letter queue after 3 days and the ApproximateAgeOfOldestMessage is 3 days."

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-dead-letter-queues.html#understanding-message-retention-periods>

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

C is the only option with dead letter queue which meets the requirement of retaining messages that fail to process without impacting other messages.

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: C

Implement an AWS service to handle messages between the two applications = Amazon Simple Queue Service

If the messages fail to process, they must be retained = a dead-letter queue

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

SQS provides a fully managed message queuing service that meets all the requirements:

SQS can handle the sending and processing of 1,000 messages per hour

Messages can be retained for up to 14 days to allow the full 2 days for processing

Using a dead-letter queue will retain failed messages without impacting other processing

SQS requires minimal operational overhead compared to running your own message queue server

upvoted 3 times

MutiverseAgent 2 years, 10 months ago

Selected Answer: B

Answer is B), the reason is:

- Because messages might up to 2 days to be processed. Visibility timeout of SQS is 12 hours, so after 12 hours another consumer might take a message from the queue which is currently being processed.

upvoted 1 times

Jeeva28 3 years ago

Selected Answer: C

Answer C, In Question if Keyword have Processing Failed >> SQS

upvoted 2 times

Bmarodi 3 years ago

Selected Answer: C

solution that meets these requirements and is the MOST operationally efficient will be option C. SQS is buffer between 2 APPs.

upvoted 2 times

norris81 3 years ago

The visibility timeout must not be more than 12 hours. (For SQS)

Jobs may take 2 days to process

upvoted 2 times

studynoplay 3 years ago

Selected Answer: C

operationally efficient = Serverless

SQS is serverless

upvoted 2 times

studynoplay 3 years ago

SNS too is serverless, but it is obvious that it is not the correct answer in this case

upvoted 1 times

apchandana 3 years, 1 month ago

Selected Answer: C

more realistic option is C.

only problem with this is the limit of the visibility timeout is 12H max. as the second application take 2 days to process, there will be a duplicate of processing messages in the queue. this might complicate things.

upvoted 3 times

nilandd44gg 2 years, 10 months ago

Amazon SQS automatically deletes messages that have been in a queue for more than the maximum message retention period. The default message retention period is 4 days. However, you can set the message retention period to a value from 60 seconds to 1,209,600 seconds (14 days) using the SetQueueAttributes action.

<https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-basic-architecture.html>

upvoted 2 times

vherman 3 years, 3 months ago

SQS has a limit 12h for visibility time out

upvoted 1 times

bullrem 3 years, 4 months ago

Selected Answer: B

Option C, using Amazon SQS, is a valid solution that meets the requirements of the company. However, it may not be the most operationally efficient solution because SQS is a managed message queue service that requires additional operational overhead to handle the retention of messages that failed to process. Option B, using Amazon Kinesis Data Streams, is more operationally efficient for this use case because it can handle the retention of messages that failed to process automatically and provides the ability to process and analyze streaming data in real-time.

upvoted 1 times

UnluckyDucky 3 years, 3 months ago

Kinesis stream save data for up to 24 hours, doesn't meet the 2 day requirement.

Kinesis streams don't have fail-safe for failed processing, unlike SQS.

The correct answer is C - SQS.

upvoted 4 times

apchandana 3 years, 1 month ago

this is not a correct statement.

A data stream is a logical grouping of shards. There are no bounds on the number of shards within a data stream (request a limit increase if you

need more). A data stream will retain data for 24 hours by default, or optionally up to 365 days.

Shard

<https://aws.amazon.com/kinesis/data-streams/getting-started/>

upvoted 1 times

LuckyAro 3 years, 4 months ago

There's no way for kinesis to know whether the message processing failed.

upvoted 1 times

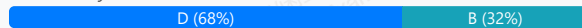
A solutions architect must design a solution that uses Amazon CloudFront with an Amazon S3 origin to store a static website. The company's security policy requires that all website traffic be inspected by AWS WAF.

How should the solutions architect comply with these requirements?

- A. Configure an S3 bucket policy to accept requests coming from the AWS WAF Amazon Resource Name (ARN) only.
- B. Configure Amazon CloudFront to forward all incoming requests to AWS WAF before requesting content from the S3 origin.
- C. Configure a security group that allows Amazon CloudFront IP addresses to access Amazon S3 only. Associate AWS WAF to CloudFront.
- D. Configure Amazon CloudFront and Amazon S3 to use an origin access identity (OAI) to restrict access to the S3 bucket. Enable AWS WAF on the distribution. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

Nigma **Highly Voted** 3 years, 6 months ago

Answer D. Use an OAI to lockdown CloudFront to S3 origin & enable WAF on CF distribution
upvoted 36 times

FNJ1111 3 years, 5 months ago

<https://aws.amazon.com/premiumsupport/knowledge-center/cloudfront-access-to-amazon-s3/> confirms use of OAI (and option D).
upvoted 14 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: B

By configuring CloudFront to forward all incoming requests to AWS WAF, the traffic will be inspected by AWS WAF before reaching the S3 origin, complying with the security policy requirement. This approach ensures that all website traffic is inspected by AWS WAF, providing an additional layer of security before accessing the content stored in the S3 origin.

Option A is not the correct choice as configuring an S3 bucket policy to accept requests from the AWS WAF ARN only would bypass the inspection of traffic by AWS WAF. It does not ensure that all website traffic is inspected.

Option C is not the optimal solution as it focuses on controlling access to S3 using a security group. Although it associates AWS WAF with CloudFront, it does not guarantee that all incoming requests are inspected by AWS WAF.

Option D is not the recommended solution as configuring an OAI in CloudFront and restricting access to the S3 bucket does not ensure that all website traffic is inspected by AWS WAF. The OAI is used for restricting direct access to S3 content, but the traffic should still pass through AWS WAF for inspection.

upvoted 11 times

bogobob 2 years, 7 months ago

Apparently you can only point to a custom host that is "not an Amazon Simple Storage Service (Amazon S3) bucket" (other than for static hosting).
<https://aws.amazon.com/blogs/security/how-to-enhance-amazon-cloudfront-origin-security-with-aws-waf-and-aws-secrets-manager/>. Answer should be D
upvoted 4 times

SinghJagdeep 2 years, 5 months ago

agreed. Must be D as per above security blog
upvoted 4 times

escalibrn 2 years, 3 months ago

Option B does use the WAF through Cloudfront, but it does not mention anything to prevent direct access to the objects without going through Cloudfront.

upvoted 4 times

Tsige 1 year, 7 months ago

CloudFront does not "forward" requests to AWS WAF. Instead, AWS WAF integrates directly with CloudFront to inspect traffic as it passes through the distribution. There is no manual forwarding process involved.

so, my Answer is D.

upvoted 5 times

Vandaman Most Recent 1 year, 3 months ago

Selected Answer: B

The requirement is for the website traffic to be inspected by WAF, not to ensure that the S3 bucket is only accessible through CloudFront. Hence, I say B

upvoted 1 times

satyaamm 1 year, 4 months ago

Selected Answer: D

Using OAI for CloudFront is the most suitable option here.

upvoted 1 times

jayessh 1 year, 7 months ago

Selected Answer: D

option D is correct

This is the correct approach because it ensures that the S3 bucket is only accessible through CloudFront, and AWS WAF can inspect all incoming traffic to the CloudFront distribution.

why option B is incorrect

because AWS WAF is integrated with CloudFront at the edge locations. CloudFront does not forward requests to AWS WAF; instead, AWS WAF inspects the requests as they come into CloudFront

upvoted 4 times

PaulGa 1 year, 7 months ago

Selected Answer: B

Ans B - configure CloudFront to forward incoming requests to AWS WAF for inspection before sending to S3. This provides an additional layer of security before accessing the content stored in the S3 origin.

D: not ideal because configuring an OAI in CloudFront and restricting access to the S3 bucket does not guarantee website traffic is inspected by WAF

upvoted 1 times

maryam_sh 1 year, 10 months ago

Selected Answer: D

B is incorrect, it misrepresents how AWS WAF works with CloudFront. AWS WAF is not an intermediary service that CloudFront forwards requests to. Instead, AWS WAF is directly integrated with CloudFront as a layer to inspect incoming requests. The correct configuration is to associate AWS WAF with the CloudFront distribution, not to forward requests separately.

upvoted 5 times

jatric 1 year, 11 months ago

Selected Answer: D

Cloud Front allows configuration to enable AWS WAF and restrict direct access to S3 through OAI will meet the requirements.

upvoted 3 times

Duckydoo 1 year, 11 months ago

Selected Answer: D

The requirements indicate that S3 is used to "store" a static website, not that it must be configured as a static website (which does not make any sense if it's to be used with CF anyway). Furthermore, the requirements also indicate that all traffic must be inspected by WAF. If you do not setup OAI/OAC, you can potentially bypass CF and access S3 directly. So option B does not satisfy the second requirement.

upvoted 4 times

shil_31 2 years ago

Selected Answer: D

Using an OAI (Origin Access Identity) restricts access to the S3 bucket, ensuring that only CloudFront can access the content.

Enabling AWS WAF on the CloudFront distribution allows you to inspect website traffic and filter out malicious requests before they reach your S3 origin.

upvoted 2 times

ManikRoy 2 years, 1 month ago

Selected Answer: D

OAI is required so that S3 bucket is not accessed directly.

upvoted 4 times

Hkayne 2 years, 1 month ago

I guess D

upvoted 1 times

keehua 2 years, 2 months ago

Selected Answer: D

There are two ways you can serve static websites on AWS S3 origin, either using website endpoints or REST API endpoints. Website endpoints does not support HTTPS. Note that the question does not mention which endpoint is used.

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/WebsiteEndpoints.html#WebsiteRestEndpointDiff>

B is incorrect because we do not 'forward' requests to AWS WAF, we attach WAF on the Cloudfront distribution itself. Could be bad wording of the question.

D is totally valid because it doesn't mention using website endpoints. D also uses OAI to restrict direct access to objects in AWS S3. Although OAI is still viable in this point of time, it is still a legacy method and it is more recommended to use OAC instead.

<https://repost.aws/knowledge-center/cloudfront-serve-static-website>

upvoted 4 times

Uzbekistan 2 years, 2 months ago

Selected Answer: B

Option B ensures that all incoming requests to the static website served through Amazon CloudFront are first forwarded to AWS WAF for inspection before the content is requested from the S3 origin. This ensures that all website traffic is inspected by AWS WAF as required by the company's security policy.

upvoted 2 times

drdz13 2 years, 3 months ago

D is not possible since you cannot set OAC or OAI if S3 bucket is used as static website host

upvoted 1 times

bujuman 2 years, 4 months ago

Selected Answer: D

WAF is associated to a Cloudfront Distribution

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

A: Doesn't make sense in context with CF.

B: You configure WAF on CF for HTTP status handling so this may be right be is badly worded

C: You might as well re-engineer S3 and CloudFront!

D: The requirement for WAF usage is met with this. Doesn't have to be smart usage, just enabled.

upvoted 3 times

Organizers for a global event want to put daily reports online as static HTML pages. The pages are expected to generate millions of views from users around the world. The files are stored in an Amazon S3 bucket. A solutions architect has been asked to design an efficient and effective solution.

Which action should the solutions architect take to accomplish this?

- A. Generate presigned URLs for the files.
- B. Use cross-Region replication to all Regions.
- C. Use the geoproximity feature of Amazon Route 53.
- D. Use Amazon CloudFront with the S3 bucket as its origin. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: D

The most effective and efficient solution would be Option D (Use Amazon CloudFront with the S3 bucket as its origin.)

Amazon CloudFront is a content delivery network (CDN) that speeds up the delivery of static and dynamic web content, such as HTML pages, images and videos. By using CloudFront, the HTML pages will be served to users from the edge location that is closest to them, resulting in faster delivery and a better user experience. CloudFront can also handle the high traffic and large number of requests expected for the global event, ensuring that the HTML pages are available and accessible to users around the world.

upvoted 11 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: D

CloudFront is well-suited for efficiently serving static HTML pages to users around the world. By using it with the S3 as its origin, the static HTML pages can be cached and distributed globally to edge locations, reducing latency and improving performance for users accessing the pages from different regions. This solution ensures efficient and effective delivery of the daily reports to millions of users worldwide, providing a scalable and high-performance solution for the global event.

A would allow temporary access to the files, but it does not address the scalability and performance requirements of serving millions of views globally.

B is not necessary for this scenario as the goal is to distribute the static HTML pages efficiently to users worldwide, not replicate the files across multiple Regions.

C is primarily used for routing DNS traffic based on the geographic location of users, but it does not provide the caching and content delivery capabilities required for this use case.

upvoted 5 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: D

CloudFront is the most suitable option here.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: D

Ans D - CloudFront content delivery network is designed for the job: speeds delivery of static/ dynamic content (HTML, images, vids). Pages will be served from the closest edge location = faster delivery; improved UX; managed high vol traffic; high vol requests

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: D

Global users = Amazon CloudFront
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

CloudFront is the best solution for this use case because:

CloudFront is a content delivery network (CDN) that caches content at edge locations around the world. This brings content closer to users for fast performance.

For high traffic global events with millions of viewers, a CDN is necessary for effective distribution. Using the S3 bucket as the origin, CloudFront can fetch the files once and cache them globally.

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: D

Option D
upvoted 1 times

k1kavi1 3 years, 5 months ago

Selected Answer: D

Agreed
upvoted 1 times

Sahilbhai 3 years, 6 months ago

answer is D agree with Shasha1
upvoted 1 times

Shasha1 3 years, 6 months ago

D
CloudFront is a content delivery network (CDN) offered by Amazon Web Services (AWS). It functions as a reverse proxy service that caches web content across AWS's global data centers, improving loading speeds and reducing the strain on origin servers. CloudFront can be used to efficiently deliver large amounts of static or dynamic content anywhere in the world.

upvoted 3 times

Wpcorgan 3 years, 6 months ago

D is correct
upvoted 2 times

Nigma 3 years, 6 months ago

D
Static content on S3 and hence Cloudfront is the best way
upvoted 2 times

Pamban 3 years, 6 months ago

Selected Answer: D

D is the correct answer
upvoted 2 times

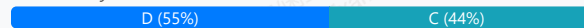
A company runs a production application on a fleet of Amazon EC2 instances. The application reads the data from an Amazon SQS queue and processes the messages in parallel. The message volume is unpredictable and often has intermittent traffic. This application should continually process messages without any downtime.

Which solution meets these requirements MOST cost-effectively?

- A. Use Spot Instances exclusively to handle the maximum capacity required.
- B. Use Reserved Instances exclusively to handle the maximum capacity required.
- C. Use Reserved Instances for the baseline capacity and use Spot Instances to handle additional capacity.
- D. Use Reserved Instances for the baseline capacity and use On-Demand Instances to handle additional capacity. **Most Voted**

Correct Answer: D

Community vote distribution



Comments

HayLLIHuK **Highly Voted** 3 years, 5 months ago

Selected Answer: C

"without any downtime" - Reserved Instances for the baseline capacity

"MOST cost-effectively" - Spot Instances to handle additional capacity

upvoted 51 times

LuckyAro 3 years, 4 months ago

Dude, read the question, cost consideration was not mentioned in the question.

upvoted 3 times

ShinobiGrappler 3 years, 4 months ago

Dude, read the question, "Which solution meets these requirements MOST cost-effectively?"

upvoted 48 times

kraken21 3 years, 2 months ago

I am leaning towards C because the idea of having a queue is to decouple the processing. If an instance goes down(spot) while processing will it not show up back after the visibility timeout? So using spot meets the cost-effective objective.

upvoted 5 times

Sutariya 2 years, 9 months ago

Intermediate data stored in SQS queue so once free then it take data and process.

upvoted 1 times

MrSaint 3 years, 1 month ago

cost-effectively means, Cheapest solution (cost) that achieve all the requirements (effectively). Its not cost-effectively if is just cheapest solution that fail to address all the requirements, in this case. (This application should continually process messages without any downtime) no matter the volume, since it is unpredictable. B for example, address the requirement but not the cheapest solution that achieve it. D is the cheaper choice that address the requirement (without any downtime). and C is cheaper than D but do not guarantee that you wont have downtime since it is SPOT instances.

upvoted 19 times

kraken21 3 years, 2 months ago

How can you have baseline capacity when your message volume is unpredictable and often has intermittent traffic?

upvoted 4 times

MutiverseAgent 2 years, 10 months ago

For this reason I think correct answer is A

upvoted 1 times

Macadam 2 years, 7 months ago

Spot instances cannot be an option as it is unreliable and the question requires the messages to be continuously processed

upvoted 5 times

allanm 10 months, 1 week ago

No way you're considering Spot instances! What if an instance goes down? Then your job is ruined which is unacceptable for the requirement

upvoted 1 times

taer **Highly Voted** 3 years, 6 months ago

Selected Answer: D

D is the correct answer

upvoted 28 times

Drayen25 3 years, 4 months ago

C is correct, read for cost effectiveness

upvoted 7 times

diabloexodia 2 years, 10 months ago

AWS has stopped issuing spot instances so i think C

upvoted 1 times

diabloexodia 2 years, 10 months ago

so i think C is incorrect*. the Correct ans is D.

upvoted 2 times

sezer 3 years, 2 months ago

if you cannot find enough spot instance you will have downtime

you cannot always find spot instance

upvoted 14 times

Kumaran1508 3 years ago

Why downtime when there are baseline reserved instances?

upvoted 3 times

Sutariya 2 years, 9 months ago

Baseline reserved instances and ondemand Spot instance is cost saver

upvoted 2 times

Neeraja2023 **Most Recent** 7 months, 1 week ago

Selected Answer: C

C. Use Reserved Instances for the baseline capacity and use Spot Instances to handle additional capacity.

- Baseline capacity is handled by Reserved Instances, ensuring no downtime for the minimum required processing.

- Spot Instances handle traffic spikes cost-effectively.

Combines reliability (RIs) with cost savings (Spot) for unpredictable workloads.

Recommended by AWS for workloads with variable traffic and parallel processing tasks.

upvoted 1 times

Darius_W 7 months, 4 weeks ago

Selected Answer: C

Answer: C

Use Reserved Instances for the steady baseline and Spot Instances for burst capacity.

Cost-effective: RIs cover always-on load at discounted rates; Spots are cheapest for intermittent spikes.

Continuous processing: If Spot capacity is interrupted, baseline RIs keep consumers running; the SQS queue buffers until Spot capacity returns.

upvoted 1 times

Darius_W 8 months ago

Selected Answer: C

C. Use Reserved Instances for the baseline capacity and Spot Instances for burst capacity.

Baseline (RIs): Guarantees steady, always-on processing so there's no downtime during normal traffic.

Bursts (Spot): Adds cheap, interruptible capacity for unpredictable spikes; if Spot is reclaimed, baseline RIs keep the system running and SQS buffers excess messages.

upvoted 1 times

Jarranymo 9 months ago

Selected Answer: C

Use Reserved Instances to handle predictable base load, and Spot Instances to handle unpredictable spikes — that's the most cost-effective way to ensure availability and performance.

upvoted 1 times

111380c 10 months ago

Selected Answer: C

no downtime

upvoted 2 times

lalaraj 11 months, 1 week ago

Selected Answer: C

Reserved capacity ensures no downtime and additional capacity can be handled with spot instances cost effectively

upvoted 1 times

network_enthusiast 1 year, 1 month ago

Selected Answer: C

C is the correct answer

upvoted 1 times

Yak_Yeti 1 year, 2 months ago

Selected Answer: C

"MOST cost-effectively" - Spot Instances to handle additional capacity

The application will process the messages with the reserved instances - slowly - with no downtime.

upvoted 1 times

chest_jd 1 year, 2 months ago

Selected Answer: C

Cost-effective - C.

upvoted 1 times

Faraz999 1 year, 2 months ago

Selected Answer: C

I am leaning towards C because the idea of having a queue is to decouple the processing.

upvoted 1 times

triptimandal01 1 year, 2 months ago

Selected Answer: C

Spot Instances can be used to handle additional capacity to accommodate intermittent traffic. Spot Instances are ideal for variable and unpredictable workloads because they allow you to take advantage of unused EC2 capacity at a significantly lower cost than On-Demand Instances.

upvoted 1 times

tch 1 year, 3 months ago

Selected Answer: D

baseline capacity" refers to the normal or expected capacity performance of a system or process, so use reserved instance - this to fulfil without any downtime and cost-effective

if "additional capacity" refers to resources or capabilities beyond the standard or baseline capacity needed to handle normal operations, allowing for flexibility and the ability to handle surges in demand or unexpected events - then we should use on-demand instances

upvoted 2 times

iamroyalty_k 1 year, 4 months ago

Selected Answer: C

C is correct because it balances cost and reliability using a combination of:

- Reserved Instances for steady-state traffic (ensuring continuous processing).
- Spot Instances for handling spikes in traffic (reducing costs significantly).

A - ✗ Not Reliable – Spot Instances can be interrupted at any time. Using them exclusively could cause processing delays or failures.

B - ✗ Not Cost-Effective – RIs are best for predictable workloads, but over-provisioning for peak loads leads to unnecessary costs when traffic is low.

D - ✗ More Expensive Than Spot Instances – On-Demand Instances are more costly than Spot Instances for handling additional capacity.

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: D

I believe the best fit for the scenario is D. D has a portion that is about 90% more expensive than C. However, Spot instances can be interrupted which means that some of the processes can be interrupted, which is not acceptable. So considering that I am choosing D.

upvoted 4 times

Mrigraj12 1 year, 4 months ago

Selected Answer: C

Easily the most cost effective

upvoted 1 times

A security team wants to limit access to specific services or actions in all of the team's AWS accounts. All accounts belong to a large organization in AWS Organizations. The solution must be scalable and there must be a single point where permissions can be maintained.

What should a solutions architect do to accomplish this?

- A. Create an ACL to provide access to the services or actions.
- B. Create a security group to allow accounts and attach it to user groups.
- C. Create cross-account roles in each account to deny access to the services or actions.
- D. Create a service control policy in the root organizational unit to deny access to the services or actions. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Nigma **Highly Voted** 3 years, 6 months ago

D. Service control policies (SCPs) are one type of policy that you can use to manage your organization. SCPs offer central control over the maximum available permissions for all accounts in your organization, allowing you to ensure your accounts stay within your organization's access control guidelines. See https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scp.html.

upvoted 22 times

cookieMr **Highly Voted** 2 years, 11 months ago

By creating an SCP in the root organizational unit, the security team can define and enforce fine-grained permissions that limit access to specific services or actions across all member accounts. The SCP acts as a guardrail, denying access to specified services or actions, ensuring that the permissions are consistent and applied uniformly across the organization. SCPs are scalable and provide a single point of control for managing permissions, allowing the security team to centrally manage access restrictions without needing to modify individual account settings.

Option A and option B are not suitable for controlling access across multiple accounts in AWS Organizations. ACLs and security groups are typically used for managing network traffic and access within a single account or a specific resource.

Option C is not the recommended approach. Cross-account roles are used for granting access, and denying access through cross-account roles can be complex and less manageable compared to using SCPs.

upvoted 10 times

awashenko 2 years, 8 months ago

This was a good explanation of why A and B are not correct. I was thinking A but after reading this I agree with you D is correct.

upvoted 2 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: D

SCP are most suitable with AWS Organizations

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: D

D is the best fit among the options. Better fit would be using AWS Control tower.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: D

Ans D - "A service control policy in the root organizational unit to deny access to the services or actions" does it at source

upvoted 2 times

Ruffyit 2 years, 6 months ago

D. Service control policies (SCPs) are one type of policy that you can use to manage your organization. SCPs offer central control over the maximum available permissions for all accounts in your organization, allowing you to ensure your accounts stay within your organization's access control guidelines. See https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scp.html.

upvoted 2 times

mach2022 2 years, 7 months ago

is D because of Deeznuts

upvoted 2 times

the_bong_lord 2 years, 4 months ago

gottem

upvoted 1 times

xplusfb 2 years, 7 months ago

Selected Answer: D

Its very clear question answer is D

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: D

Service control policies (SCPs) are a type of organization policy that you can use to manage permissions in your organization. SCPs offer central control over the maximum available permissions for all accounts in your organization. SCPs help you to ensure your accounts stay within your organization's access control guidelines.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

D. Service control policies (SCPs) are one type of policy that you can use to manage your organization. SCPs offer central control over the maximum available permissions for all accounts in your organization, allowing you to ensure your accounts stay within your organization's access control guidelines. See https://docs.aws.amazon.com/organizations/latest/userguide/orgs_manage_policies_scp.html.

upvoted 2 times

Bmarodi 3 years ago

Selected Answer: D

I vote for option D by Creating a service control policy (SCP) in the root organizational unit to deny access to the services or actions, meets the requirements.

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: D

To limit access to specific services or actions in all of the team's AWS accounts and maintain a single point where permissions can be managed, the solutions architect should create a service control policy (SCP) in the root organizational unit to deny access to the services or actions (Option D).

Service control policies (SCPs) are policies that you can use to set fine-grained permissions for your AWS accounts within your organization. SCPs are attached to the root of the organizational unit (OU) or to individual accounts, and they specify the permissions that are allowed or denied for the accounts within the scope of the policy. By creating an SCP in the root organizational unit, the security team can set permissions for all of the accounts in the organization from a single location, ensuring that the permissions are consistently applied across all accounts.

upvoted 6 times

career360guru 3 years, 5 months ago

Selected Answer: D

Option D

upvoted 1 times

Wpcorgan 3 years, 6 months ago

D is incorrect

upvoted 1 times

babaxoxo 3 years, 6 months ago

an organization and requires single point place to manage permissions

upvoted 2 times

goatbernard 3 years, 6 months ago

Selected Answer: D

SCP for organization

upvoted 4 times

A company is concerned about the security of its public web application due to recent web attacks. The application uses an Application Load Balancer (ALB). A solutions architect must reduce the risk of DDoS attacks against the application.

What should the solutions architect do to meet this requirement?

- A. Add an Amazon Inspector agent to the ALB.
- B. Configure Amazon Macie to prevent attacks.
- C. Enable AWS Shield Advanced to prevent attacks. **Most Voted**
- D. Configure Amazon GuardDuty to monitor the ALB.

Correct Answer: C

Community vote distribution

C (100%)

Comments

studynoplay **Highly Voted** 3 years ago

What's going on, suddenly the questions are so easy
upvoted 12 times

Sutariya 2 years, 10 months ago

Its due to confidence level going up after experience.
upvoted 8 times

awashenko **Highly Voted** 2 years, 8 months ago

Selected Answer: C

When you see DDOS immediately think Shield
upvoted 7 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: C

AWS Shield is most suitable for DDoS attacks.
upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: C

Ans C - DDoS under AWS = Shield/ Shield Advanced
upvoted 2 times

Ruffyit 2 years, 6 months ago

When you see DDOS immediately think Shield
upvoted 6 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: C

AWS Shield is a managed DDoS protection service that safeguards applications running on AWS.
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

Enable AWS Shield Advanced to prevent attacks.
upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: C

By enabling Shield Advanced, the web application benefits from automatic protection against common and sophisticated DDoS attacks. It utilizes advanced detection and mitigation techniques, including ML algorithms and traffic analysis, to provide effective DDoS protection. It also includes features like real-time monitoring, attack notifications, and detailed attack reports.

A is not related to DDoS protection. Amazon Inspector is a security assessment service that helps identify vulnerabilities and security issues in applications and EC2.

B is also not the appropriate solution. Macie is a service that uses machine learning to discover, classify, and protect sensitive data stored in AWS. It focuses on data security and protection, not specifically on DDoS prevention.

D is not the most effective solution. GuardDuty is a threat detection service that analyzes events and network traffic to identify potential security threats and anomalies. While it can provide insights into potential DDoS attacks, it does not actively prevent or mitigate them.

upvoted 4 times

techhb 3 years, 5 months ago

Explained in details here <https://medium.com/@tshemku/aws-waf-vs-firewall-manager-vs-shield-vs-shield-advanced-4c86911e94c6>

upvoted 3 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: C

To reduce the risk of DDoS attacks against the application, the solutions architect should enable AWS Shield Advanced (Option C).

AWS Shield is a managed Distributed Denial of Service (DDoS) protection service that helps protect web applications running on AWS from DDoS attacks. AWS Shield Advanced is an additional layer of protection that provides enhanced DDoS protection capabilities, including proactive monitoring and automatic inline mitigations, to help protect against even the largest and most sophisticated DDoS attacks. By enabling AWS Shield Advanced, the solutions architect can help protect the application from DDoS attacks and reduce the risk of disruption to the application.

upvoted 5 times

career360guru 3 years, 5 months ago

Selected Answer: C

C is right answer

upvoted 2 times

Wpcorgan 3 years, 6 months ago

C is correct

upvoted 2 times

goatbernard 3 years, 6 months ago

Selected Answer: C

AWS Shield Advanced

upvoted 4 times

Nigma 3 years, 6 months ago

DDOS = AWS Shield

upvoted 5 times

A company's web application is running on Amazon EC2 instances behind an Application Load Balancer. The company recently changed its policy, which now requires the application to be accessed from one specific country only.

Which configuration will meet this requirement?

- A. Configure the security group for the EC2 instances.
- B. Configure the security group on the Application Load Balancer.
- C. Configure AWS WAF on the Application Load Balancer in a VPC. **Most Voted**
- D. Configure the network ACL for the subnet that contains the EC2 instances.

Correct Answer: C

Community vote distribution

C (100%)

Comments

handyplazt **Highly Voted** 3 years, 6 months ago

Selected Answer: C

Geographic (Geo) Match Conditions in AWS WAF. This new condition type allows you to use AWS WAF to restrict application access based on the geographic location of your viewers. With geo match conditions you can choose the countries from which AWS WAF should allow access.

<https://aws.amazon.com/about-aws/whats-new/2017/10/aws-waf-now-supports-geographic-match/>

upvoted 30 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: C

To meet the requirement of allowing the web application to be accessed from one specific country only, the company should configure AWS WAF (Web Application Firewall) on the Application Load Balancer in a VPC (Option C).

AWS WAF is a web application firewall service that helps protect web applications from common web exploits that could affect application availability, compromise security, or consume excessive resources. AWS WAF allows you to create rules that block or allow traffic based on the values of specific request parameters, such as IP address, HTTP header, or query string value. By configuring AWS WAF on the Application Load Balancer and creating rules that allow traffic from a specific country, the company can ensure that the web application is only accessible from that country.

upvoted 7 times

Dharmarajan **Most Recent** 1 year, 4 months ago

Selected Answer: C

AWF's Geo Match rule helps in this scenario.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: C

Ans C - WAF with geo-match (region or country).

<https://docs.aws.amazon.com/waf/latest/developerguide/waf-rule-statement-type-geo-match.html>

upvoted 2 times

PoolDead 1 year, 10 months ago

C --> One of the feature of WAF is Access Control: Implement IP whitelisting and blacklisting to allow or block traffic from specific IP addresses or address ranges. This can be useful for restricting access to your web application to trusted users or regions.

upvoted 3 times

Ruffyt 2 years, 6 months ago

Geographic (Geo) Match Conditions in AWS WAF. This new condition type allows you to use AWS WAF to restrict application access based on the geographic location of your viewers. With geo match conditions you can choose the countries from which AWS WAF should allow access.

<https://aws.amazon.com/about-aws/whats-new/2017/10/aws-waf-now-supports-geographic-match/>

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

C. Configure AWS WAF on the Application Load Balancer in a VPC

upvoted 1 times

Sutariya 2 years, 10 months ago

We can use AWS WAF to configure access control rule to access from specific location.

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: C

By configuring AWS WAF on the ALB in a VPC, you can apply access control rules based on the geographic location of the incoming requests. AWS WAF allows you to create rules that include conditions based on the IP addresses' country of origin. You can specify the desired country and deny access to requests originating from any other country by leveraging AWS WAF's Geo Match feature.

Option A and option B focus on network-level access control and do not provide country-specific filtering capabilities.

Option D is not the ideal solution for restricting access based on country. Network ACLs primarily control traffic at the subnet level based on IP addresses and port numbers, but they do not have built-in capabilities for country-based filtering.

upvoted 5 times

Abrar2022 3 years ago

Configure AWS WAF for Geo Match Policy

upvoted 3 times

aba2s 3 years, 5 months ago

Selected Answer: C

Source from an AWS link

Geographic (Geo) Match Conditions in AWS WAF. This condition type allows you to use AWS WAF to restrict application access based on the geographic location of your viewers.

With geo match conditions you can choose the countries from which AWS WAF should allow access.

upvoted 3 times

techhb 3 years, 5 months ago

Selected Answer: C

WAF Shield Advanced for DDOS,

GuardDuty is a continuous monitoring service that alerts you of potential threats, while Inspector is a one-time assessment service that provides a report of vulnerabilities and deviations from best practices.

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: C

OptionC. Configure WAF for Geo Match Policy

upvoted 2 times

Wpcorgan 3 years, 6 months ago

C is correct

upvoted 2 times

mricee9 3 years, 6 months ago

Selected Answer: C

C

<https://aws.amazon.com/about-aws/whats-new/2017/10/aws-waf-now-supports-geographic-match/>

upvoted 3 times

Nigma 3 years, 6 months ago

C. WAF with ALB is the right option

upvoted 2 times

A company provides an API to its users that automates inquiries for tax computations based on item prices. The company experiences a larger number of inquiries during the holiday season only that cause slower response times. A solutions architect needs to design a solution that is scalable and elastic.

What should the solutions architect do to accomplish this?

A. Provide an API hosted on an Amazon EC2 instance. The EC2 instance performs the required computations when the API request is made.

B. Design a REST API using Amazon API Gateway that accepts the item names. API Gateway passes item names to AWS Lambda for tax computations. **Most Voted**

C. Create an Application Load Balancer that has two Amazon EC2 instances behind it. The EC2 instances will compute the tax on the received item names.

D. Design a REST API using Amazon API Gateway that connects with an API hosted on an Amazon EC2 instance. API Gateway accepts and passes the item names to the EC2 instance for tax computations.

Correct Answer: B

Community vote distribution

B (98%) 3%

Comments

bullrem **Highly Voted** 3 years, 4 months ago

Selected Answer: B

Option D is similar to option B in that it uses Amazon API Gateway to handle the API requests, but it also includes an EC2 instance to perform the tax computations. However, using an EC2 instance in this way is less scalable and less elastic than using AWS Lambda to perform the computations. An EC2 instance is a fixed resource and requires manual scaling and management, while Lambda is an event-driven, serverless compute service that automatically scales with the number of requests, making it more suitable for handling variable workloads and reducing response times during high traffic periods. Additionally, Lambda is more cost-efficient than EC2 instances, as you only pay for the compute time consumed by your functions, making it a more cost-effective solution.

upvoted 29 times

career360guru **Highly Voted** 3 years, 5 months ago

Selected Answer: B

Option B. Though D is also possible B is more scalable as Lambda will autoscale to meet the dynamic load.

upvoted 6 times

pentium75 2 years, 5 months ago

Even though you can scale EC2 instances (which is less flexible as Lambda), D does not mention scaling but rather a single EC2 instance. Thus it's clearly out.

upvoted 3 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: B

Both API Gateway and Lambda are suitable as they are scalable.

upvoted 1 times

PaulGa 1 year, 7 months ago

Selected Answer: B

Ans B - I was going for option D... but then I saw bullrem's response (1 year, 9 mth ago):

"Option D is similar to option B in that it uses Amazon API Gateway to handle the API requests, but it also includes an EC2 instance to perform the tax computations. However, using an EC2 instance in this way is less scalable and less elastic than using AWS Lambda to perform the computations."

upvoted 2 times

lofzee 2 years ago

Selected Answer: B

b. easy

upvoted 2 times

TruthWS 2 years, 2 months ago

Option B: Lambda = scalable and elastic

upvoted 3 times

Uzbekistan 2 years, 2 months ago

Selected Answer: B

Option B leverages AWS Lambda, which is a serverless compute service that automatically scales in response to incoming requests. When a request is made to the API hosted on Amazon API Gateway, API Gateway triggers the associated AWS Lambda function, passing the item names as input parameters. The Lambda function then performs the tax computations based on the provided item names. AWS Lambda automatically manages the compute capacity, ensuring that there is no need to provision or manage servers. This serverless architecture offers scalability and elasticity, as Lambda functions can scale out to handle a larger number of inquiries during the holiday season and scale in during periods of lower demand. Additionally, AWS Lambda is a fully managed service, reducing operational overhead for the company.

upvoted 2 times

app12 2 years, 4 months ago

The thing that bothers me about B is that the request sends the name and then based on the name, the tax is calculated. How do you calculate a value e.g. tax if you just have a name...

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

EC2 without autoscaling is not elastic so A, C & D won't be suitable. B uses AWS Lambda which is elastic and scalable by design.

upvoted 4 times

pentium75 2 years, 5 months ago

Selected Answer: B

Though EC2 can scale (even if less flexible than Lambda), neither A, C nor D involve scaling. All these answers are about a single EC2 instance or a pair of EC2 instances. The only answer that includes scaling and elasticity is B.

upvoted 5 times

LoXoL 2 years, 4 months ago

I agree.

upvoted 1 times

Ruffyit 2 years, 6 months ago

scalable and elastic = serverless = API gateway and AWS Lambda

upvoted 3 times

paniya93 2 years, 8 months ago

Selected Answer: B

in 002 answer is B. Why is that?

upvoted 1 times

vijaykamal 2 years, 8 months ago

Selected Answer: B

Options A, C, and D involve EC2 instances, which are not as inherently scalable and elastic as serverless AWS Lambda functions, and they would require more manual management and operational overhead. Therefore, option B is the most appropriate choice for a scalable and elastic API solution.

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

REST API using Amazon API Gateway and integrating it with AWS Lambda (option B) is the recommended approach to achieve a scalable and elastic solution for the company's API during the holiday season.

No good EC2 in this case

using an EC2 instance in this way is less scalable and less elastic than using AWS Lambda to perform the computations

upvoted 3 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

scalable and elastic = serverless = API gateway and AWS Lambda

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

B) Design a REST API using Amazon API Gateway that accepts the item names. API Gateway passes item names to AWS Lambda for tax computations

This option provides the most scalable and elastic solution:

API Gateway handles creating the REST API frontend to receive requests

Lambda functions scale automatically to handle spikes in traffic during peak seasons

No servers to manage for the computations, providing high scalability

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: B

Option A (hosting an API on an Amazon EC2 instance) would require manual management and scaling of the EC2 instances, making it less scalable and elastic compared to a serverless solution.

Option C (creating an Application Load Balancer with EC2 instances for tax computations) also involves manual management of the instances and does not offer the same level of scalability and elasticity as a serverless solution.

Option D (designing a REST API using API Gateway and connecting it with an API hosted on an EC2 instance) adds unnecessary complexity and management overhead. It is more efficient to directly integrate API Gateway with AWS Lambda for tax computations.

Therefore, designing a REST API using Amazon API Gateway and integrating it with AWS Lambda (option B) is the recommended approach to achieve a scalable and elastic solution for the company's API during the holiday season.

upvoted 3 times

A solutions architect is creating a new Amazon CloudFront distribution for an application. Some of the information submitted by users is sensitive. The application uses HTTPS but needs another layer of security. The sensitive information should be protected throughout the entire application stack, and access to the information should be restricted to certain applications.

Which action should the solutions architect take?

- A. Configure a CloudFront signed URL.
- B. Configure a CloudFront signed cookie.
- C. Configure a CloudFront field-level encryption profile. **Most Voted**
- D. Configure CloudFront and set the Origin Protocol Policy setting to HTTPS Only for the Viewer Protocol Policy.

Correct Answer: C

Community vote distribution

C (83%)

Other

Comments

Bobbybash **Highly Voted** 3 years, 6 months ago

CCCCCCCC

Field-level encryption allows you to enable your users to securely upload sensitive information to your web servers. The sensitive information provided by your users is encrypted at the edge, close to the user, and remains encrypted throughout your entire application stack. This encryption ensures that only applications that need the data—and have the credentials to decrypt it—are able to do so.

upvoted 60 times

jatric **Most Recent** 1 year, 11 months ago

Selected Answer: C

field level encryption allow to protect sensitive information throughout the application stack

upvoted 2 times

zinabu 2 years, 2 months ago

With Amazon CloudFront, you can enforce secure end-to-end connections to origin servers by using HTTPS. Field-level encryption adds an additional layer of security that lets you protect specific data throughout system processing so that only certain applications can see it.

Field-level encryption allows you to enable your users to securely upload sensitive information to your web servers. The sensitive information provided by your users is encrypted at the edge, close to the user, and remains encrypted throughout your entire application stack. This encryption ensures that only applications that need the data—and have the credentials to decrypt it—are able to do so.

upvoted 4 times

huzafaharoun 2 years, 3 months ago

C:

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

upvoted 3 times

NayeraB 2 years, 3 months ago

Selected Answer: C

C is the only one that addresses handling sensitive information.

upvoted 3 times

bujuman 2 years, 4 months ago

Selected Answer: C

Reviewing my first vote after research. It seems that C is the best answer:

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

A if for fetch. B requires cookies. D just enforces HTTPS which is already mentioned for the solution (CloudFront only allows HTTPS) and does not add another layer of security.

C provides field level encryption security which is another layer of security.

upvoted 2 times

master9 2 years, 5 months ago

Selected Answer: A

Please go through below link:

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-signed-urls.html>

upvoted 2 times

pentium75 2 years, 5 months ago

This is about controlling access for downloads (making sure that the download request is coming from an authenticated user), it has nothing to do with protecting data that is sent to the application.

upvoted 3 times

Leo1688 2 years, 6 months ago

cccc,this link <https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

upvoted 3 times

vijaykamal 2 years, 8 months ago

Selected Answer: C

Options A and B (signed URL and signed cookie) are used for controlling access to specific resources and are typically used for restricting access based on URLs or cookies. They do not provide field-level encryption for sensitive data within HTTP requests.

Option D (configuring CloudFront with the Origin Protocol Policy set to HTTPS Only for the Viewer Protocol Policy) is related to enforcing HTTPS communication between CloudFront and the viewer (end-user). While important for security, it doesn't address the specific requirement of protecting sensitive data within the application stack.

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

C) Configure a CloudFront field-level encryption profile.

Field-level encryption allows you to encrypt sensitive information at the edge before distributing content through CloudFront. It provides an additional layer of security for sensitive user-submitted data.

The other options would not provide field-level encryption

upvoted 2 times

mr_D3v1n3 2 years, 10 months ago

Would the HTTPS imply that the cert was signed by a CA

upvoted 1 times

cookieMr 2 years, 11 months ago

Selected Answer: C

Option A and Option B are used for controlling access to specific resources or content based on signed URLs or cookies. While they provide security and access control, they do not provide field-level encryption for sensitive data within the requests.

Option D ensures that communication between the viewer and CloudFront is encrypted with HTTPS. However, it does not specifically address the protection and encryption of sensitive information within the application stack.

Therefore, the most appropriate action to protect sensitive information throughout the entire application stack and restrict access to certain applications is to configure a CloudFront field-level encryption profile (Option C).

upvoted 3 times

Jeeva28 3 years ago

Selected Answer: C

With Amazon CloudFront, you can enforce secure end-to-end connections to origin servers by using HTTPS. Field-level encryption adds an additional layer of security that lets you protect specific data throughout system processing so that only certain applications can see it.

upvoted 2 times

WheracanIstart 3 years, 3 months ago

Selected Answer: C

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-level-encryption.html>

"Field-level encryption allows you to enable your users to securely upload sensitive information to your web servers. The sensitive information provided by your users is encrypted at the edge, close to the user, and remains encrypted throughout your entire application stack".

upvoted 4 times

bdp123 3 years, 4 months ago

Selected Answer: C

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/field-levelencryption.html>

html

"With Amazon CloudFront, you can enforce secure end-to-end connections to origin servers by using HTTPS. Field-level encryption adds an additional layer of security that lets you protect specific data throughout system processing so that only certain applications can see it."

upvoted 4 times

ProfXsamson 3 years, 4 months ago

C, field-level encryption should be used when necessary to protect sensitive data.

upvoted 2 times

A gaming company hosts a browser-based application on AWS. The users of the application consume a large number of videos and images that are stored in Amazon S3. This content is the same for all users.

The application has increased in popularity, and millions of users worldwide accessing these media files. The company wants to provide the files to the users while reducing the load on the origin.

Which solution meets these requirements MOST cost-effectively?

- A. Deploy an AWS Global Accelerator accelerator in front of the web servers.
- B. Deploy an Amazon CloudFront web distribution in front of the S3 bucket. **Most Voted**
- C. Deploy an Amazon ElastiCache for Redis instance in front of the web servers.
- D. Deploy an Amazon ElastiCache for Memcached instance in front of the web servers.

Correct Answer: B

Community vote distribution

B (93%) 7%

Comments

Nigma **Highly Voted** 3 years ago

B. Cloud front is best for content delivery. Global Accelerator is best for non-HTTP (TCP/UDP) cases and supports HTTP cases as well but with static IP (elastic IP) or anycast IP address only.

upvoted 24 times

cookieMr **Highly Voted** 2 years, 5 months ago

Selected Answer: B

Option A is not the most cost-effective solution for this scenario. While Global Accelerator can improve global application performance, it is primarily used for accelerating TCP and UDP traffic, such as gaming and real-time applications, rather than serving static media files.

Options C and D are used for caching frequently accessed data in-memory to improve application performance. However, they are not specifically designed for caching and serving media files like CloudFront, and therefore, may not provide the same cost-effectiveness and scalability for this use case.

Hence, deploying an CloudFront web distribution in front of the S3 is the most cost-effective solution for delivering media files to millions of users worldwide while reducing the load on the origin.

upvoted 6 times

rlamberti **Most Recent** 2 years, 1 month ago

Selected Answer: B

CloudFront will cache the data in Edge Locations, offloading it partially from the source location (s3)
B looks good to me.

upvoted 3 times

TariqKipkemei 2 years, 3 months ago

Selected Answer: B

Deploy an Amazon CloudFront web distribution in front of the S3 bucket

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B) Deploy an Amazon CloudFront web distribution in front of the S3 bucket.

CloudFront is the most cost-effective solution for this use case because:

CloudFront can cache static assets like videos and images at edge locations closer to users. This improves performance. Serving files from the CloudFront cache reduces load on the S3 origin.

CloudFront pricing is very low for data transfer and requests.

upvoted 2 times

Kiki_Pass 2 years, 4 months ago

Selected Answer: B

ElasticCache is for DB Cache(RDS) nor for S3

upvoted 1 times

kruasan 2 years, 7 months ago

Selected Answer: B

ElasticCache, enhances the performance of web applications by quickly retrieving information from fully-managed in-memory data stores. It utilizes Memcached and Redis, and manages to considerably reduce the time your applications would, otherwise, take to read data from disk-based databases.

Amazon CloudFront supports dynamic content from HTTP and WebSocket protocols, which are based on the Transmission Control Protocol (TCP) protocol. Common use cases include dynamic API calls, web pages and web applications, as well as an application's static files such as audio and images. It also supports on-demand media streaming over HTTP.

AWS Global Accelerator supports both User Datagram Protocol (UDP) and TCP-based protocols. It is commonly used for non-HTTP use cases, such as gaming, IoT and voice over IP. It is also good for HTTP use cases that need static IP addresses or fast regional failover

upvoted 4 times

LuckyAro 2 years, 11 months ago

Selected Answer: C

The company wants to provide the files to the users while reducing the load on the origin.

Cloudfront speeds-up content delivery but I'm not sure it reduces the load on the origin.

Some form of caching would cache content and deliver to users without going to the origin for each request.

upvoted 2 times

awsgeek75 1 year, 11 months ago

CloudFront's only main purpose is to serve these kind of scenarios!

upvoted 3 times

pentium75 1 year, 11 months ago

"Some form of caching would cache content and deliver to users without going to the origin for each request", isn't this EXACTLY what CloudFront does?

upvoted 4 times

pentium75 1 year, 11 months ago

ElasticCache for Redis (C) can be used by an application to store key-value pairs. It does not cache videos or images and it is not an automatic process (the application can put and retrieve values).

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Selected Answer: B

To provide media files to users while reducing the load on the origin and meeting the requirements cost-effectively, the gaming company should deploy an Amazon CloudFront web distribution in front of the S3 bucket (Option B).

CloudFront is a content delivery network (CDN) that speeds up the delivery of static and dynamic web content, such as images and videos, to users. By using CloudFront, the media files will be served to users from the edge location that is closest to them, resulting in faster delivery and a better user experience. CloudFront can also handle the high traffic and large number of requests expected from the millions of users, ensuring that the media files are available and accessible to users around the world.

upvoted 4 times

techhb 2 years, 11 months ago

Please dont post ChatGPT answers here, chatgpt keeps on changing its answers, its not the right way to copy paste, thanks.

upvoted 2 times

ocbn3wby 2 years, 10 months ago

Woaaaa! I always wondered where this kind of logic and explanation came from in this guy's answers. Nice catch TECHHB!

upvoted 3 times

ocbn3wby 2 years, 10 months ago

Answers are mostly correct. Only a small percentage were wrong

upvoted 2 times

Bofi 2 years, 9 months ago

why not? if the answers are correct and offer best possible explanation for the wrong options, I see no reason why it shouldn't be posted here.

Also, most of his answers were right, although reasons for the wrong options were sometimes lacking, but all in all, his responses were very good.

upvoted 2 times

career360guru 2 years, 11 months ago

Selected Answer: B

Option B

upvoted 1 times

k1kavi1 2 years, 12 months ago

Selected Answer: B

Agreed

upvoted 1 times

rewdboy 3 years ago

Selected Answer: B

B is the correct answer

upvoted 1 times

Wpcorgan 3 years ago

B is correct

upvoted 2 times

A company has a multi-tier application that runs six front-end web servers in an Amazon EC2 Auto Scaling group in a single Availability Zone behind an Application Load Balancer (ALB). A solutions architect needs to modify the infrastructure to be highly available without modifying the application.

Which architecture should the solutions architect choose that provides high availability?

- A. Create an Auto Scaling group that uses three instances across each of two Regions.
- B. Modify the Auto Scaling group to use three instances across each of two Availability Zones. **Most Voted**
- C. Create an Auto Scaling template that can be used to quickly create more instances in another Region.
- D. Change the ALB in front of the Amazon EC2 instances in a round-robin configuration to balance traffic to the web tier.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Nigma **Highly Voted** 2 years, 6 months ago
B. auto scaling groups can not span multi region
upvoted 30 times

cookieMr **Highly Voted** 1 year, 11 months ago

Selected Answer: B

Option A (creating an Auto Scaling group across two Regions) introduces additional complexity and potential replication challenges, which may not be necessary for achieving high availability within a single Region.

Option C (creating an Auto Scaling template for another Region) suggests multi-region redundancy, which may not be the most straightforward solution for achieving high availability without modifying the application.

Option D (changing the ALB to a round-robin configuration) does not provide the desired high availability. Round-robin configuration alone does not ensure fault tolerance and does not leverage multiple Availability Zones for resilience.

Hence, modifying the Auto Scaling group to use three instances across each of two Availability Zones is the appropriate choice to provide high availability for the multi-tier application.

upvoted 11 times

sreedharveluru557 **Most Recent** 3 weeks ago

Selected Answer: B

ASG's can't span regions. B is the right answer

upvoted 1 times

satyaamm 1 year, 4 months ago

Selected Answer: B

Modifying the ALB is the most suitable option here.

upvoted 1 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

Modify the Auto Scaling group to use three instances across each of two Availability Zones

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

Option B. Modify the Auto Scaling group to use three instances across each of the two Availability Zones.

upvoted 3 times

techhb 2 years, 5 months ago

B. auto scaling groups cannot span multi region

upvoted 6 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: B

Option B. Modify the Auto Scaling group to use three instances across each of the two Availability Zones.

This option would provide high availability by distributing the front-end web servers across multiple Availability Zones. If there is an issue with one Availability Zone, the other Availability Zone would still be available to serve traffic. This would ensure that the application remains available and highly available even if there is a failure in one of the Availability Zones.

upvoted 5 times

career360guru 2 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

k1kavi1 2 years, 5 months ago

Selected Answer: B

Agreed

upvoted 1 times

Shasha1 2 years, 6 months ago

B

option B This architecture provides high availability by having multiple Availability Zones hosting the same application. This allows for redundancy in case one Availability Zone experiences downtime, as traffic can be served by the other Availability Zone. This solution also increases scalability and performance by allowing traffic to be spread across two Availability Zones.

upvoted 2 times

mricee9 2 years, 6 months ago

Selected Answer: B

B is rightt

upvoted 2 times

Wpcorgan 2 years, 6 months ago

B is correct

upvoted 1 times

xua81376 2 years, 6 months ago

B auto scaling i multiple AZ

upvoted 1 times

An ecommerce company has an order-processing application that uses Amazon API Gateway and an AWS Lambda function. The application stores data in an Amazon Aurora PostgreSQL database. During a recent sales event, a sudden surge in customer orders occurred. Some customers experienced timeouts, and the application did not process the orders of those customers.

A solutions architect determined that the CPU utilization and memory utilization were high on the database because of a large number of open connections. The solutions architect needs to prevent the timeout errors while making the least possible changes to the application.

Which solution will meet these requirements?

A. Configure provisioned concurrency for the Lambda function. Modify the database to be a global database in multiple AWS Regions.

B. Use Amazon RDS Proxy to create a proxy for the database. Modify the Lambda function to use the RDS Proxy endpoint instead of the database endpoint. **Most Voted**

C. Create a read replica for the database in a different AWS Region. Use query string parameters in API Gateway to route traffic to the read replica.

D. Migrate the data from Aurora PostgreSQL to Amazon DynamoDB by using AWS Database Migration Service (AWS DMS). Modify the Lambda function to use the DynamoDB table.

Correct Answer: B

Community vote distribution

B (100%)

Comments

handyplazt **Highly Voted** 3 years, 6 months ago

Selected Answer: B

Many applications, including those built on modern serverless architectures, can have a large number of open connections to the database server and may open and close database connections at a high rate, exhausting database memory and compute resources. Amazon RDS Proxy allows applications to pool and share connections established with the database, improving database efficiency and application scalability.
<https://aws.amazon.com/id/rds/proxy/>

upvoted 38 times

rexxxx_x 2 years, 4 months ago

Are you sure?

upvoted 2 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: B

Option A (configuring provisioned concurrency and creating a global database) does not directly address the high connection utilization issue on the database, and creating a global database may introduce additional complexity without immediate benefit to solving the timeout errors.

Option C (creating a read replica in a different AWS Region) introduces additional data replication and management complexity, which may not be necessary to address the timeout errors.

Option D (migrating to Amazon DynamoDB) involves a significant change in the data storage technology and requires modifying the application to use DynamoDB instead of Aurora PostgreSQL. This may not be the most suitable solution when the goal is to make minimal changes to the application.

Therefore, using Amazon RDS Proxy and modifying the Lambda function to use the RDS Proxy endpoint is the recommended solution to prevent timeout errors and reduce the impact on the database during peak loads.

upvoted 11 times

satyaamm Most Recent 1 year, 4 months ago

Selected Answer: B

RDS proxy helps tackle the problem of sudden hikes as it plays a medium between the users and the application
upvoted 1 times

lofzee 2 years ago

Selected Answer: B

Amazon RDS Proxy is a fully managed, highly available database proxy for Amazon Relational Database Service (RDS) that makes applications more scalable, more resilient to database failures, and more secure.

B
upvoted 3 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

Connection problems causing high CPU and Memory usage? Use RDS proxy!
upvoted 3 times

tonybuivannghia 1 year, 8 months ago

Improving database efficiency by reducing the stress on database resources (e.g., CPU, RAM) and minimize open connections (and timeouts).
upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

Use Amazon RDS Proxy to create a proxy for the database. Modify the Lambda function to use the RDS Proxy endpoint instead of the database endpoint.
upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

using Amazon RDS Proxy and modifying the Lambda function to use the RDS Proxy endpoint is the recommended solution to prevent timeout errors and reduce the impact on the database during peak loads.
upvoted 2 times

sairam 3 years, 4 months ago

I also think the answer is B. However can RDS Proxy be used with Amazon Aurora PostgreSQL database?
upvoted 2 times

everfly 3 years, 3 months ago

RDS Proxy can be used with Aurora
<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/rds-proxy.html>
upvoted 6 times

gustavtd 3 years, 5 months ago

Selected Answer: B

I expect a answer with database replica but there is not, so B is most suitable
upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: B

Option B. Use Amazon RDS Proxy to create a proxy for the database. Modify the Lambda function to use the RDS Proxy endpoint instead of the database endpoint.

Using Amazon RDS Proxy can help reduce the number of connections to the database and improve the performance of the application. RDS Proxy establishes a connection pool to the database and routes connections to the available connections in the pool. This can help reduce the number of open connections to the database and improve the performance of the application. The Lambda function can be modified to use the RDS Proxy endpoint instead of the database endpoint to take advantage of this improvement.
upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A is not a valid solution because configuring provisioned concurrency for the Lambda function does not address the issue of high CPU utilization and memory utilization on the database.

Option C is not a valid solution because creating a read replica in a different Region does not address the issue of high CPU utilization and memory utilization on the database.

Option D is not a valid solution because migrating the data from Aurora PostgreSQL to DynamoDB would require significant changes to the application and may not be the best solution for this particular problem.

upvoted 3 times

BENICE 3 years, 5 months ago

Option --- B

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: B

As it is mentioned that issue was due to high CPU and Memory due to many open connections to DB, B is the right answer.

upvoted 2 times

Shasha1 3 years, 6 months ago

B

Using Amazon RDS Proxy will allow the application to handle more connections and higher loads without timeouts, while making the least possible changes to the application. The RDS Proxy will enable connection pooling, allowing multiple connections from the Lambda function to be served from a single proxy connection. This will reduce the number of open connections on the database, which is causing high CPU and memory utilization

upvoted 4 times

Wpcorgan 3 years, 6 months ago

B is correct

upvoted 2 times

xua81376 3 years, 6 months ago

B - Proxy to manage connections

upvoted 3 times

babaxoxo 3 years, 6 months ago

Selected Answer: B

Issue related to opening many connections and the solution requires least code changes so B satisfies the conditions

upvoted 8 times

Nigma 3 years, 6 months ago

Correct B

upvoted 2 times

An application runs on Amazon EC2 instances in private subnets. The application needs to access an Amazon DynamoDB table.

What is the MOST secure way to access the table while ensuring that the traffic does not leave the AWS network?

- A. Use a VPC endpoint for DynamoDB. **Most Voted**
- B. Use a NAT gateway in a public subnet.
- C. Use a NAT instance in a private subnet.
- D. Use the internet gateway attached to the VPC.

Correct Answer: A

Community vote distribution

A (100%)

Comments

mabotega **Highly Voted** 2 years, 6 months ago

Selected Answer: A

VPC endpoints for service in private subnets

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/vpc-endpoints-dynamodb.html>

upvoted 18 times

cookieMr **Highly Voted** 1 year, 11 months ago

Option B (using a NAT gateway in a public subnet) and option C (using a NAT instance in a private subnet) are not the most secure options because they involve routing traffic through a network address translation (NAT) device, which requires an internet gateway and traverses the public internet.

Option D (using the internet gateway attached to the VPC) would require routing traffic through the internet gateway, which would result in the traffic leaving the AWS network.

Therefore, the recommended and most secure approach is to use a VPC endpoint for DynamoDB to ensure private and secure access to the DynamoDB table from your EC2 instances in private subnets, without the need to traverse the internet or leave the AWS network.

upvoted 7 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: A

A gateway VPC endpoint is the most suitable for accessing DynamoDB and S3.

upvoted 1 times

8935983 8 months, 4 weeks ago

Adding to your point VPC gateways are also suitable for accessing two components

upvoted 1 times

vijaykamal 1 year, 8 months ago

Selected Answer: A

Using an internet gateway (Option D) is used for enabling outbound internet connectivity from resources in your VPC. It's not the appropriate choice for securely accessing DynamoDB within your VPC.

upvoted 3 times

Ramdi1 1 year, 9 months ago

Selected Answer: A

A gateway VPC Endpoint is designed for supported AWS service such as dynamo db or s3 in this case i assume the endpoint is still the valid option

upvoted 2 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: A

Use a VPC endpoint for DynamoDB. A VPC endpoint enables customers to privately connect to supported AWS services: Amazon DynamoDB or Amazon Simple Storage Service (Amazon S3).

upvoted 3 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

A VPC endpoint enables private connectivity between VPCs and AWS services without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect. Traffic remains within the AWS network.

upvoted 2 times

MikeDu 1 year, 10 months ago

Selected Answer: A

VPC endpoints for service in private subnets

upvoted 2 times

RashiJaiswal 1 year, 11 months ago

Selected Answer: A

VPC endpoint for dynamodb and S3

upvoted 2 times

markw92 1 year, 11 months ago

VPC endpoints for DynamoDB can alleviate these challenges. A VPC endpoint for DynamoDB enables Amazon EC2 instances in your VPC to use their private IP addresses to access DynamoDB with no exposure to the public internet. Your EC2 instances do not require public IP addresses, and you don't need an internet gateway, a NAT device, or a virtual private gateway in your VPC. You use endpoint policies to control access to DynamoDB. Traffic between your VPC and the AWS service does not leave the Amazon network.

upvoted 4 times

dmt6263 2 years ago

AAAAAAAAAA

upvoted 2 times

gx2222 2 years, 2 months ago

Selected Answer: A

Option A: Use a VPC endpoint for DynamoDB - This is the correct option. A VPC endpoint for DynamoDB allows communication between resources in your VPC and Amazon DynamoDB without traversing the internet or a NAT instance, which is more secure.

upvoted 3 times

GalileoEC2 2 years, 3 months ago

A

The most secure way to access an Amazon DynamoDB table from Amazon EC2 instances in private subnets while ensuring that the traffic does not leave the AWS network is to use Amazon VPC Endpoints for DynamoDB.

Amazon VPC Endpoints enable private communication between Amazon EC2 instances in a VPC and Amazon services such as DynamoDB, without the need for an internet gateway, NAT device, or VPN connection. When you create a VPC endpoint for DynamoDB, traffic from the EC2 instances to the DynamoDB table remains within the AWS network and does not traverse the public internet.

upvoted 2 times

AIIGOD 2 years, 3 months ago

private...backend Answer A

upvoted 2 times

bdp123 2 years, 4 months ago

Selected Answer: A

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/vpc-endpointsdynamodb.html>

A VPC endpoint for DynamoDB enables Amazon EC2 instances in your VPC to use their private IP addresses to access DynamoDB with no exposure to the public internet. Your EC2 instances do not require public IP addresses, and you don't need an internet gateway, a NAT device, or a virtual private gateway in your VPC. You use endpoint policies to control access to DynamoDB. Traffic between your VPC and the AWS service does not leave the Amazon network.

upvoted 3 times

ProfXsamson 2 years, 4 months ago

ExamTopics.com should be sued for this answer tagged as Correct answer.

upvoted 4 times

mp165 2 years, 5 months ago

Selected Answer: A

A is correct. VPC end point. D exposed to the internet

upvoted 4 times

An entertainment company is using Amazon DynamoDB to store media metadata. The application is read intensive and experiencing delays. The company does not have staff to handle additional operational overhead and needs to improve the performance efficiency of DynamoDB without reconfiguring the application.

What should a solutions architect recommend to meet this requirement?

- A. Use Amazon ElastiCache for Redis.
- B. Use Amazon DynamoDB Accelerator (DAX). **Most Voted**
- C. Replicate data by using DynamoDB global tables.
- D. Use Amazon ElastiCache for Memcached with Auto Discovery enabled.

Correct Answer: B

Community vote distribution

B (100%)

Comments

techhb **Highly Voted** 3 years, 5 months ago

Selected Answer: B

DAX stands for DynamoDB Accelerator, and it's like a turbo boost for your DynamoDB tables. It's a fully managed, in-memory cache that speeds up the read and write performance of your DynamoDB tables, so you can get your data faster than ever before.

upvoted 26 times

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: B

A. Using Amazon ElastiCache for Redis would require modifying the application code and is not specifically designed to enhance DynamoDB performance.

C. Replicating data with DynamoDB global tables would require additional configuration and operational overhead.

D. Using Amazon ElastiCache for Memcached with Auto Discovery enabled would also require application code modifications and is not specifically designed for improving DynamoDB performance.

In contrast, option B, using Amazon DynamoDB Accelerator (DAX), is the recommended solution as it is purpose-built for enhancing DynamoDB performance without the need for application reconfiguration. DAX provides a managed caching layer that significantly reduces read latency and offloads traffic from DynamoDB tables.

upvoted 13 times

raghav2704 **Most Recent** 1 year ago

Selected Answer: B

elastic cache redis and memcached as mentioned in option A and D though supported by , it is not recommended with non sql database as Dynamo DB as it simply leads to more application changes with more hassle.

And replicating data with global tables will make your job even cumbersome, so dont do it.

i will go with option B

upvoted 1 times

1e22522 1 year, 10 months ago

Selected Answer: B

Use the turbo

upvoted 2 times

awsgeek75 2 years, 5 months ago

B: <https://aws.amazon.com/dynamodb/dax/>

improve 10x performance (marketing pitch on above link) with fully managed service so no reconfiguration or operational overhead involved.

upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: B

Amazon DynamoDB Accelerator (DAX) is a fully managed, highly available caching service built for Amazon DynamoDB. DAX delivers up to a 10 times performance improvement—from milliseconds to microseconds—even at millions of requests per second.

[https://aws.amazon.com/dynamodb/dax/#:~:text=Amazon%20DynamoDB%20Accelerator%20\(-,DAX\),-is%20a%20fully](https://aws.amazon.com/dynamodb/dax/#:~:text=Amazon%20DynamoDB%20Accelerator%20(-,DAX),-is%20a%20fully)

upvoted 2 times

Abrar2022 3 years ago

Selected Answer: B

improve the performance efficiency of DynamoDB

upvoted 2 times

gx2222 3 years, 2 months ago

Selected Answer: B

Amazon DynamoDB Accelerator (DAX) is a fully managed, highly available, in-memory cache for DynamoDB that helps improve the read performance of DynamoDB tables. DAX provides a caching layer between the application and DynamoDB, reducing the number of read requests made directly to DynamoDB. This can significantly reduce read latencies and improve overall application performance.

upvoted 3 times

osmk 3 years, 2 months ago

B-->Applications that are read-intensive==><https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/DAX.html#DAX.use-cases>

upvoted 2 times

LuckyAro 3 years, 4 months ago

Selected Answer: B

DynamoDB Accelerator, less over head.

upvoted 3 times

wmp7039 3 years, 4 months ago

Option B is incorrect as the constraint in the question is not to reconfigure the application. DAX requires application to be reconfigured and point to DAX instead of DynamoDB

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/DAX.client.modify-your-app.html>

Answer should be A

upvoted 3 times

LoXoL 2 years, 4 months ago

DAX does not require application logic modification (compatible with existing DynamoDB APIs). ElastiCache would work after changes on app's code.

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: B

To improve the performance efficiency of DynamoDB without reconfiguring the application, a solutions architect should recommend using Amazon DynamoDB Accelerator (DAX) which is Option B as the correct answer.

DAX is a fully managed, in-memory cache that can be used to improve the performance of read-intensive workloads on DynamoDB. DAX stores frequently accessed data in memory, allowing the application to retrieve data from the cache rather than making a request to DynamoDB. This can significantly reduce the number of read requests made to DynamoDB, improving the performance and reducing the latency of the application.

upvoted 4 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A, using Amazon ElastiCache for Redis, would not be a good fit because it is not specifically designed for use with DynamoDB and would require reconfiguring the application to use it.

Option C, replicating data using DynamoDB global tables, would not directly improve the performance of reading requests and would require additional operational overhead to maintain the replication.

Option D, using Amazon ElastiCache for Memcached with Auto Discovery enabled, would also not be a good fit because it is not specifically designed for use with DynamoDB and would require reconfiguring the application to use it.

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B

upvoted 3 times

k1kavi1 3 years, 5 months ago

Selected Answer: B

Agreed

upvoted 3 times

Shasha1 3 years, 6 months ago

B

DAX is a fully managed, highly available, in-memory cache for DynamoDB that delivers lightning-fast performance and consistent low-latency responses. It provides fast performance without requiring any application reconfiguration

upvoted 4 times

Wpcorgan 3 years, 6 months ago

B is correct

upvoted 2 times

goatbernard 3 years, 6 months ago

Selected Answer: B

DAX is the cache for this

upvoted 2 times

A company's infrastructure consists of Amazon EC2 instances and an Amazon RDS DB instance in a single AWS Region. The company wants to back up its data in a separate Region.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Use AWS Backup to copy EC2 backups and RDS backups to the separate Region. **Most Voted**
- B. Use Amazon Data Lifecycle Manager (Amazon DLM) to copy EC2 backups and RDS backups to the separate Region.
- C. Create Amazon Machine Images (AMIs) of the EC2 instances. Copy the AMIs to the separate Region. Create a read replica for the RDS DB instance in the separate Region.
- D. Create Amazon Elastic Block Store (Amazon EBS) snapshots. Copy the EBS snapshots to the separate Region. Create RDS snapshots. Export the RDS snapshots to Amazon S3. Configure S3 Cross-Region Replication (CRR) to the separate Region.

Correct Answer: A

Community vote distribution

A (98%)

24

Comments

cookieMr **Highly Voted** 2 years, 11 months ago

Selected Answer: A

Using AWS Backup to copy EC2 and RDS backups to the separate Region is the solution that meets the requirements with the least operational overhead. AWS Backup simplifies the backup process and automates the copying of backups to another Region, reducing the manual effort and operational complexity involved in managing separate backup processes for EC2 instances and RDS databases.

Option B is incorrect because Amazon Data Lifecycle Manager (Amazon DLM) is not designed for directly copying RDS backups to a separate region.

Option C is incorrect because creating Amazon Machine Images (AMIs) and read replicas adds complexity and operational overhead compared to a dedicated backup solution.

Option D is incorrect because using Amazon EBS snapshots, RDS snapshots, and S3 Cross-Region Replication (CRR) involves multiple manual steps and additional configuration, increasing complexity.

upvoted 13 times

vtbk **Highly Voted** 3 years, 5 months ago

Selected Answer: A

Cross-Region backup

Using AWS Backup, you can copy backups to multiple different AWS Regions on demand or automatically as part of a scheduled backup plan. Cross-Region backup is particularly valuable if you have business continuity or compliance requirements to store backups a minimum distance away from your production data.

<https://docs.aws.amazon.com/aws-backup/latest/devguide/whatisbackup.html>

upvoted 5 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: A

AWS Backups are the most suitable here.

upvoted 1 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: A

A for sure

upvoted 2 times

thewalker 2 years, 4 months ago

Selected Answer: A

The easiest way to backup an EC2 instance and RDS Database would be to use AWS Backup. With AWS Backup you can:

Create a backup plan and select both the EC2 volume and RDS database for backup.

Choose a backup schedule that meets your requirements, such as daily or weekly backups.

AWS Backup will automatically take snapshots of the EC2 volume and backups of the RDS database as per the configured schedule.

The backups will be stored in S3 for long term retention based on your backup plan configuration.

upvoted 4 times

thewalker 2 years, 4 months ago

You can easily restore the EC2 volume or RDS database from these backups in case of data loss or corruption.

AWS manages the entire backup process so there is no operational overhead for you.

Some other options include using cron jobs to trigger snapshots and backups. But AWS Backup provides a fully managed service to centrally backup both EC2 and RDS with minimal effort.

The above is the output from Amazon Q.

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

AWS Backup provides a fully managed, centralized backup service across AWS services. It can be configured to automatically copy backups across Regions.

This requires minimal operational overhead compared to the other options:

upvoted 3 times

oguzbeliren 2 years, 10 months ago

D would have been a great option but the questions requires less manual effort. So, A is better.

upvoted 2 times

cheese929 3 years, 1 month ago

Selected Answer: A

A is correct

upvoted 3 times

kruasan 3 years, 1 month ago

Selected Answer: A

Option B, using Amazon Data Lifecycle Manager (Amazon DLM) to copy EC2 backups and RDS backups to the separate Region, would require more operational overhead because DLM is primarily designed for managing the lifecycle of Amazon EBS snapshots, and would require additional configuration to manage RDS backups.

Option C, creating AMIs of the EC2 instances and read replicas of the RDS DB instance in the separate Region, would require more manual effort to manage the backup and disaster recovery process, as it requires manual creation and management of AMIs and read replicas.

upvoted 4 times

kruasan 3 years, 1 month ago

Option D, creating EBS snapshots and RDS snapshots, exporting them to Amazon S3, and configuring S3 Cross-Region Replication (CRR) to the separate Region, would require more configuration and management effort. Additionally, S3 CRR can have additional charges for data transfer and storage in the destination region.

Therefore, option A is the best choice for meeting the company's requirements with the least operational overhead.

upvoted 4 times

gx2222 3 years, 2 months ago

Selected Answer: A

Option A, using AWS Backup to copy EC2 backups and RDS backups to the separate region, is the correct answer for the given scenario.

Using AWS Backup is a simple and efficient way to backup EC2 instances and RDS databases to a separate region. It requires minimal operational overhead and can be easily managed through the AWS Backup console or API. AWS Backup can also provide automated scheduling and retention management for backups, which can help ensure that backups are always available and up to date.

upvoted 4 times

dan80 3 years, 5 months ago

A is correct - you need to find a backup solution for EC2 and RDS. DLM doesn't work with RDS, only with snapshots.

upvoted 2 times

techhb 3 years, 5 months ago

Selected Answer: A

using Amazon DLM to copy EC2 backups and RDS backups to the separate region, is not a valid solution because Amazon DLM does not support backing up data across regions.

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Selected Answer: B

Option B. Use Amazon Data Lifecycle Manager (Amazon DLM) to copy EC2 backups and RDS backups to the separate Region.

Amazon DLM is a fully managed service that helps automate the creation and retention of Amazon EBS snapshots and RDS DB snapshots. It can be used to create and manage backup policies that specify when and how often snapshots should be created, as well as how long they should be retained. With Amazon DLM, you can easily and automatically create and manage backups of your EC2 instances and RDS DB instances in a separate Region, with minimal operational overhead.

upvoted 1 times

HayLLIHuK 3 years, 5 months ago

Buruguduystunstugudunstuy, sorry, but I haven't found any info about copying RDS backups by DLM. The DLM works only with EBS. So the only answer is A - AWS Backup

upvoted 3 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option A, using AWS Backup to copy EC2 backups and RDS backups to the separate Region, would also work, but it may require more manual configuration and management.

Option C, creating AMIs of the EC2 instances and copying them to the separate Region, and creating a read replica for the RDS DB instance in the separate Region, would work, but it may require more manual effort to set up and maintain.

Option D, creating EBS snapshots and copying them to the separate Region, creating RDS snapshots, and exporting them to Amazon S3, and configuring S3 CRR to the separate Region, would also work, but it would involve multiple steps and may require more manual effort to set up and maintain. Overall, using Amazon DLM is likely to be the easiest and most efficient option for meeting the requirements with the least operational overhead.

upvoted 1 times

PassNow1234 3 years, 5 months ago

Some of your answers are very detailed. Can you back them up with a reference?

upvoted 2 times

jwu413 3 years, 4 months ago

All of their answers are from ChatGPT

upvoted 5 times

techhb 3 years, 5 months ago

using Amazon DLM to copy EC2 backups and RDS backups to the separate region, is not a valid solution because Amazon DLM does not support backing up data across regions.

upvoted 5 times

PassNow1234 3 years, 5 months ago

Thanks techhb

upvoted 2 times

egmiranda 3 years, 4 months ago

I choose A, but DLM support cross regions. DLM doesn't support RDS. Cross region copy rules it's a feature of DLM ("For each schedule, you can define the frequency, fast snapshot restore settings (snapshot lifecycle policies only), cross-Region copy rules, and tags")

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/snapshot-lifecycle.html>

upvoted 2 times

Kruiz29 3 years, 4 months ago

This guy is giving wrong answers in detail...lol

upvoted 5 times

YogK 3 years ago

AWS DLM does not support RDS backups, only works with EBS storage. <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/snapshot-lifecycle.html>

upvoted 2 times

career360guru 3 years, 5 months ago

Selected Answer: A

Option A as it is fully managed service with least operational overhead

upvoted 2 times

Shasha1 3 years, 6 months ago

A

AWS Backup is a fully managed service that handles the process of copying backups to a separate Region automatically
upvoted 2 times

babaxoxo 3 years, 6 months ago

Selected Answer: A

Ans A with least operational overhead
upvoted 2 times

rjam 3 years, 6 months ago

AWS Backup supports Supports cross-region backups
upvoted 4 times

A solutions architect needs to securely store a database user name and password that an application uses to access an Amazon RDS DB instance. The application that accesses the database runs on an Amazon EC2 instance. The solutions architect wants to create a secure parameter in AWS Systems Manager Parameter Store.

What should the solutions architect do to meet this requirement?

A. Create an IAM role that has read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM role to the EC2 instance.

Most Voted

B. Create an IAM policy that allows read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM policy to the EC2 instance.

C. Create an IAM trust relationship between the Parameter Store parameter and the EC2 instance. Specify Amazon RDS as a principal in the trust policy.

D. Create an IAM trust relationship between the DB instance and the EC2 instance. Specify Systems Manager as a principal in the trust policy.

Correct Answer: A

Community vote distribution

A (97%)

3%

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 11 months ago

Selected Answer: A

CORRECT Option A

To securely store a database user name and password in AWS Systems Manager Parameter Store and allow an application running on an EC2 instance to access it, the solutions architect should create an IAM role that has read access to the Parameter Store parameter and allow Decrypt access to an AWS KMS key that is used to encrypt the parameter. The solutions architect should then assign this IAM role to the EC2 instance.

This approach allows the EC2 instance to access the parameter in the Parameter Store and decrypt it using the specified KMS key while enforcing the necessary security controls to ensure that the parameter is only accessible to authorized parties.

upvoted 25 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Option B, would not be sufficient, as IAM policies cannot be directly attached to EC2 instances.

Option C, would not be a valid solution, as the Parameter Store parameter and the EC2 instance are not entities that can be related through an IAM trust relationship.

Option D, would not be a valid solution, as the trust policy would not allow the EC2 instance to access the parameter in the Parameter Store or decrypt it using the specified KMS key.

upvoted 9 times

sdasdawa **Highly Voted** 3 years ago

Selected Answer: A

Agree with A, IAM role is for services (EC2 for example)

IAM policy is more for users and groups

upvoted 10 times

lofzee **Most Recent** 1 year, 6 months ago

A all day. Don't even need to read the other answers.

You can't attach a policy to EC2. You have to attach a role.

upvoted 3 times

awsgeek75 1 year, 11 months ago

Selected Answer: A

policy needs to be assigned to something so B is inaccurate
CD are just made up things
upvoted 3 times

TariqKipkemei 2 years, 3 months ago

Selected Answer: A

Create an IAM role that has read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM role to the EC2 instance
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

CORRECT Option A
upvoted 2 times

cookieMr 2 years, 5 months ago

Selected Answer: A

By creating an IAM role with read access to the Parameter Store parameter and Decrypt access to the associated AWS KMS key, the EC2 will have the necessary permissions to securely retrieve and decrypt the database user name and password from the Parameter Store. This approach ensures that the sensitive information is protected and can be accessed only by authorized entities.

Answers B, C, and D are not correct because they do not provide a secure way to store and retrieve the database user name and password from the Parameter Store. IAM policies, trust relationships, and associations with the DB instance are not the appropriate mechanisms for securely managing sensitive credentials in this scenario. Answer A is the correct choice as it involves creating an IAM role with the necessary permissions and assigning it to the EC2 instance to access the Parameter Store securely.

upvoted 3 times

cheese929 2 years, 7 months ago

Selected Answer: A

A is correct
upvoted 2 times

kruasan 2 years, 7 months ago

Selected Answer: A

By creating an IAM role and assigning it to the EC2 instance, the application running on the EC2 instance can access the Parameter Store parameter securely without the need for hard-coding the database user name and password in the application code.

The IAM role should have read access to the Parameter Store parameter and Decrypt access to an AWS KMS key that is used to encrypt the parameter to ensure that the parameter is protected at rest.

upvoted 2 times

HayLLIHuK 2 years, 11 months ago

There should be the Decrypt access to KMS.

"If you choose the SecureString parameter type when you create your parameter, Systems Manager uses AWS KMS to encrypt the parameter value."
<https://docs.aws.amazon.com/systems-manager/latest/userguide/systems-manager-parameter-store.html>

IAM role - for EC2

upvoted 2 times

BENICE 2 years, 11 months ago

A -- is correct option
upvoted 2 times

career360guru 2 years, 11 months ago

Option A.
upvoted 2 times

k1kavi1 2 years, 12 months ago

Selected Answer: A

A is correct
upvoted 2 times

Shasha1 3 years ago

Answer A

Create an IAM role that has read access to the Parameter Store parameter. Allow Decrypt access to an AWS Key Management Service (AWS KMS) key that is used to encrypt the parameter. Assign this IAM role to the EC2 instance. This solution will allow the application to securely access the database user name and password stored in the parameter store.

upvoted 2 times

EKA_CloudGod 3 years ago

Selected Answer: A

A. Attach IAM role to EC2 Instance

<https://aws.amazon.com/blogs/security/digital-signing-asymmetric-keys-aws-kms/>

upvoted 2 times

babaxoxo 3 years ago

Selected Answer: A

Attach IAM role to EC2 Instance profile

upvoted 4 times

goatbernard 3 years ago

Selected Answer: B

IAM policy

upvoted 2 times

A company is designing a cloud communications platform that is driven by APIs. The application is hosted on Amazon EC2 instances behind a Network Load Balancer (NLB). The company uses Amazon API Gateway to provide external users with access to the application through APIs. The company wants to protect the platform against web exploits like SQL injection and also wants to detect and mitigate large, sophisticated DDoS attacks.

Which combination of solutions provides the MOST protection? (Choose two.)

- A. Use AWS WAF to protect the NLB.
- B. Use AWS Shield Advanced with the NLB. **Most Voted**
- C. Use AWS WAF to protect Amazon API Gateway. **Most Voted**
- D. Use Amazon GuardDuty with AWS Shield Standard
- E. Use AWS Shield Standard with Amazon API Gateway.

Correct Answer: BC

Community vote distribution

BC (93%)

2%

Comments

babaxoxo **Highly Voted** 3 years ago

Selected Answer: BC

Shield - Load Balancer, CF, Route53

AWF - CF, ALB, API Gateway

upvoted 50 times

Ouk 2 years, 11 months ago

Thank u U meant WAF* - CloudFormation, right? haha

upvoted 5 times

YogK 2 years, 6 months ago

Shield - Amazon Elastic Compute Cloud (EC2), Elastic Load Balancing (ELB), Amazon CloudFront, AWS Global Accelerator, and Route 53.

WAF - Amazon CloudFront, the Application Load Balancer (ALB), Amazon API Gateway, and AWS AppSync

upvoted 14 times

rjam **Highly Voted** 3 years ago

Selected Answer: BC

AWS Shield Advanced - DDos attacks

AWS WAF to protect Amazon API Gateway, because WAF sits before the API Gateway and then comes NLB.

upvoted 8 times

studynoplay 2 years, 7 months ago

don't agree that NLB sits before API gateway. it should be other way around

upvoted 3 times

aadityaravi8 2 years, 5 months ago

yes.. coming from outside to inside... first of all DDos protection is required so the outer most NLB with Shield Advanced and then filter particular request doing SQL injection and all i.e API Gateway with WAF

upvoted 3 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: BC

AWS Shield protects against Layer 3 and 4 attacks through NLB and AWS WAF protects against Layer 7 attacks through API Gateway.

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: BC

I did not get this right initially, but when I looked it up, it became clear.

Basically any device that acts at network level - Shield (Layer 4 devices, maybe capable of layer 7 as well). For example, NLB, ELB, CF, Route 53.

Any device that works on HTTP/HTTPS/SFTP level(Layer 7 services) ==> WAF, which is ALB, API Gateway.

a Layer 7 device are not capable of layer 4 sevice, and they rely on underlying hardware/firmware(OR in this case of Software defined networking, Software) to do that.

upvoted 2 times

Dharmarajan 1 year, 4 months ago

Definitely a very good question, to be noted for the exam.

upvoted 2 times

EMPERBACH 1 year, 7 months ago

Selected Answer: BD

B- (Shield Advance) PROTECT the platform against web exploits like SQL injection

D- (GuardDuty) also wants to DETECT mitigate large, sophisticated DDoS attacks

WAF use for filter traffic, not make sense here.

upvoted 1 times

lofzee 1 year, 6 months ago

Shield advanced does not protect against SQL injection. That is what WAF is for.

GuardDuty is not the right tool here.

Answers are B and C bro.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: BC

B) Use AWS Shield Advanced with the NLB

C) Use AWS WAF to protect Amazon API Gateway

The key reasons are:

AWS Shield Advanced provides expanded DDoS protection against larger and more sophisticated attacks

Using it with the NLB helps protect against network floods

WAF still provides critical protection against exploits at the API lay

upvoted 7 times

Sat897 2 years, 3 months ago

Selected Answer: BC

WAF - can't support NLB and its supports API Gateway

AWS Shield Advanced - NLB - DDOS

upvoted 2 times

cookieMr 2 years, 5 months ago

B. AWS Shield Advanced provides advanced DDoS protection for the NLB, making it the appropriate choice for protecting against large and sophisticated DDoS attacks at the network layer.

C. AWS WAF is designed to provide protection at the application layer, making it suitable for securing the API Gateway against web exploits like SQL injection.

A. AWS WAF is not compatible with NLB as it operates at the application layer, whereas NLB operates at the transport layer.

D. While GuardDuty helps detect threats, it does not directly protect against web exploits or DDoS attacks. Shield Standard focuses on edge resources, not specifically NLBs.

E. Shield Standard provides basic DDoS protection for edge resources, but it does not directly protect the NLB or address web exploits at the application layer.

upvoted 8 times

cheese929 2 years, 7 months ago

Selected Answer: BC

B and C is correct

upvoted 2 times

kruasan 2 years, 7 months ago

Selected Answer: BC

NLB is a Layer 3/4 component while WAF is a Layer 7 protection component.

That is why WAF is only available for Application Load Balancer in the ELB portfolio. NLB does not terminate the TLS session therefore WAF is not capable of acting on the content. I would consider using AWS Shield at Layer 3/4.

<https://repost.aws/questions/QU2fYXwSWUS0q9vZiWDoaEzA/nlb-need-to-attach-aws-waf>

upvoted 5 times

jdr75 2 years, 8 months ago

Selected Answer: C

- A. Use AWS WAF to protect the NLB.
INCORRECT, cos' WAF not integrate with network LB
- B. Use AWS Shield Advanced with the NLB.

YES. AWS Shield Advanced provides additional protections against more sophisticated and larger attacks for your applications running in AWS. The doubt is : why apply the protection in the NLB when the facing of the app. is the API Gateway?, because Shield should be in front of the communications, not behind.

Nevertheless, this is the best option.

- C. Use AWS WAF to protect Amazon API Gateway.

YES, <https://aws.amazon.com/es/waf/faqs/>

- D. Use Amazon GuardDuty with AWS Shield Standard

INCORRECT, GuardDuty not prevent attacks.

- E. Use AWS Shield Standard with Amazon API Gateway.

INCORRECT. It could be, in principle, a good option, cos' it's in front of the gateway, but the questions said explicitly:

"wants to detect and mitigate large, sophisticated DDoS attacks",

and Standard not provide this feature.

upvoted 2 times

kerl 2 years, 10 months ago

for those who select A, it is wrong, WAF is Layer 7, it only support ABL, APIGateway, CloudFront,COgnito User Pool and AppSync graphql API (<https://docs.aws.amazon.com/waf/latest/developerguide/waf-chapter.html>). NLB is NOT supported. Answer is BC

upvoted 4 times

bullrem 2 years, 10 months ago

Selected Answer: AB

A and B are the best options to provide the greatest protection for the platform against web vulnerabilities and large, sophisticated DDoS attacks.

Option A: Use AWS WAF to protect the NLB. This will provide protection against common web vulnerabilities such as SQL injection.

Option B: Use AWS Shield Advanced with the NLB. This will provide additional protection against large and sophisticated DDoS attacks.

upvoted 2 times

bullrem 2 years, 10 months ago

The best protection for the platform would be to use A and C together because it will protect both the NLB and the API Gateway from web vulnerabilities and DDoS attacks.

upvoted 1 times

omoakin 2 years, 6 months ago

correct

upvoted 1 times

bullrem 2 years, 10 months ago

A and C are the best options for protecting the platform against web vulnerabilities and detecting and mitigating large and sophisticated DDoS attacks.

A: AWS WAF can be used to protect the NLB from web vulnerabilities such as SQL injection.

C: AWS WAF can be used to protect Amazon API Gateway and also provide protection against DDoS attacks.

B: AWS Shield Advanced is used to protect resources from DDoS attacks, but it is not specific to the NLB and may not provide the same level of protection as using WAF specifically on the NLB.

D and E: Amazon GuardDuty and AWS Shield Standard are primarily used for threat detection and may not provide the same level of protection as using WAF and Shield Advanced.

upvoted 1 times

Arifzefen 2 years, 5 months ago

A is not correct as WAF doesn't support Network Load Balancer

upvoted 2 times

drabi 2 years, 11 months ago

Selected Answer: BC

WS Shield Advanced can help protect your Amazon EC2 instances and Network Load Balancers against infrastructure-layer Distributed Denial of Service (DDoS) attacks. Enable AWS Shield Advanced on an AWS Elastic IP address and attach the address to an internet-facing EC2 instance or Network Load Balancer.<https://aws.amazon.com/blogs/security/tag/network-load-balancers/>

upvoted 2 times

duriselman 2 years, 11 months ago

Regional resources

You can protect regional resources in all Regions where AWS WAF is available. You can see the list at AWS WAF endpoints and quotas in the Amazon Web Services General Reference.

You can use AWS WAF to protect the following regional resource types:

Amazon API Gateway REST API

Application Load Balancer

AWS AppSync GraphQL API

Amazon Cognito user pool

You can only associate a web ACL to an Application Load Balancer that's within AWS Regions. For example, you cannot associate a web ACL to an Application Load Balancer that's on AWS Outposts.

upvoted 1 times

duriselvan 2 years, 11 months ago

Ans:-a and C

upvoted 1 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Selected Answer: AC

CORRECT

A. Use AWS WAF to protect the NLB.

C. Use AWS WAF to protect Amazon API Gateway.

AWS WAF is a web application firewall that helps protect web applications from common web exploits such as SQL injection and cross-site scripting attacks. By using AWS WAF to protect the NLB and Amazon API Gateway, the company can provide an additional layer of protection for its cloud communications platform against these types of web exploits.

upvoted 1 times

PassNow1234 2 years, 11 months ago

Your answer is wrong.

Sophisticated DDOS = Shield Advanced (DDoS attacks the front!) What happens if your load balances goes down?

Your API gateway is on the BACK further behind the NLB. SQL Protect that with the WAF

B and C are right.

upvoted 5 times

jwu413 2 years, 10 months ago

This guy just copies and pastes from ChatGPT.

upvoted 5 times

Buruguduystunstugudunstuy 2 years, 11 months ago

About AWS Shield Advanced and Amazon GuardDuty

AWS Shield Advanced is a managed DDoS protection service that provides additional protection for Amazon EC2 instances, Amazon RDS DB instances, Amazon Elastic Load Balancers, and Amazon CloudFront distributions. It can help detect and mitigate large, sophisticated DDoS attacks, "but it does not provide protection against web exploits like SQL injection."

Amazon GuardDuty is a threat detection service that uses machine learning and other techniques to identify potentially malicious activity in your AWS accounts. It can be used in conjunction with AWS Shield Standard, which provides basic DDoS protection for Amazon EC2 instances, Amazon RDS DB instances, and Amazon Elastic Load Balancers. However, neither Amazon GuardDuty nor AWS Shield Standard provides protection against web exploits like SQL injection.

Overall, the combination of using AWS WAF to protect the NLB and Amazon API Gateway provides the most protection against web exploits and large, sophisticated DDoS attacks.

upvoted 1 times

BENICE 2 years, 11 months ago

Option B and C

upvoted 1 times

A company has a legacy data processing application that runs on Amazon EC2 instances. Data is processed sequentially, but the order of results does not matter. The application uses a monolithic architecture. The only way that the company can scale the application to meet increased demand is to increase the size of the instances.

The company's developers have decided to rewrite the application to use a microservices architecture on Amazon Elastic Container Service (Amazon ECS).

What should a solutions architect recommend for communication between the microservices?

- A. Create an Amazon Simple Queue Service (Amazon SQS) queue. Add code to the data producers, and send data to the queue. Add code to the data consumers to process data from the queue. **Most Voted**
- B. Create an Amazon Simple Notification Service (Amazon SNS) topic. Add code to the data producers, and publish notifications to the topic. Add code to the data consumers to subscribe to the topic.
- C. Create an AWS Lambda function to pass messages. Add code to the data producers to call the Lambda function with a data object. Add code to the data consumers to receive a data object that is passed from the Lambda function.
- D. Create an Amazon DynamoDB table. Enable DynamoDB Streams. Add code to the data producers to insert data into the table. Add code to the data consumers to use the DynamoDB Streams API to detect new table entries and retrieve the data.

Correct Answer: A

Community vote distribution

A (95%)

5%

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 11 months ago

Selected Answer: A

Option B, using Amazon Simple Notification Service (SNS), would not be suitable for this use case, as SNS is a pub/sub messaging service that is designed for one-to-many communication, rather than point-to-point communication between specific microservices.

Option C, using an AWS Lambda function to pass messages, would not be suitable for this use case, as it would require the data producers and data consumers to have a direct connection and invoke the Lambda function, rather than being decoupled through a message queue.

Option D, using an Amazon DynamoDB table with DynamoDB Streams, would not be suitable for this use case, as it would require the data consumers to continuously poll the DynamoDB Streams API to detect new table entries, rather than being notified of new data through a message queue.

upvoted 21 times

jerry1 1 year, 2 months ago

I think A is obvious the most suitable, but i dont understand how is this related to ECS mentioned in the question

upvoted 1 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Hence, Option A is the correct answer.

Create an Amazon Simple Queue Service (Amazon SQS) queue. Add code to the data producers, and send data to the queue. Add code to the data consumers to process data from the queue.

upvoted 10 times

cookieMr **Highly Voted** 2 years, 5 months ago

Selected Answer: A

A. Creating an Amazon SQS queue allows for asynchronous communication between microservices, decoupling the data producers and consumers. It provides scalability, flexibility, and ensures that data processing can happen independently and at a desired pace.

B. Amazon SNS is more suitable for pub/sub messaging, where multiple subscribers receive the same message. It may not be the best fit for sequential data processing.

C. Using AWS Lambda functions for communication introduces unnecessary complexity and may not be the optimal solution for sequential data processing.

D. Amazon DynamoDB with DynamoDB Streams is primarily designed for real-time data streaming and change capture scenarios. It may not be the most efficient choice for sequential data processing in a microservices architecture.

upvoted 7 times

satyaamm Most Recent 1 year, 4 months ago

Selected Answer: A

SQS Queues are most suitable as they provide decoupling between microservices.

upvoted 1 times

scar0909 1 year, 9 months ago

Selected Answer: A

A for sure

upvoted 2 times

reviewmine 1 year, 9 months ago

Selected Answer: A

To Decouple a monolithic application - SQS

- SQS standard - not in order

- SQS FIFO - in order

upvoted 5 times

upliftinghut 1 year, 10 months ago

Selected Answer: A

Data is processed sequentially, but the order of results does not matter => SQS; if order matters => SQL FIFO

upvoted 3 times

Cloud_A 1 year, 11 months ago

Selected Answer: A

A is the answer.

upvoted 2 times

TariqKipkemei 2 years, 3 months ago

Selected Answer: A

Data is processed sequentially, but the order of results does not matter = Amazon Simple Queue Service

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

A) Create an Amazon Simple Queue Service (Amazon SQS) queue. Add code to the data producers, and send data to the queue. Add code to the data consumers to process data from the queue.

For asynchronous communication between decoupled microservices, an SQS queue is the most appropriate service to use.

SQS provides a scalable, highly available queue to buffer messages between producers and consumers.

The order of processing does not matter, so a queue model fits well.

The consumers can scale independently to process messages from the queue.

upvoted 4 times

omoakin 2 years, 6 months ago

BBBBBBBBB

upvoted 1 times

Bmarodi 2 years, 6 months ago

Selected Answer: A

SQS for decoupling a monolithic architecture, hence option A is the right answer.

upvoted 2 times

Madhuaws 2 years, 8 months ago

it also says 'the order of results does not matter'. Option B is correct.

upvoted 1 times

asoli 2 years, 8 months ago

Selected Answer: A

The answer is A.

B is wrong because SNS cannot send events "directly" to ECS.

<https://docs.aws.amazon.com/sns/latest/dg/sns-event-destinations.html>

upvoted 2 times

user_deleted 2 years, 9 months ago

Selected Answer: B

it doesn't say it is one-to-one relationships, SNS is better

upvoted 3 times

markw92 2 years, 5 months ago

watch out for this sentence in the question... "Data needs to process sequentially...."

upvoted 3 times

career360guru 2 years, 11 months ago

Selected Answer: A

Best answer is A.

Though C or D is possible it requires additional components and integration and so they are not efficient. Assuming that rate of incoming requests is within limits that SQS can handle A is best option.

upvoted 2 times

k1kavi1 2 years, 12 months ago

Selected Answer: A

A is correct

upvoted 2 times

Shasha1 3 years ago

answer is B.

An Amazon Simple Notification Service (Amazon SNS) topic can be used for communication between the microservices in this scenario. The data producers can be configured to publish notifications to the topic, and the data consumers can be configured to subscribe to the topic and receive notifications as they are published. This allows for asynchronous communication between the microservices. Question here focus on communication between microservices

upvoted 2 times

A company wants to migrate its MySQL database from on premises to AWS. The company recently experienced a database outage that significantly impacted the business. To ensure this does not happen again, the company wants a reliable database solution on AWS that minimizes data loss and stores every transaction on at least two nodes.

Which solution meets these requirements?

- A. Create an Amazon RDS DB instance with synchronous replication to three nodes in three Availability Zones.
- B. Create an Amazon RDS MySQL DB instance with Multi-AZ functionality enabled to synchronously replicate the data.
- C. Create an Amazon RDS MySQL DB instance and then create a read replica in a separate AWS Region that synchronously replicates the data.
- D. Create an Amazon EC2 instance with a MySQL engine installed that triggers an AWS Lambda function to synchronously replicate the data to an Amazon RDS MySQL DB instance.

Most Voted

Correct Answer: B

Community vote distribution

B (98%)

2

Comments

rjam **Highly Voted** 3 years, 6 months ago

Selected Answer: B

Amazon RDS MySQL DB instance with Multi-AZ functionality enabled to synchronously replicate the data
Standby DB in Multi-AZ- synchronous replication

Read Replica always asynchronous. so option C is ignored.
upvoted 21 times

studynoplay **Highly Voted** 3 years ago

Selected Answer: B

RDS Multi-AZ = Synchronous = Disaster Recovery (DR)
Read Replica = Asynchronous = High Availability
upvoted 16 times

pentium75 2 years, 5 months ago

B is correct but the explanation is flawed ;)

RDS Multi-AZ = Synchronous = High Availability
Read Replica = Asynchronous = Disaster Recovery (DR)
upvoted 7 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: B

RDS Multi AZ is the most suitable here.
upvoted 1 times

PoolDead 1 year, 10 months ago

"Minimizes Data Loss" Therefore answer is B.

Amazon RDS read replicas use asynchronous replication, not synchronous. Therefore, this option does not meet the requirement for minimizing data loss as asynchronous replication can result in data lag.
upvoted 2 times

Nawaff 1 year, 11 months ago

Selected Answer: B

Answer is B

Find the below URL for the perfect explanation for the differences between:

- Multi-AZ DB
- Multi-Region DB
- Read replicas DB

<https://aws.amazon.com/rds/features/read-replicas/>

upvoted 2 times

scar0909 2 years, 3 months ago

Selected Answer: B

Multi AZ for availability

upvoted 2 times

riyasara 2 years, 7 months ago

Option A is incorrect because Amazon RDS does not support synchronous replication to three nodes in three Availability Zones.

Option C is incorrect because while you can create a read replica in a separate AWS Region¹, the replication from the primary DB instance to the read replica is asynchronous, not synchronous.

upvoted 4 times

cookieMr 2 years, 11 months ago

Selected Answer: B

B. Create an Amazon RDS MySQL DB instance with Multi-AZ functionality enabled to synchronously replicate the data.

Enabling Multi-AZ functionality in Amazon RDS ensures synchronous replication of data to a standby replica in a different Availability Zone. This provides high availability and minimizes data loss in the event of a database outage.

A. Creating an Amazon RDS DB instance with synchronous replication to three nodes in three Availability Zones would provide even higher availability but is not necessary for the stated requirements.

C. Creating a read replica in a separate AWS Region would provide disaster recovery capabilities but does not ensure synchronous replication or meet the requirement of storing every transaction on at least two nodes.

D. Using an EC2 instance with a MySQL engine and triggering an AWS Lambda function for replication introduces unnecessary complexity and is not the most suitable solution for ensuring reliable and synchronous replication.

upvoted 3 times

channn 3 years, 2 months ago

Selected Answer: B

B

since all other answers r wrong

upvoted 3 times

jayce5 3 years, 2 months ago

Selected Answer: B

B

Since read replica is async.

upvoted 2 times

LuckyAro 3 years, 4 months ago

Selected Answer: C

Multi AZ is not as protected as Multi-Region Read Replica.

upvoted 1 times

pentium75 2 years, 5 months ago

But is IS protected. Read replica is asynchronous, fails to meet the "store EVERY transaction on at least two nodes" requirement.

upvoted 1 times

JayBee65 3 years, 5 months ago

I curios to know why A isn't right. Is it just that it would take more effort?

upvoted 3 times

pentium75 2 years, 5 months ago

How would you implement A?

upvoted 1 times

techhb 3 years, 5 months ago

B is correct C requires more wokr.

upvoted 2 times

BENICE 3 years, 5 months ago

Option B

upvoted 2 times

bammy 3 years, 5 months ago

Multi-AZ will give at least two nodes as required by the question. The answer is B.

Amazon RDS provides high availability and failover support for DB instances using Multi-AZ deployments with a single standby DB instance.
<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Concepts.MultiAZSingleStandby.html>

upvoted 5 times

career360guru 3 years, 5 months ago

Selected Answer: B

Option B

upvoted 2 times

Shasha1 3 years, 6 months ago

Option A is the correct answer in this scenario because it meets the requirements specified in the question. It creates an Amazon RDS DB instance with synchronous replication to three nodes in three Availability Zones, which will provide high availability and durability for the database, ensuring that the data is stored on multiple nodes and automatically replicated across Availability Zones.

Option B is not a correct answer because it creates an Amazon RDS MySQL DB instance with Multi-AZ functionality enabled, which only provides failover capabilities. It does not enable synchronous replication to multiple nodes, which is required in this scenario.

upvoted 2 times

JayBee65 3 years, 5 months ago

Option B is not incorrect: "The primary DB instance is synchronously replicated across Availability Zones to a standby replica to provide data redundancy and minimize latency spikes during system backups" from

<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Concepts.MultiAZSingleStandby.html>

upvoted 2 times

Buruguduystunstugudunstuy 3 years, 5 months ago

I would go with Option B since it meets the company's requirements and is the most suitable solution.

By creating an Amazon RDS MySQL DB instance with Multi-AZ functionality enabled, the solutions architect will ensure that data is automatically synchronously replicated across multiple AZs within the same Region. This provides high availability and data durability, minimizing the risk of data loss and ensuring that every transaction is stored on at least two nodes.

upvoted 3 times

A company is building a new dynamic ordering website. The company wants to minimize server maintenance and patching. The website must be highly available and must scale read and write capacity as quickly as possible to meet changes in user demand.

Which solution will meet these requirements?

A. Host static content in Amazon S3. Host dynamic content by using Amazon API Gateway and AWS Lambda. Use Amazon DynamoDB with on-demand capacity for the database. Configure Amazon CloudFront to deliver the website content.

Most Voted

B. Host static content in Amazon S3. Host dynamic content by using Amazon API Gateway and AWS Lambda. Use Amazon Aurora with Aurora Auto Scaling for the database. Configure Amazon CloudFront to deliver the website content.

C. Host all the website content on Amazon EC2 instances. Create an Auto Scaling group to scale the EC2 instances. Use an Application Load Balancer to distribute traffic. Use Amazon DynamoDB with provisioned write capacity for the database.

D. Host all the website content on Amazon EC2 instances. Create an Auto Scaling group to scale the EC2 instances. Use an Application Load Balancer to distribute traffic. Use Amazon Aurora with Aurora Auto Scaling for the database.

Correct Answer: A

Community vote distribution

A (95%)

5%

Comments

romko **Highly Voted** 3 years, 6 months ago

Selected Answer: A

A - is correct, because Dynamodb on-demand scales write and read capacity

B - Aurora auto scaling scales only read replicas

upvoted 53 times

klayytech 3 years, 2 months ago

That's not correct. Amazon Aurora with Aurora Auto Scaling can scale both read and write replicas. Is there anything else you would like me to help you with?

upvoted 7 times

Yadav_Sanjay 2 years, 11 months ago

That's why Dynamo DB is best suited option

upvoted 3 times

SteveNguyen 1 year, 6 months ago

Dynamo is serverless, so you don't need to worry about maintenance and patching.

upvoted 2 times

Duckydoo 1 year, 12 months ago

Could you point us to a source where it says that Aurora Auto Scaling can scale write replicas? The AWS documentation specifically mentions that it supports only read replicas (e.g. <https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Integrating.AutoScaling.html>):

To meet your connectivity and workload requirements, Aurora Auto Scaling dynamically adjusts the number of Aurora Replicas (reader DB instances) provisioned for an Aurora DB cluster. Aurora Auto Scaling is available for both Aurora MySQL and Aurora PostgreSQL.

upvoted 3 times

Yadav_Sanjay 2 years, 11 months ago

Correct...Both can serve purpose but note the keyword "must scale read and write capacity as quickly as possible to meet changes in user demand". DynamoDB can scale quickly than Aurora. Remember "PUSH BUTTON SCALING FEATURE" of Dynamo DB.

upvoted 8 times

Manlikeleke **Highly Voted** 3 years, 6 months ago

please is this dump enough to pass the exam?

upvoted 16 times

Bobbybash 3 years, 6 months ago

I HOPE SO

upvoted 9 times

LuckyAro 3 years, 4 months ago

You can tell us now ? Going by the date of your post I guess you would have challenged the exam by now ? so how did it go ?

upvoted 9 times

soufiyane 2 years, 2 months ago

did you pass ?

upvoted 2 times

tch **Most Recent** 1 year, 3 months ago

Selected Answer: A

Aurora is more complex to set up and manage....

DynamoDB is simpler to use, less configuration needed, the most important is highly scalable, auto-scaling

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/AutoScaling.html>

upvoted 1 times

satyaamm 1 year, 4 months ago

Selected Answer: A

DynamoDB provides the best read and write capacity here.

upvoted 1 times

Dharmarajan 1 year, 4 months ago

Selected Answer: A

I also initially thought Aurora is better suited since it gives auto scaling as well as auto scaling. However, it doesn't auto scale for writer instances. there is a note in Aurora doc:

"Aurora Auto Scaling doesn't apply to the workload on the writer DB instance. Aurora Auto Scaling helps with the workload only on the reader instances."

<https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Integrating.AutoScaling.html>

Hence the right answer is A. But man, how devious! a small detail can derail you from making the right choice!

upvoted 3 times

devarajchidambaram 1 year, 5 months ago

Selected Answer: A

Dynamodb - No need to provision, patch, or manage servers.

upvoted 1 times

ManikRoy 2 years, 1 month ago

Selected Answer: A

Dynamo DB Push Scaling

upvoted 2 times

Cloud_A 2 years, 5 months ago

Selected Answer: A

[https://aws.amazon.com/blogs/database/how-to-determine-if-amazon-dynamodb-is-appropriate-for-your-needs-and-then-plan-your-migration/#:~:text=Are%20working%20with%20an%20online%20transaction%20processing%20\(OLTP\)%20workload.%20High%2Dperformance%20reads%20and%20writes%20are%20easy%20to%20manage%20with%20DynamoDB%2C%20and%20you%20can%20expect%20performance%20that%20is%20effectively%20constant%20across%20widely%20varying%20loads.](https://aws.amazon.com/blogs/database/how-to-determine-if-amazon-dynamodb-is-appropriate-for-your-needs-and-then-plan-your-migration/#:~:text=Are%20working%20with%20an%20online%20transaction%20processing%20(OLTP)%20workload.%20High%2Dperformance%20reads%20and%20writes%20are%20easy%20to%20manage%20with%20DynamoDB%2C%20and%20you%20can%20expect%20performance%20that%20is%20effectively%20constant%20across%20widely%20varying%20loads.)

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

C,D are out due to EC2 scaling which is not ideal for static content scaling.

A and B are logical choices. B uses Aurora which is more for relational database and comes with the baggage and limitations of RDBMS scaling.

DynamoDB (no SQL) is easier to scale for both read and write. A is simply better than B for an ordering website so that is the better option. Note that B would have been good if A wasn't a choice.

upvoted 3 times

ChaBum 1 year, 9 months ago

Where do you see "static content scaling"?

In the question, I can read "dynamic ordering website", I fail to see how dynamic website can be hosted on S#.

upvoted 2 times

tom_cruise 2 years, 8 months ago

Selected Answer: A

dynamodb is serverless
upvoted 5 times

Angryasianxd 2 years, 8 months ago

Selected Answer: A

Hi all! The answer is A and NOT B on this one as the company is building an ordering website (OLTP). DynamoDB's high performance read and writes are perfect for an OLTP use case.

<https://aws.amazon.com/blogs/database/how-to-determine-if-amazon-dynamodb-is-appropriate-for-your-needs-and-then-plan-your-migration/>
upvoted 4 times

n0pz 2 years, 9 months ago

S3 is discarded since the question says: A company is building a new dynamic ordering website,
upvoted 2 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

minimize server maintenance and patching, highly available, scale read and write = serverless = Amazon S3, Amazon API Gateway, AWS Lambda, Amazon DynamoDB
upvoted 3 times

DebAwsAccount 2 years, 9 months ago

Selected Answer: A

Key phrase in the Question is must scale read and write capacity. Aurora is only for Read.
Amazon DynamoDB has two read/write capacity modes for processing reads and writes on your tables:
On-demand
Provisioned (default, free-tier eligible)
<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.ReadWriteCapacityMode.html>
upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

Minimize maintenance & Patching = Serverless
S3, DynamoDB are serverless
upvoted 2 times

ravindrabagale 2 years, 10 months ago

Minimize maintenance & Patching = Serverless services
Serverless services with no sql database is perfect combination
upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: A

B. This solution leverages serverless technologies like API Gateway and Lambda for hosting dynamic content, reducing server maintenance and patching. Aurora with Aurora Auto Scaling provides a highly available and scalable database solution. Hosting static content in S3 and configuring CloudFront for content delivery ensures high availability and efficient scaling.

A. Using DynamoDB with on-demand capacity may provide scalability, but it does not offer the same level of flexibility and performance as Aurora. Additionally, it does not address the hosting of dynamic content using serverless technologies.

C. Hosting all the website content on EC2 instances requires server maintenance and patching. While using ASG and an ALB helps with availability and scalability, it does not minimize server maintenance as requested.

D. Hosting all the website content on EC2 instances introduces server maintenance and patching. Using Aurora with Aurora Auto Scaling is a good choice for the database, but it does not address the need to minimize server maintenance and patching for the overall infrastructure.

upvoted 1 times

A company has an AWS account used for software engineering. The AWS account has access to the company's on-premises data center through a pair of AWS Direct Connect connections. All non-VPC traffic routes to the virtual private gateway.

A development team recently created an AWS Lambda function through the console. The development team needs to allow the function to access a database that runs in a private subnet in the company's data center.

Which solution will meet these requirements?

- A. Configure the Lambda function to run in the VPC with the appropriate security group. **Most Voted**
- B. Set up a VPN connection from AWS to the data center. Route the traffic from the Lambda function through the VPN.
- C. Update the route tables in the VPC to allow the Lambda function to access the on-premises data center through Direct Connect.
- D. Create an Elastic IP address. Configure the Lambda function to send traffic through the Elastic IP address without an elastic network interface.

Correct Answer: A

Community vote distribution

A (76%)

C (24%)

Comments

Gil80 **Highly Voted** 3 years, 6 months ago

Selected Answer: A

To configure a VPC for an existing function:

1. Open the Functions page of the Lambda console.
2. Choose a function.
3. Choose Configuration and then choose VPC.
4. Under VPC, choose Edit.
5. Choose a VPC, subnets, and security groups. <-- **That's why I believe the answer is A**.

Note:

If your function needs internet access, use network address translation (NAT). Connecting a function to a public subnet doesn't give it internet access or a public IP address.

upvoted 24 times

markw92 2 years, 11 months ago

The question says on-prem database...how do we create a SG for that instance in AWS? C make sense. my 2 cents..

upvoted 7 times

allanm 10 months, 1 week ago

On-prem databases is already connected to AWS through direct connection. This is about deploying Lambda in the VPC

upvoted 1 times

AZ_Master 2 years, 6 months ago

A is correct. To configure SG for Lambda, go to Lambda function -> Configure -> Edit VPC and scroll down to see "security groups" where you can configure Lambda for VPC.

Also see here

<https://repost.aws/questions/QU5aj1a6jBQ92Kp56klbZFNw/aws-lambda-to-on-premise-via-direct-connect-and-aws-privatelink>

upvoted 2 times

SSadiq 1 year, 12 months ago

SG is for Lambda and not for the on-prem database. A is the correct option

upvoted 2 times

javitech83 **Highly Voted** 3 years, 6 months ago

Selected Answer: A

it is A. C is not correct at all as in the question it mentions that the VPC already has connectivity with on-premises
upvoted 11 times

LuckyAro 3 years, 4 months ago

C says to "update the route table" not create a new connection. C is correct.
upvoted 6 times

ruqui 3 years ago

C is wrong. Lambda can't connect by default to resources in a private VPC, so you have to do some specific setup steps to run in a private VPC, Answer A is correct
upvoted 3 times

Adios_Amigo 3 years, 1 month ago

No need to do route updates. This is because the route to the destination on-premises is already set.
upvoted 4 times

jerry1 1 year, 2 months ago

the question didn't mention where the direct connect is set up but mention that all non vpc traffic are private
so i think it makes sense you need to config route table
and i think in C, update the vpc already imply that the lambda is going to be run in that vpc
upvoted 1 times

ChhatwaniB **Most Recent** 1 year, 2 months ago

Selected Answer: C

Same question in exam topic #524 and highly voted answer is C there

Since database on company's data center can not have security group. my vote goes to C
upvoted 2 times

Dharmarajan 1 year, 4 months ago

Selected Answer: A

The language is confusing, but the option A essentially says that "Configure the Lambda to execute in your own VPC (instead of AWS's own Lambda VPC) and since your VPC already is configured with the route to On prem data center, it will be able to access the on prem data center resources.
upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: A

By default, Lambda functions operate outside the customer's VPC. To access on-premises resources via Direct Connect, the Lambda function must be deployed inside the VPC.

Once attached to the VPC, Lambda can use the VPC's networking setup (route tables, security groups, and the virtual private gateway) to communicate with the on-premises database over Direct Connect.
upvoted 2 times

ensbrvsns 1 year, 9 months ago

Selected Answer: C

either A or C
upvoted 1 times

jatric 1 year, 11 months ago

Selected Answer: C

C is correct as lambda already in VPC and AWS account already has connection setup with on-premise database in private subnet
upvoted 2 times

lofzee 2 years ago

Selected Answer: A

B,C,D don't have any logic behind them.
A is the most logical answer as you need to connect a function to a VPC. The VPC will be connected to the on-prem database.
upvoted 2 times

MehulKapadia 2 years, 1 month ago

Selected Answer: A

Answer A: During Lambda function creation select "Advanced Settings" select "Enable VPC", this will allow you to select VPC, Subnets and SecurityGroup for your Lambda function. This is the way Lambda can get controlled access to resources in your VPC.

Default Lambda Settings:

When you create a Lambda function without specifying a VPC, the Lambda function does not get associated with any particular VPC. By default, Lambda functions are not deployed within a VPC and do not have access to resources within a VPC, such as EC2 instances, RDS databases, or Elasticache clusters, unless you explicitly configure the Lambda function to connect to a VPC.

upvoted 5 times

Uzbekistan 2 years, 2 months ago

Selected Answer: C

Update the route tables in the VPC to allow the Lambda function to access the on-premises data center through Direct Connect.

By updating the route tables in the VPC to allow the Lambda function to access the on-premises data center through Direct Connect, is the most appropriate solution. By updating the route tables, you can specify the route for traffic from the Lambda function to the IP address range of the on-premises data center via the Direct Connect connection. This ensures that the Lambda function can securely communicate with the database in the private subnet of the data center.

upvoted 2 times

awsgeek75 2 years, 4 months ago

Every time I read this question the badly phrased options make no sense at all. I now want to vote for A but it makes no sense.

Question says: All non-VPC traffic routes to the virtual private gateway

So Lambda is technically a non VPC traffic too. This means it already goes through the VPGW but we don't know what it connects. Assuming it connect the data-centre to AWS then A makes sense. BUT all this is based on different interpretation now for me.

upvoted 9 times

pentium75 2 years, 5 months ago

Selected Answer: A

The wording is strange because technically, the Lambda function does not "run in the VPC", rather it is connected to the VPC, but otherwise A is what relevant documentation says - connect the Lambda function to the VPN and allow traffic in the security group.

Not B, we have Direct Connect, no need for VPN.

Not C, route is already in place. And route alone does not help - the "route tables in the VPC" are completely irrelevant as long as we don't connect the Lambda function to the VPC.

Not D, an "Elastic IP address" is always connected to an "elastic network interface", such is created automatically with A.

upvoted 7 times

Kanagarajd 2 years, 3 months ago

I agree with explanation!

upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

The question and options are very badly worded so it makes C a possible candidate (unconvincingly though!).

B: VPN is not needed as Direct Connect is already there

D: Irrelevant

A is too generic (appropriate security group for what?) Lambda has fixed VPC or ENI

C is logically relevant

upvoted 1 times

pentium75 2 years, 5 months ago

A says "configure the Lambda function to RUN IN the VPC", but "a Lambda function ALWAYS runs inside a VPC owned by the Lambda service" (<https://docs.aws.amazon.com/lambda/latest/dg/foundation-networking.html>). "You can configure a Lambda function to CONNECT TO private subnets in a virtual private cloud (VPC) in your AWS account", but "connect to" is not the same as "run in"

(<https://docs.aws.amazon.com/lambda/latest/dg/configuration-vpc.html>). Otherwise A would make sense (you CAN assign a security group to the Elastic Network Interface that Lambda uses to connect to your VPC).

upvoted 2 times

pentium75 2 years, 5 months ago

B We already have Direct Connect, so why set up VPN

C doesn't make sense because "all non-VPC traffic [already] routes to the virtual private gateway" (which is obviously connected to the Direct Connect gateway), so why should you "update the route tables"?

D sounds plausible; however, an Elastic IP address is associated with an Elastic Network Interface (though that is automatically provided by AWS). So the "without an elastic network interface" makes D wrong.

My best guess is that there's a typo or misunderstanding in the answers. It's either A but it should read "connect to the VPC" instead of "run in the VPC", or it's D but it should read "without CREATING an elastic network interface" or "WITH an elastic network interface".

upvoted 2 times

xdkonorek2 2 years, 7 months ago

Selected Answer: C

it's not A:

A Lambda function always runs inside a VPC owned by the Lambda service.

<https://docs.aws.amazon.com/lambda/latest/dg/foundation-networking.html>

upvoted 2 times

liux99 2 years, 7 months ago

The answer is C. The question is to allow lambda to access the database running in private subnet in the corporate data center. The only connectivity with the data center is Direct connect.

upvoted 2 times

lgogor 2 years, 7 months ago

Answer C is correct:

<https://repost.aws/questions/QUJaj1a6jBQ92Kp56klbZFNw/aws-lambda-to-on-premise-via-direct-connect-and-aws-privatelink>

upvoted 2 times

A company runs an application using Amazon ECS. The application creates resized versions of an original image and then makes Amazon S3 API calls to store the resized images in Amazon S3.

How can a solutions architect ensure that the application has permission to access Amazon S3?

- A. Update the S3 role in AWS IAM to allow read/write access from Amazon ECS, and then relaunch the container.
- B. Create an IAM role with S3 permissions, and then specify that role as the taskRoleArn in the task definition. **Most Voted**
- C. Create a security group that allows access from Amazon ECS to Amazon S3, and update the launch configuration used by the ECS cluster.
- D. Create an IAM user with S3 permissions, and then relaunch the Amazon EC2 instances for the ECS cluster while logged in as this account.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 5 months ago

Selected Answer: B

To ensure that an Amazon Elastic Container Service (ECS) application has permission to access Amazon Simple Storage Service (S3), the correct solution is to create an AWS Identity and Access Management (IAM) role with the necessary S3 permissions and specify that role as the taskRoleArn in the task definition for the ECS application.

Option B, creating an IAM role with S3 permissions and specifying that role as the taskRoleArn in the task definition, is the correct solution to meet the requirement.

upvoted 15 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Option A, updating the S3 role in IAM to allow read/write access from ECS and relaunching the container, is not the correct solution because the S3 role is not associated with the ECS application.

Option C, creating a security group that allows access from ECS to S3 and updating the launch configuration used by the ECS cluster, is not the correct solution because security groups are used to control inbound and outbound traffic to resources, and do not grant permissions to access resources.

Option D, creating an IAM user with S3 permissions and relaunching the EC2 instances for the ECS cluster while logged in as this account, is not the correct solution because it is generally considered best practice to use IAM roles rather than IAM users to grant permissions to resources.

upvoted 9 times

Guru4Cloud **Most Recent** 1 year, 9 months ago

Selected Answer: B

B. Create an IAM role with S3 permissions, and then specify that role as the taskRoleArn in the task definition

upvoted 4 times

cookieMr 1 year, 11 months ago

Selected Answer: B

Option B: Create an IAM role with S3 permissions and specify that role as the taskRoleArn in the task definition. This approach allows the ECS task to assume the specified role and gain the necessary permissions to access Amazon S3.

Option A is incorrect because updating the S3 role in IAM and relaunching the container does not associate the updated role with the ECS task.

Option C is incorrect because creating a security group that allows access from Amazon ECS to Amazon S3 does not grant the necessary permissions to the ECS task.

Option D is incorrect because creating an IAM user with S3 permissions and relaunching the EC2 instances for the ECS cluster does not associate the IAM user with the ECS task.

upvoted 4 times

dydzah 2 years ago

<https://repost.aws/knowledge-center/ecs-fargate-access-aws-services>

upvoted 2 times

k1kavi1 2 years, 5 months ago

Selected Answer: B

<https://www.examttopics.com/discussions/amazon/view/27954-exam-aws-certified-solutions-architect-associate-saa-c02/>

<https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/aws-resource-ecs-taskdefinition.html>

upvoted 3 times

techhb 2 years, 5 months ago

Selected Answer: B

The short name or full Amazon Resource Name (ARN) of the AWS Identity and Access Management role that grants containers in the task permission to call AWS APIs on your behalf.

upvoted 3 times

BENICE 2 years, 5 months ago

Option B

upvoted 2 times

career360guru 2 years, 5 months ago

Selected Answer: B

Option B.

upvoted 3 times

k1kavi1 2 years, 5 months ago

Selected Answer: B

Agreed

upvoted 1 times

lighrz 2 years, 6 months ago

Selected Answer: B

B is the best answer

upvoted 1 times

Wpcorgan 2 years, 6 months ago

B is correct

upvoted 1 times

taer 2 years, 6 months ago

Selected Answer: B

The answer is B.

upvoted 1 times

Nigma 2 years, 6 months ago

B is the answer

upvoted 2 times

A company has a Windows-based application that must be migrated to AWS. The application requires the use of a shared Windows file system attached to multiple Amazon EC2 Windows instances that are deployed across multiple Availability Zone:

What should a solutions architect do to meet this requirement?

- A. Configure AWS Storage Gateway in volume gateway mode. Mount the volume to each Windows instance.
- B. Configure Amazon FSx for Windows File Server. Mount the Amazon FSx file system to each Windows instance. **Most Voted**
- C. Configure a file system by using Amazon Elastic File System (Amazon EFS). Mount the EFS file system to each Windows instance.
- D. Configure an Amazon Elastic Block Store (Amazon EBS) volume with the required size. Attach each EC2 instance to the volume. Mount the file system within the volume to each Windows instance.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Nigma **Highly Voted** 2 years, 6 months ago

Correct is B

FSx --> shared Windows file system (SMB)

EFS --> Linux NFS

upvoted 10 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: B

FSx is most suitable here for windows file systems.

upvoted 1 times

TariqKipkemei 1 year, 9 months ago

Selected Answer: B

Windows file system = Amazon FSx for Windows File Server

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

Configure Amazon FSx for Windows File Server. Mount the Amazon FSx file system to each Windows instance.

upvoted 2 times

cookieMr 1 year, 11 months ago

Selected Answer: B

Option B: Configure Amazon FSx for Windows File Server. This service provides a fully managed Windows file system that can be easily shared across multiple EC2 Windows instances. It offers high performance and supports Windows applications that require file storage.

Option A is incorrect because AWS Storage Gateway in volume gateway mode is not designed for shared file systems.

Option C is incorrect because while Amazon EFS can be mounted to multiple instances, it is a Linux-based file system and may not be suitable for Windows applications.

Option D is incorrect because attaching and mounting an Amazon EBS volume to multiple instances simultaneously is not supported.

upvoted 3 times

Bmarodi 2 years ago

Selected Answer: B

Option B is right answer.

upvoted 1 times

k1kavi1 2 years, 5 months ago

Selected Answer: B

References :

<https://www.examttopics.com/discussions/amazon/view/28006-exam-aws-certified-solutions-architect-associate-saa-c02/>

<https://docs.aws.amazon.com/AmazonECS/latest/developerguide/wfsx-volumes.html>

upvoted 2 times

techhb 2 years, 5 months ago

Selected Answer: B

EFS is not compatible with Windows.

<https://pilotcoresystems.com/insights/ebs-efs-fsx-s3-how-these-storage-options-differ/#:~:text=EFS%20works%20with%20Linux%20and,with%20all%20Window%20Server%20platforms.>

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 5 months ago

Selected Answer: B

A. Configure AWS Storage Gateway in volume gateway mode. Mount the volume to each Windows instance.

This option is incorrect because AWS Storage Gateway is not a file storage service. It is a hybrid storage service that allows you to store data in the cloud while maintaining low-latency access to frequently accessed data. It is designed to integrate with on-premises storage systems, not to provide file storage for Amazon EC2 instances.

B. Configure Amazon FSx for Windows File Server. Mount the Amazon FSx file system to each Windows instance.

This is the correct answer. Amazon FSx for Windows File Server is a fully managed file storage service that provides a native Windows file system that can be accessed over the SMB protocol. It is specifically designed for use with Windows-based applications, and it can be easily integrated with existing applications by mounting the file system to each EC2 instance.

upvoted 4 times

Buruguduystunstugudunstuy 2 years, 5 months ago

C. Configure a file system by using Amazon Elastic File System (Amazon EFS). Mount the EFS file system to each Windows instance.

This option is incorrect because Amazon EFS is a file storage service that is designed for use with Linux-based applications. It is not compatible with Windows-based applications, and it cannot be accessed over the SMB protocol.

D. Configure an Amazon Elastic Block Store (Amazon EBS) volume with the required size. Attach each EC2 instance to the volume. Mount the file system within the volume to each Windows instance.

This option is incorrect because Amazon EBS is a block storage service, not a file storage service. It is designed for storing raw block-level data that can be accessed by a single EC2 instance at a time. It is not designed for use as a shared file system that can be accessed by multiple instances.

upvoted 1 times

BENICE 2 years, 5 months ago

B - is correct

upvoted 1 times

career360guru 2 years, 5 months ago

Selected Answer: B

Option B

upvoted 1 times

Wpcorgan 2 years, 6 months ago

B is correct

upvoted 1 times

xua81376 2 years, 6 months ago

B FSx for windows

upvoted 2 times

BENICE 2 years, 6 months ago

B is correct option

upvoted 1 times

rjam 2 years, 6 months ago

Selected Answer: B

Amazon FSx for Windows File Server

upvoted 4 times

A company is developing an ecommerce application that will consist of a load-balanced front end, a container-based application, and a relational database. A solutions architect needs to create a highly available solution that operates with as little manual intervention as possible.

Which solutions meet these requirements? (Choose two.)

- A. Create an Amazon RDS DB instance in Multi-AZ mode. **Most Voted**
- B. Create an Amazon RDS DB instance and one or more replicas in another Availability Zone.
- C. Create an Amazon EC2 instance-based Docker cluster to handle the dynamic application load.
- D. Create an Amazon Elastic Container Service (Amazon ECS) cluster with a Fargate launch type to handle the dynamic application load. **Most Voted**
- E. Create an Amazon Elastic Container Service (Amazon ECS) cluster with an Amazon EC2 launch type to handle the dynamic application load.

Correct Answer: AD

Community vote distribution

AD (100%)

Comments

techhb **Highly Voted** 2 years, 11 months ago

Selected Answer: AD

<https://containersonaws.com/introduction/ec2-or-aws-fargate/>

A.(O) multi-az <= 'little intervention'

B.(X) read replica <= Promoting a read replica to be a standalone DB instance

You can promote a read replica into a standalone DB instance. When you promote a read replica, the DB instance is rebooted before it becomes available.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_ReadRepl.html

C.(X) use Amazon ECS instead of EC2-based docker for little human intervention

D.(O) Amazon ECS on AWS Fargate : AWS Fargate is a technology that you can use with Amazon ECS to run containers without having to manage servers or clusters of Amazon EC2 instances.

E.(X) EC2 launch type

The EC2 launch type can be used to run your containerized applications on Amazon EC2 instances that you register to your Amazon ECS cluster and manage yourself.

upvoted 14 times

Danny23132412141_2312 **Highly Voted** 3 years ago

AD for sure

Link: <https://www.examttopics.com/discussions/amazon/view/43729-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 5 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: AD

RDS Multi AZ and AWS Fargate are the most suitable here.

upvoted 1 times

lostmagnet001 1 year, 10 months ago

Selected Answer: AD

Highly available application - Amazon RDS DB instance in Multi-AZ

little manual intervention - Fargate

upvoted 2 times

TariqKipkemei 2 years, 3 months ago

Selected Answer: AD

highly available application, little manual intervention = serverless = Amazon Elastic Container Service with Fargate and Amazon RDS DB instance in Multi-AZ mode

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: AD

The correct answers are A and D.

A) Creating an RDS DB instance in Multi-AZ mode provides automatic failover to a standby replica in another Availability Zone, providing high availability.

D) Using ECS Fargate removes the need to provision and manage EC2 instances, allowing the service to scale dynamically based on demand. ECS handles load balancing and availability out of the box.

upvoted 2 times

jkirancdev 2 years, 4 months ago

Selected Answer: AD

AD is the correct answer

upvoted 2 times

cookieMr 2 years, 5 months ago

Selected Answer: AD

A. Create an Amazon RDS DB instance in Multi-AZ mode. This ensures that the database is highly available with automatic failover to a standby replica in another Availability Zone.

D. Create an Amazon Elastic Container Service (Amazon ECS) cluster with a Fargate launch type to handle the dynamic application load. Fargate abstracts the underlying infrastructure, automatically scaling and managing the containers, making it a highly available and low-maintenance option.

Option B is not the best choice as it only creates replicas in another Availability Zone without the automatic failover capability provided by Multi-AZ mode.

Option C is not the best choice as managing a Docker cluster on EC2 instances requires more manual intervention compared to using the serverless capabilities of Fargate in option D.

Option E is not the best choice as it uses the EC2 launch type, which requires managing and scaling the EC2 instances manually. Fargate, as mentioned in option D, provides a more automated and scalable solution.

upvoted 4 times

studynoplay 2 years, 6 months ago

Selected Answer: AD

little manual intervention = Serverless

upvoted 2 times

career360guru 2 years, 11 months ago

Selected Answer: AD

Option A&D

upvoted 1 times

k1kavi1 2 years, 12 months ago

Selected Answer: AD

A and D

upvoted 2 times

Gabs90 3 years ago

Selected Answer: AD

A and D

upvoted 1 times

Wpcorgan 3 years ago

A and D

upvoted 1 times

BENICE 3 years ago

A and D are the options

upvoted 1 times

A company uses Amazon S3 as its data lake. The company has a new partner that must use SFTP to upload data files. A solutions architect needs to implement a highly available SFTP solution that minimizes operational overhead.

Which solution will meet these requirements?

A. Use AWS Transfer Family to configure an SFTP-enabled server with a publicly accessible endpoint. Choose the S3 data lake as the destination. **Most Voted**

B. Use Amazon S3 File Gateway as an SFTP server. Expose the S3 File Gateway endpoint URL to the new partner. Share the S3 File Gateway endpoint with the new partner.

C. Launch an Amazon EC2 instance in a private subnet in a VPC. Instruct the new partner to upload files to the EC2 instance by using a VPN. Run a cron job script, on the EC2 instance to upload files to the S3 data lake.

D. Launch Amazon EC2 instances in a private subnet in a VPC. Place a Network Load Balancer (NLB) in front of the EC2 instances. Create an SFTP listener port for the NLB. Share the NLB hostname with the new partner. Run a cron job script on the EC2 instances to upload files to the S3 data lake.

Correct Answer: A

Community vote distribution

A (100%)

Comments

roxx529 **Highly Voted** 3 years ago

For Exam :

Whenever you see SFTP , FTP look for "Transfer" in options available

upvoted 69 times

LoXoL 2 years, 4 months ago

+ FTPS

upvoted 5 times

Chirantan **Highly Voted** 3 years, 5 months ago

Answer is A

AWS Transfer Family securely scales your recurring business-to-business file transfers to AWS Storage services using SFTP, FTPS, FTP, and AS2 protocols.

<https://aws.amazon.com/aws-transfer-family/>

upvoted 17 times

oguzbeliren 2 years, 10 months ago

Answer A is not an answer because it requires more manual effort. While AWS Transfer Family simplifies the setup of an SFTP server, it still requires management and monitoring. This includes handling scaling, backups, patching, and other administrative tasks associated with managing an SFTP server.

upvoted 2 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: A

AWS Transfer Family is for FTP,SFTP,FTPS

upvoted 1 times

JohnZh 2 years, 2 months ago

Option A is the most suitable choice for implementing a highly available SFTP solution with minimal operational overhead in this scenario

upvoted 2 times

thewalker 2 years, 4 months ago

Selected Answer: A

The key advantages of AWS Transfer Family are:

It provides a fully managed file transfer service that eliminates the need to manage your own file transfer infrastructure. This reduces operational overhead.

It supports multiple protocols like SFTP, FTPS, FTP and AS2, allowing easy and secure exchange of data with business partners and customers.

File transfers happen directly into Amazon S3 buckets or Amazon EFS file systems, so the transferred data can be easily accessed by other AWS services for analytics, processing etc.

AWS Transfer Family maintains existing client-side configurations, so file transfer workflows remain unchanged for end users and partners.

It provides high availability and auto-scaling capabilities to handle varying transfer workloads.

upvoted 7 times

thewalker 2 years, 4 months ago

Storing transferred files in AWS allows using a broad range of services for compliance, archiving and deriving insights from the data.

AWS manages the file transfer infrastructure so you don't have to provision, operate and maintain file transfer servers.

For more details on AWS Transfer Family features, pricing and quotas, please refer to the documentation at <https://aws.amazon.com/transfer-family>

upvoted 2 times

ansagr 2 years, 6 months ago

Selected Answer: A

Amazon S3 File Gateway, involves deploying an on-premises gateway that interfaces with S3. While it's a valid solution, it introduces a level of on-premises infrastructure that may require more operational management.

upvoted 2 times

JA2018 1 year, 6 months ago

just to add on, stem did not manage if the other party is accessing the owner's S3 from an on-prem site.

upvoted 1 times

TariqKipkemei 2 years, 9 months ago

Selected Answer: A

AWS Transfer Family securely scales your recurring business-to-business file transfers to AWS Storage services using SFTP, FTPS, FTP, and AS2 protocols.

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

A is the correct answer.

AWS Transfer Family provides a fully managed SFTP service that can integrate directly with S3. It handles scaling, availability, and security automatically with minimal overhead.

upvoted 3 times

oguzbeliren 2 years, 10 months ago

AWS Transfer Family is a fully managed service that makes it easy to set up and manage secure file transfers. It provides a high-availability SFTP server that can be accessed from the public internet. However, this solution does not minimize operational overhead, as it requires the solutions architect to manage the SFTP server.

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: A

This solution provides a highly available SFTP solution without the need for manual management or operational overhead. AWS Transfer Family allows you to easily set up an SFTP server with authentication, authorization, and integration with S3 as the storage backend.

Option B is not the best choice as it suggests using Amazon S3 File Gateway, which is primarily used for file-based access to S3 storage over NFS or SMB protocols, not for SFTP access.

Option C is not the best choice as it requires manual management of an EC2 instance, VPN setup, and cron job script for uploading files, introducing operational overhead and potential complexity.

Option D is not the best choice as it also requires manual management of EC2 instances, Network Load Balancer, and cron job scripts for file uploads. It is more complex and involves additional components compared to the simpler and fully managed solution provided by AWS Transfer Family in option A.

upvoted 4 times

cookieMr 2 years, 11 months ago

This solution provides a highly available SFTP solution without the need for manual management or operational overhead. AWS Transfer Family allows you to easily set up an SFTP server with authentication, authorization, and integration with S3 as the storage backend.

Option B is not the best choice as it suggests using Amazon S3 File Gateway, which is primarily used for file-based access to S3 storage over NFS or SMB protocols, not for SFTP access.

Option C is not the best choice as it requires manual management of an EC2 instance, VPN setup, and cron job script for uploading files, introducing operational overhead and potential complexity.

Option D is not the best choice as it also requires manual management of EC2 instances, Network Load Balancer, and cron job scripts for file uploads. It is more complex and involves additional components compared to the simpler and fully managed solution provided by AWS Transfer Family in option A.

upvoted 4 times

cookieMr 2 years, 11 months ago

A is correct

upvoted 2 times

markw92 2 years, 11 months ago

I can't wrap my head around why the answer is D? this is so frustrating to see where i went wrong. I vote for A.

upvoted 3 times

studynoplay 3 years ago

Selected Answer: A

minimizes operational overhead = Serverless

AWS Transfer Family is serverless

upvoted 2 times

Rahulbit34 3 years, 1 month ago

AWS Transfer Family is compatible for SFTP<FTPS<FTP. A is the answer

upvoted 2 times

kruasan 3 years, 1 month ago

Selected Answer: A

AWS Transfer Family is a fully managed AWS service that you can use to transfer files into and out of Amazon Simple Storage Service (Amazon S3) storage or Amazon Elastic File System (Amazon EFS) file systems over the following protocols:

Secure Shell (SSH) File Transfer Protocol (SFTP): version 3

File Transfer Protocol Secure (FTPS)

File Transfer Protocol (FTP)

Applicability Statement 2 (AS2)

upvoted 4 times

Oyz 3 years, 2 months ago

Selected Answer: A

A - is the correct answer.

upvoted 3 times

BENICE 3 years, 5 months ago

A -- is the option

upvoted 3 times

A company needs to store contract documents. A contract lasts for 5 years. During the 5-year period, the company must ensure that the documents cannot be overwritten or deleted. The company needs to encrypt the documents at rest and rotate the encryption keys automatically every year.

Which combination of steps should a solutions architect take to meet these requirements with the LEAST operational overhead? (Choose two.)

- A. Store the documents in Amazon S3. Use S3 Object Lock in governance mode.
- B. Store the documents in Amazon S3. Use S3 Object Lock in compliance mode. **Most Voted**
- C. Use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Configure key rotation.
- D. Use server-side encryption with AWS Key Management Service (AWS KMS) customer managed keys. Configure key rotation. **Most Voted**
- E. Use server-side encryption with AWS Key Management Service (AWS KMS) customer provided (imported) keys. Configure key rotation.

Correct Answer: BD

Community vote distribution

BD (79%)

BC (20%)

Comments

[Removed] **Highly Voted** 3 years, 6 months ago

Selected Answer: BD

Originally answered B and C due to least operational overhead. after research its bugging me that the s3 key rotation is determined based on AWS master Key rotation which cannot guarantee the key is rotated with in a 365 day period. stated as "varies" in the documentation. also its impossible to configure this in the console.

KMS-C is a tick box in the console to turn on annual key rotation but requires more operational overhead than SSE-S3.

C - will not guarantee the questions objectives but requires little overhead.

D - will guarantee the questions objective with more overhead.

upvoted 26 times

vadiminski_a 3 years, 5 months ago

I'd have to disagree on that. It states here that aws managed keys are rotated every year which is what the question asks:

<https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html> so C would be correct.

However, it also states that you cannot enable or disable rotation for aws managed keys which would again point towards D

upvoted 4 times

jdr75 3 years, 2 months ago

You can't use this link

<https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html>

to said that "sse-s3" rotates every year, cos' precisely that link refers to "KMS", that is covered with option D.

That the reason the solution is B+D.

upvoted 3 times

Dharmarajan 1 year, 4 months ago

The doc says:

"By default, when you enable automatic key rotation for a KMS key, AWS KMS generates new cryptographic material for the KMS key every year. You can also specify a custom rotation-period to define the number of days after you enable automatic key rotation that AWS KMS will rotate your key material, and the number of days between each automatic rotation thereafter. If you need to immediately initiate key material rotation, you can perform on-demand rotation, regardless of whether or not automatic key rotation is enabled. On-demand rotations do not change existing automatic rotation schedules."

Per this, the KMS does allow annual rotation and with an option to set frequency.

upvoted 1 times

pentium75 **Highly Voted** 2 years, 5 months ago

Selected Answer: BD

A - Governance mode allows exceptions

B - Yes

C - SSE-S3 rotates keys when AWS thinks is right, not when customer wants ("every year")

D - Yes

E - "customer provided (imported) keys" can obviously not be 'rotated automatically', the customer would have to provide/import new keys.

upvoted 9 times

celestial39 2 years, 4 months ago

KMS indeed rotates keys every year, but the reason why C is wrong is that the Amazon managed keys can't be configured to rotate or not.

REF: <https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html#rotate-keys-how-it-works>

upvoted 2 times

LoXoL 2 years, 4 months ago

Agree with pentium75

upvoted 3 times

Yak_Yeti **Most Recent** 1 year, 2 months ago

Selected Answer: BC

SSE-S3 has the least operational overhead for encryption at rest with automatic key rotation because AWS fully manages the keys and their rotation (even if the schedule isn't user-configurable to be strictly annual).

However, if the strict annual rotation is a non-negotiable requirement, then you would have to accept the slightly higher (but still relatively low) operational overhead of managing a KMS customer managed key with automatic annual rotation enabled. Option D.

Given the wording of the original question prioritizing "least operational overhead," the intended answer likely leans towards the fully managed nature of SSE-S3's automatic rotation, even without a guaranteed annual cycle. If strict annual rotation was paramount, the question should have been phrased differently.

IMHO

upvoted 1 times

bignatov 1 year, 9 months ago

Selected Answer: BC

For me it is B and C. C because it does everything automatically and this is the requirement. (with LEAST operational overhead)

upvoted 1 times

tch 1 year, 2 months ago

I think Amazon S3 managed keys (SSE-S3) are designed for and exclusively support encryption of objects within Amazon S3 storage only....

upvoted 1 times

Americanman 1 year, 9 months ago

S3 Object Lock can help prevent Amazon S3 objects from being deleted or overwritten for a fixed amount of time or indefinitely.

In compliance mode, a protected object version can't be overwritten or deleted by any user, including the root user in your AWS account.

upvoted 2 times

ChymKuBoy 1 year, 11 months ago

Selected Answer: BD

BD for sure

upvoted 2 times

lofzee 2 years ago

Selected Answer: BD

basically what that pentium75 guy said - correct.

upvoted 2 times

sudohogan 2 years, 1 month ago

"Least operational overhead": C

upvoted 2 times

huangyou2003 2 years, 1 month ago

Selected Answer: BD

C- you don't have control over rotation schedule for SSE-S3

upvoted 2 times

Tralfalgarlaw 2 years, 1 month ago

Selected Answer: BD

B. Using S3 Object Lock in compliance mode ensures that the documents cannot be substituted or deleted during the specified retention period, which in this case is 5 years. This helps meet the requirement of ensuring the documents remain immutable for the duration of the contract.

D. Using server-side encryption with AWS Key Management Service (AWS KMS) customer managed keys allows for encryption of the documents at rest. Additionally, configuring key rotation for the customer managed keys ensures that the encryption keys are automatically rotated every year, meeting the requirement of rotating encryption keys automatically.

upvoted 3 times

MehulKapadia 2 years, 1 month ago

Selected Answer: BD

Answer: BD:

B: S3 Compliance Mode ensures no one can overwrite or delete the object.

D: Customer-managed KMS Key: (must be enabled) automatic every 1 year

Options not right:

A: Governance mode allows override and delete.

C: SSE-S3 customer do not have control on rotation of keys(Which is once a year in our requirement)

E: As per AWS Documentation, Customer Imported keys cannot be auto rotated.

upvoted 2 times

scar0909 2 years, 3 months ago

Selected Answer: BD

<https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html>

upvoted 1 times

thewalker 2 years, 4 months ago

Selected Answer: BD

The best option to encrypt data at rest in Amazon S3 and rotate the keys every year is to use AWS KMS (Key Management Service).

With AWS KMS:

You can create a customer master key (CMK) and schedule automatic key rotation every year. This ensures the data is encrypted with a new key annually.

When storing objects in S3, you can choose server-side encryption with AWS KMS (SSE-KMS). This will encrypt the data with the CMK you created.

Even if the encrypted data is copied or transferred, it will remain encrypted since the keys are managed by KMS.

You have full control over the keys and can define IAM policies for key access.

AWS manages the encryption, key operations and auditing through integrated services like CloudTrail.

It provides an end-to-end encryption solution within AWS without needing to handle encryption/decryption yourself.

upvoted 1 times

omarshaban 2 years, 4 months ago

THIS WAS IN MY EXAM

upvoted 3 times

Mikado211 2 years, 5 months ago

File cannot be overwritten = s3 compliance mode

encryption AT REST = user-side encryption

upvoted 1 times

Mikado211 2 years, 5 months ago

so the correct answer is BD

upvoted 1 times

awsgeek75 2 years, 4 months ago

user side encryption?

upvoted 1 times

Mikado211 2 years, 5 months ago

Selected Answer: BD

File cannot be overwritten = compliance mode

Encryption AT REST = user-side encryption

upvoted 3 times

ale_brd_111 2 years, 6 months ago

Selected Answer: BD

Question might be outdated.

Amazon S3 now automatically applies server-side encryption with Amazon S3 managed keys (SSE-S3) as the default encryption for all buckets since January 5, 2023.

Additionally, it encrypts the key itself with another key that undergoes regular rotation, enhancing security. Regarding key rotation, the document specifies that the key used to encrypt the S3 Encryption Key undergoes regular rotation. However, it does not explicitly mention the rotation frequency or the ability to customize it. Therefore, considering the requirement for key rotation and the lack of explicit details about rotation frequency, options B and D would be suitable choices.

upvoted 4 times

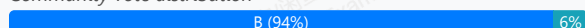
A company has a web application that is based on Java and PHP. The company plans to move the application from on premises to AWS. The company needs the ability to test new site features frequently. The company also needs a highly available and managed solution that requires minimum operational overhead.

Which solution will meet these requirements?

- A. Create an Amazon S3 bucket. Enable static web hosting on the S3 bucket. Upload the static content to the S3 bucket. Use AWS Lambda to process all dynamic content.
- B. Deploy the web application to an AWS Elastic Beanstalk environment. Use URL swapping to switch between multiple Elastic Beanstalk environments for feature testing. **Most Voted**
- C. Deploy the web application to Amazon EC2 instances that are configured with Java and PHP. Use Auto Scaling groups and an Application Load Balancer to manage the website's availability.
- D. Containerize the web application. Deploy the web application to Amazon EC2 instances. Use the AWS Load Balancer Controller to dynamically route traffic between containers that contain the new site features for testing.

Correct Answer: B

Community vote distribution



Comments

cookieMr **Highly Voted** 2 years, 5 months ago

Selected Answer: B

B. Provides a highly available and managed solution with minimum operational overhead. By deploying the web application to EBS, the infrastructure and platform management are abstracted, allowing easy deployment and scalability. With URL swapping, different environments can be created for testing new site features, and traffic can be routed between these environments without any downtime.

A. Suggests using S3 for static content hosting and Lambda for dynamic content. While it offers simplicity for static content, it does not provide the necessary flexibility and dynamic functionality required by a Java and PHP-based web application.

C. Involves manual management of EC2, ASG, and ELB, which requires more operational overhead and may not provide the desired level of availability and ease of testing.

D. Introduces containerization, which adds complexity and operational overhead for managing containers and infrastructure, making it less suitable for a requirement of minimum operational overhead.

upvoted 16 times

Shasha1 **Highly Voted** 3 years ago

B
Elastic Beanstalk is a fully managed service that makes it easy to deploy and run applications in the AWS; To enable frequent testing of new site features, you can use URL swapping to switch between multiple Elastic Beanstalk environments.

upvoted 14 times

oguzbeliren 2 years, 4 months ago

The correct answer is D.

AWS Elastic Beanstalk is a service that makes it easy to deploy and manage web applications in the AWS cloud. However, it is not a good solution for testing new site features frequently, as it can be difficult to switch between multiple Elastic Beanstalk environments.

upvoted 3 times

satyaamm **Most Recent** 1 year, 4 months ago

Selected Answer: B

AWS Elastic Beanstalk is most suitable for deploying and managing web applications.

upvoted 1 times

sudo Hogan 1 year, 7 months ago

Containers allow you test your app in isolated environments, therefore D is the correct option.

upvoted 1 times

hro 1 year, 8 months ago

B - Because AWS Elastic Beanstalk performs an in-place update when you update your application versions, your application might become unavailable to users for a short period of time. To avoid this, perform a blue/green deployment. To do this, deploy the new version to a separate environment, and then swap the CNAMEs of the two environments to redirect traffic to the new version instantly.

upvoted 4 times

reviewmine 1 year, 9 months ago

Selected Answer: B

Elastic Beanstalk can test Blue/Green deployment. Switching Dev to prod/ prod to dev easily.

upvoted 3 times

awsgeek75 1 year, 11 months ago

Selected Answer: B

A and C are not allowing for feature testing.

B and D allow feature testing. D requires overhead of containerisation as well as the LB controller to selectively chose containers for features (assuming on how this might be implemented). EBS allows switching between environment like A/B testing but on whole site. Expensive but cost is not a concern for this question.

upvoted 3 times

master9 1 year, 11 months ago

Selected Answer: D

WS Elastic Beanstalk supports multiple environments, but each environment can only run one platform at a time. A platform is a combination of an operating system, runtime, and web server, and in this case, Java and PHP would be considered different platforms.

So, if you want to use both Java and PHP, you would need to create two separate environments, one for each. You can then link these environments together using AWS services like Route 53 for routing traffic, or use an Application Load Balancer to distribute incoming traffic between the two environments.

upvoted 1 times

ale_brd_111 1 year, 12 months ago

Selected Answer: B

Option B (AWS Elastic Beanstalk): Elastic Beanstalk is a fully managed service that makes it easy to deploy and run applications in multiple languages (including Java and PHP) with MINIMAL OPERATION OVERHEAD. It abstracts the infrastructure management, allowing you to focus on your application. URL swapping in Elastic Beanstalk allows you to easily switch between different environments, making it convenient for testing new features.

upvoted 2 times

Po_chih 2 years, 2 months ago

Selected Answer: B

B

Elastic Beanstalk is a fully managed service that makes it easy to deploy and run applications in the AWS; To enable frequent testing of new site features, you can use URL swapping to switch between multiple Elastic Beanstalk environments.

https://docs.aws.amazon.com/zh_tw/whitepapers/latest/blue-green-deployments/swap-the-environment-of-an-elastic-beanstalk-application.html

upvoted 4 times

Po_chih 2 years, 2 months ago

Selected Answer: B

B

Elastic Beanstalk is a fully managed service that makes it easy to deploy and run applications in the AWS; To enable frequent testing of new site features, you can use URL swapping to switch between multiple Elastic Beanstalk environments.

https://docs.aws.amazon.com/zh_tw/amazondynamodb/latest/developerguide/vpc-endpoints-dynamodb.html

upvoted 2 times

TariqKipkemei 2 years, 3 months ago

Selected Answer: B

AWS Elastic Beanstalk URL swapping is the main ask of this question.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B is the correct answer.

Using AWS Elastic Beanstalk provides a fully managed platform to deploy the web application. Elastic Beanstalk will handle provisioning EC2 instances, load balancing, auto scaling, and application health monitoring.

Elastic Beanstalk's ability to support multiple environments and swap URLs allows easy testing of new features before swapping into production. This requires minimal overhead compared to managing infrastructure directly.

upvoted 4 times

oguzbeliren 2 years, 4 months ago

The correct answer is D.

AWS Elastic Beanstalk is a service that makes it easy to deploy and manage web applications in the AWS cloud. However, it is not a good solution for testing new site features frequently, as it can be difficult to switch between multiple Elastic Beanstalk environments.

upvoted 1 times

Abrar2022 2 years, 6 months ago

S3 is for hosting static websites not dynamic websites or applications

Beanstalk will take care of this.

upvoted 1 times

kruasan 2 years, 7 months ago

Selected Answer: B

Frequent feature testing -

- Multiple Elastic Beanstalk environments can be created easily for development, testing and production use cases.
- Traffic can be routed between environments for A/B testing and feature iteration using simple URL swapping techniques. No complex routing rules or infrastructure changes required.

upvoted 1 times

ashu089 2 years, 8 months ago

who needs discussion in the era the of chatGPT

upvoted 3 times

aadityaravi8 2 years, 5 months ago

chatGPT always change its answer. just say wrong answer, he will come up with new answer each time with justification. chatGPT is not trusted at all

upvoted 4 times

baku98 1 year, 12 months ago

In the era of ChatGPT, individuals across education, business, content creation, healthcare, programming, language learning, innovation, and mental health benefit from discussions. Students, professionals, writers, developers, language learners, innovators, and those seeking support find ChatGPT valuable for learning, problem-solving, creative endeavors, and companionship. It serves as a versatile tool for information, collaboration, and engagement across diverse domains, enhancing communication in an accessible and interactive manner.

upvoted 1 times

A company has an ordering application that stores customer information in Amazon RDS for MySQL. During regular business hours, employees run one-time queries for reporting purposes. Timeouts are occurring during order processing because the reporting queries are taking a long time to run. The company needs to eliminate the timeouts without preventing employees from performing queries.

What should a solutions architect do to meet these requirements?

- A. Create a read replica. Move reporting queries to the read replica. **Most Voted**
- B. Create a read replica. Distribute the ordering application to the primary DB instance and the read replica.
- C. Migrate the ordering application to Amazon DynamoDB with on-demand capacity.
- D. Schedule the reporting queries for non-peak hours.

Correct Answer: A

Community vote distribution

A (100%)

Comments

BENICE **Highly Voted** 3 years, 5 months ago

A is correct answer. This was in my exam
upvoted 29 times

Grace83 3 years, 2 months ago

Did these questions help with your exam?
upvoted 5 times

zied007 **Most Recent** 1 year, 8 months ago

Selected Answer: A

The best solution to address the timeouts and eliminate the impact of long-running reporting queries without disrupting order processing is: A
upvoted 2 times

PR5577 1 year, 10 months ago

Selected Answer: A

Reporting queries will point to read replica. Application will still point to primary db for write / read operations.
upvoted 2 times

bakstorage00001 2 years, 2 months ago

Selected Answer: A

The correct answer to this question is A: Create a read replica. Move reporting queries to the read replica. This solution is designed to alleviate the load on the primary database used by the ordering application. By offloading the reporting queries to a read replica, the primary instance is freed up to handle operational transactions like order processing without contention from the resource-intensive reporting queries. This should effectively reduce or eliminate the timeouts currently experienced during order processing.
upvoted 3 times

lostmagnet001 2 years, 4 months ago

Selected Answer: A

create the replica and all the report queries get data from that read replica.
upvoted 2 times

truongtx8 2 years, 4 months ago

Selected Answer: A

B incorrect because ordering application needs to write data to the DB.
upvoted 3 times

Ruffyt 2 years, 7 months ago

A. By moving the reporting queries to the read replica, the primary DB instance used for order processing is not affected by the long-running reporting queries. This helps eliminate timeouts during order processing while allowing employees to perform their queries without impacting the application's performance.

B. While this can provide some level of load distribution, it does not specifically address the issue of timeouts caused by reporting queries during order processing.

C. While DynamoDB offers scalability and performance benefits, it may require significant changes to the application's data model and querying approach.

D. While this approach can help alleviate the impact on order processing, it does not address the requirement of eliminating timeouts without preventing employees from performing queries.

upvoted 3 times

David_Ang 2 years, 7 months ago

Selected Answer: A

"A" is correct because it does not cause problems in the primary DB

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: A

reports = read replica

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

A is the correct answer.

Creating an RDS MySQL read replica will allow the reporting queries to be isolated and run without affecting performance of the primary ordering application.

Read replicas allow read-only workloads to be scaled out while eliminating contention with the primary write workload.

upvoted 3 times

james2033 2 years, 10 months ago

Selected Answer: A

Question keyword "regular business hours" made D is incorrect.

C migrate to Amazon DynamoDB (No-SQL) is meaningless, remove C.

Answer B, create a "read replica", it is ok, but "ordering application pointed to read replica" is incorrect.

A is correct answer. Easy question.

upvoted 4 times

sickcow 2 years, 11 months ago

Selected Answer: A

A sounds right

upvoted 2 times

rauldevilla 2 years, 11 months ago

Selected Answer: A

Using the primary instance continues with the problem

upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: A

A. By moving the reporting queries to the read replica, the primary DB instance used for order processing is not affected by the long-running reporting queries. This helps eliminate timeouts during order processing while allowing employees to perform their queries without impacting the application's performance.

B. While this can provide some level of load distribution, it does not specifically address the issue of timeouts caused by reporting queries during order processing.

C. While DynamoDB offers scalability and performance benefits, it may require significant changes to the application's data model and querying approach.

D. While this approach can help alleviate the impact on order processing, it does not address the requirement of eliminating timeouts without preventing employees from performing queries.

upvoted 4 times

steev 2 years, 12 months ago

Selected Answer: A

correct

upvoted 2 times

cheese929 3 years, 1 month ago

Selected Answer: A

A is correct.

upvoted 2 times

kruasan 3 years, 1 month ago

Selected Answer: A

Creating a read replica allows the company to offload the reporting queries to a separate database instance, reducing the load on the primary database used for order processing. By moving the reporting queries to the read replica, the ordering application running on the primary DB instance can continue to process orders without timeouts due to the long-running reporting queries.

Option B is not a good solution because distributing the ordering application to the primary DB instance and the read replica does not address the issue of long-running reporting queries causing timeouts during order processing.

upvoted 2 times

A hospital wants to create digital copies for its large collection of historical written records. The hospital will continue to add hundreds of new documents each day. The hospital's data team will scan the documents and will upload the documents to the AWS Cloud.

A solutions architect must implement a solution to analyze the documents, extract the medical information, and store the documents so that an application can run SQL queries on the data. The solution must maximize scalability and operational efficiency.

Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

A. Write the document information to an Amazon EC2 instance that runs a MySQL database.

B. Write the document information to an Amazon S3 bucket. Use Amazon Athena to query the data. **Most Voted**

C. Create an Auto Scaling group of Amazon EC2 instances to run a custom application that processes the scanned files and extracts the medical information.

D. Create an AWS Lambda function that runs when new documents are uploaded. Use Amazon Rekognition to convert the documents to raw text. Use Amazon Transcribe Medical to detect and extract relevant medical information from the text.

E. Create an AWS Lambda function that runs when new documents are uploaded. Use Amazon Textract to convert the documents to raw text. Use Amazon Comprehend Medical to detect and extract relevant medical information from the text.

Most Voted

Correct Answer: BE

Community vote distribution

BE (100%)

Comments

KADSM **Highly Voted** 3 years ago

B and E are correct. Textract to extract text from files. Rekognition can also be used for text detection but after Rekognition - it's mentioned that Transcribe is used. Transcribe is used for Speech to Text. So that option D may not be valid.

upvoted 14 times

romko **Highly Voted** 2 years, 12 months ago

Selected Answer: BE

Usually documents it can be few pages with text, so storing large text in MySQL is not very sufficient + deploy it on EC2 required operation overhead, so A is out.

Only Textract is used for converting documents to text and Comprehend Medical to parse medical phrases. So E is correct.

Correct are BE

upvoted 5 times

satyaamm **Most Recent** 1 year, 3 months ago

Selected Answer: BE

Athena for SQL queries and Lambda for scalability.

upvoted 1 times

LoXoL 1 year, 10 months ago

Selected Answer: BE

no brainer: B,E

upvoted 4 times

awsgeek75 1 year, 11 months ago

Selected Answer: BE

E: Amazon Textract & Amazon Comprehend Medical obviously do the job with least operational overhead. D can do this but it will be extra work and overhead.

B for running SQL queries on S3 bucket directly without extra overhead.

upvoted 4 times

Ruffyit 2 years, 1 month ago

Write the document information to an Amazon S3 bucket. Use Amazon Athena to query the data.

Create an AWS Lambda function that runs when new documents are uploaded. Use Amazon Textract to convert the documents to raw text. Use Amazon Comprehend Medical to detect and extract relevant medical information from the text.

upvoted 2 times

David_Ang 2 years, 1 month ago

Selected Answer: BE

another mistake from the admin, should correct this one, because we all agree

upvoted 4 times

vijaykamal 2 years, 2 months ago

Answer - BE

Option D mentions using Amazon Rekognition and Amazon Transcribe Medical, which are primarily designed for image and audio analysis, respectively. While they can be part of a document processing pipeline, Amazon Textract and Amazon Comprehend Medical are more suitable for extracting structured information from documents, making option E a better choice.

upvoted 3 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: BE

Write the document information to an Amazon S3 bucket. Use Amazon Athena to query the data.

Create an AWS Lambda function that runs when new documents are uploaded. Use Amazon Textract to convert the documents to raw text. Use Amazon Comprehend Medical to detect and extract relevant medical information from the text.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: BE

B and E are the correct answers.

B is correct because storing the scanned documents in Amazon S3 provides highly scalable and durable storage. Amazon Athena allows running SQL queries directly against the data in S3 without needing to load the data into a database.

E is correct because using Lambda functions triggered by uploads provides a serverless approach to automatically process each document. Amazon Textract and Comprehend Medical can extract text and medical information without needing to manage server

upvoted 5 times

james2033 2 years, 4 months ago

Selected Answer: BE

Amazon Comprehend Medical for image reading

<https://aws.amazon.com/comprehend/medical/> .

Amazon Transcribe Medical for speech audio. Remove D. Keep E.

A is meaningless, remove A (EC2).

B use Amazon S3, Athena for querying, keep B.

Conclusion combination B and E are correct answers.

upvoted 3 times

MNotABot 2 years, 5 months ago

AC wrong as involve EC2. Either one of DE are correct so that makes B correct. Now E is obvious answer if we have read AWS FAQs

upvoted 2 times

animefan1 2 years, 5 months ago

Selected Answer: BE

Textract to extract the content and Athena to run sql queries on S3 data

upvoted 2 times

sickcow 2 years, 5 months ago

Selected Answer: BE

From a DE/ML perspective Lambda + Textract + S3 + Athena is the best way to go

upvoted 2 times

rauldevilla 2 years, 5 months ago

Selected Answer: BE

Transcribe is used. Transcribe is used for Speech to Text

upvoted 2 times

cookieMr 2 years, 5 months ago

Selected Answer: BE

B is correct because it suggests writing the document information to an Amazon S3 bucket, which provides scalable and durable object storage. Using Amazon Athena, the data can be queried using SQL, enabling efficient analysis.

E is correct because it involves creating an AWS Lambda function triggered by new document uploads. Amazon Textract is used to convert the documents to raw text, and Amazon Comprehend Medical extracts relevant medical information from the text.

A is incorrect because writing the document information to an Amazon EC2 instance with a MySQL database is not a scalable or efficient solution for analysis.

C is incorrect because creating an Auto Scaling group of Amazon EC2 instances for processing scanned files and extracting information would introduce unnecessary complexity and management overhead.

D is incorrect because using an EC2 instance with a MySQL database for storing document information is not the optimal solution for scalability and efficient analysis.

upvoted 3 times

AlankarJ 2 years, 6 months ago

It states in the question that the written documents are scanned. They are converted into images after being scanned. Rekognition would be best to analyse images.

upvoted 1 times

A company is running a batch application on Amazon EC2 instances. The application consists of a backend with multiple Amazon RDS databases. The application is causing a high number of reads on the databases. A solutions architect must reduce the number of database reads while ensuring high availability.

What should the solutions architect do to meet this requirement?

A. Add Amazon RDS read replicas. **Most Voted**

B. Use Amazon ElastiCache for Redis.

C. Use Amazon Route 53 DNS caching

D. Use Amazon ElastiCache for Memcached.

Correct Answer: A

Community vote distribution

A (60%)

B (40%)

Comments

leonnn **Highly Voted** 3 years, 6 months ago

Selected Answer: B

Use ElastiCache to reduce reading and choose redis to ensure high availability.

upvoted 49 times

Lalo 3 years, 3 months ago

Where is the high availability when the database fails and the cache time runs out?

The answer is a.

upvoted 26 times

Mia2009687 2 years, 11 months ago

They run multiple databases

upvoted 1 times

ruqui 3 years ago

A can't be an answer because the requirement is "reduce the number of database reads"

upvoted 11 times

mandragon 3 years ago

Elasticache for Redis ensures high availability by using read replicas and Multi AZ with failover. It is also faster since it uses cache.

<https://docs.aws.amazon.com/AmazonElastiCache/latest/red-ug/AutoFailover.html>

upvoted 5 times

JoeGuan 2 years, 9 months ago

Caching Frequently Accessed Data: ElastiCache allows you to store frequently accessed or computationally expensive data in-memory within the cache nodes. This means that when an application requests data, ElastiCache can provide the data directly from the cache without having to query the RDS database. This reduces the number of reads on the RDS database because the data is retrieved from the faster in-memory cache.

upvoted 6 times

channn **Highly Voted** 3 years, 2 months ago

Selected Answer: A

A vs B:

A: reduce the number of database reads on main + high availability provide

B: only reduce the number of DB reads

so A wins

upvoted 32 times

pentium75 2 years, 5 months ago

Why "only reduce the number of DB reads"? This is exactly what is asked.

Elasticache for Redis can be HA (contrary to Elasticache for Memcached).

upvoted 7 times

LoXoL 2 years, 4 months ago

pentium75 is right.

upvoted 2 times

46e1a46 **Most Recent** 1 month ago

Selected Answer: A

Two requirements need to be addressed:

Reduce database reads → offload reads to replicas

Ensure high availability → replicas provide redundancy

Caching reduces database reads, BUT:

Requires application code changes to implement cache logic. Cache misses still hit the database. Doesn't inherently provide HA for the database itself. More complex than just adding read replicas

upvoted 1 times

robst84 5 months, 2 weeks ago

Selected Answer: B

Using Replicas won't reduce the number of reads. The answer has to be ElastiCache.

upvoted 1 times

Darius_W 8 months ago

Selected Answer: B

B. Use Amazon ElastiCache for Redis.

Why: A cache reduces repeated database reads. Redis supports Multi-AZ with replication and automatic failover, meeting the high availability requirement.

A (RDS read replicas): Increases read capacity but doesn't reduce total DB reads hitting RDS.

D (Memcached): No HA/replication; a node failure loses cache and availability.

C: DNS caching is unrelated to DB read load.

upvoted 1 times

MultipleWorkerMirroredStrategy 9 months, 1 week ago

Selected Answer: B

Since the question requires us to "reduce the number of database reads while ensuring high availability", a read replica would not meet this requirement.

ElastiCache for Redis provides high availability, while Memcached doesn't. Therefore, B seems to be the right answer for me.

upvoted 2 times

Charlesvg 11 months, 4 weeks ago

Selected Answer: B

Elasticache allows to reduce the reads on the DB while assuring HA.

<https://aws.amazon.com/fr/blogs/database/introducing-99-99-availability-with-amazon-elasticache-for-redis-and-amazon-memorydb-for-redis/>

upvoted 1 times

network_enthusiast 1 year, 1 month ago

Selected Answer: B

B is the correct answer

upvoted 1 times

kg508 1 year, 3 months ago

Selected Answer: B

I voted B because it was mentioned that the number of database reads should be reduced

If I had to reduce the database read load, I would have done it to A

upvoted 1 times

satyaamm 1 year, 3 months ago

Selected Answer: A

RDS read replicas provide HA unlike Redis

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: B

Read replicas distribute read traffic but do not reduce the total number of reads. They scale read capacity but still require every read to hit a database instance.

upvoted 1 times

Qu_B 1 year, 8 months ago

This question is stupid! Both answers can be correct!

upvoted 5 times

Omariox 1 year, 8 months ago

Selected Answer: A

for high availability, the answer should be A

upvoted 4 times

jaradat02 1 year, 10 months ago

Selected Answer: B

RDS read replicas are mainly used for higher performance, and not for higher availability, that's why I chose ElastiCache using Redis.

upvoted 2 times

Aws_aspr 1 year, 10 months ago

Selected Answer: B

I Vote B since question is asking to reduce the database reads. Using Read replicas offloads the reading operation from main db but doesn't reduce it.

upvoted 2 times

freedafeng 1 year, 11 months ago

B is not correct. the effectiveness of caching depends on the application. it's not always a good solution.

upvoted 1 times

jatric 1 year, 11 months ago

Selected Answer: A

why not read replica when you get read performant with HA?

upvoted 4 times

A company needs to run a critical application on AWS. The company needs to use Amazon EC2 for the application's database. The database must be highly available and must fail over automatically if a disruptive event occurs.

Which solution will meet these requirements?

- A. Launch two EC2 instances, each in a different Availability Zone in the same AWS Region. Install the database on both EC2 instances. Configure the EC2 instances as a cluster. Set up database replication. **Most Voted**
- B. Launch an EC2 instance in an Availability Zone. Install the database on the EC2 instance. Use an Amazon Machine Image (AMI) to back up the data. Use AWS CloudFormation to automate provisioning of the EC2 instance if a disruptive event occurs.
- C. Launch two EC2 instances, each in a different AWS Region. Install the database on both EC2 instances. Set up database replication. Fail over the database to a second Region.
- D. Launch an EC2 instance in an Availability Zone. Install the database on the EC2 instance. Use an Amazon Machine Image (AMI) to back up the data. Use EC2 automatic recovery to recover the instance if a disruptive event occurs.

Correct Answer: A

Community vote distribution

A (63%)

C (37%)

Comments

Gil80 **Highly Voted** 3 years, 6 months ago

Selected Answer: A

Changing my vote to A. After reviewing a Udemy course of SAA-C03, it seems that A (multi-AZ and Clusters) is sufficient for HA.
upvoted 42 times

berks 3 years, 5 months ago

what number of class ?

upvoted 5 times

AAAWrekng 2 years, 7 months ago

Here AWS defines HA, and uses the word cluster - AWS has several methods for achieving HA through both approaches, such as through a scalable load balanced cluster or assuming an active-standby pair. - <https://docs.aws.amazon.com/whitepapers/latest/real-time-communication-on-aws/high-availability-and-scalability-on-aws.html>

upvoted 2 times

Gil80 **Highly Voted** 3 years, 6 months ago

Selected Answer: C

The question states that it is a critical app and it has to be HA. A could be the answer, but it's in the same AZ, so if the entire region fails, it doesn't cater for the HA requirement.

However, the likelihood of a failure in two different regions at the same time is 0. Therefore, to me it seems that C is the better option to cater for HA requirement.

In addition, C does state like A that the DB app is installed on an EC2 instance.

upvoted 29 times

javitech83 3 years, 6 months ago

but for C you need communication between the two VPC, which increase the complexity. With a should be enough for HA

upvoted 8 times

Steve_4542636 3 years, 3 months ago

The question doesn't ask which option is the most HA. It asks what meets the requirements.

upvoted 7 times

aussiehoa 3 years ago

Design for region failure? may as well design for AWS failure and put replica in GCP and Azure :v
upvoted 12 times

Kp88 2 years, 10 months ago

And on-prem in multiple DCs and one in mars too :D
upvoted 10 times

cyber_bedouin 2 years, 8 months ago

yep lol, even in the other questions, for HA you can use Multi-AZ
upvoted 3 times

jjcode 2 years, 6 months ago

this is reading comprehension exam not a practical exam.
upvoted 3 times

Burrito69 2 years, 2 months ago

Option C proposes launching two EC2 instances in different AWS Regions and setting up database replication, with failover to a second Region. While this solution does provide geographic redundancy, it may introduce higher latency due to cross-region communication and data replication. Additionally, failover to a different Region typically involves more complex configurations and longer recovery times compared to failover within the same Region.

While Option C may offer a level of redundancy, it might not provide the same level of responsiveness and automatic failover capabilities as Option A, which leverages Availability Zones within the same Region. In scenarios where low latency and rapid failover are critical, Option A is often preferred. However, if geographic redundancy is a top priority and the potential trade-offs in latency and failover time are acceptable, Option C could still be a viable solution.

upvoted 7 times

bora4motion **Most Recent** 1 year, 1 month ago

Selected Answer: A

I don't understand people who voted for C really. A!
upvoted 1 times

SirDNS 1 year, 2 months ago

Selected Answer: A

Failover between Regions is typically slower than within a Region.
upvoted 2 times

Anastesas 1 year, 3 months ago

Selected Answer: A

Cross-region setups are primarily used for disaster recovery rather than high availability due to higher latency and complexity. T
upvoted 1 times

bignatov 1 year, 8 months ago

Selected Answer: A

I am voting for A.
upvoted 2 times

1e22522 1 year, 10 months ago

Selected Answer: A

It's a due to being in the same region different AZ for latency purposes.
upvoted 2 times

jatric 1 year, 11 months ago

Selected Answer: A

A provides HA with 2 EC2 in two AZ with database replication
upvoted 3 times

ChymKuBoy 1 year, 11 months ago

C for sure
upvoted 1 times

Duckydoo 1 year, 11 months ago

What does "Configure the EC2 instances as a cluster" mean? The only "EC2 cluster" that I am aware of is a "cluster placement group". If that's the case, then all EC2 instances in that cluster must be in the same AZ. So option A would be invalid then.
upvoted 2 times

shil_31 2 years ago

Selected Answer: A

Option B uses an AMI for backup and CloudFormation for automation, but doesn't provide high availability or automatic failover.
Option C launches instances in different regions, which may not be necessary and may increase costs.
Option D uses EC2 automatic recovery, which can recover an instance, but doesn't provide high availability or automatic failover.
upvoted 2 times

k_ 2 years ago

Selected Answer: C

The term "disruptive event" implies it requires DR, HA is not sufficient.
upvoted 3 times

Solomon2001 2 years, 1 month ago

Selected Answer: A

Explanation:

Option C provides a solution that ensures high availability by deploying EC2 instances in different AWS Regions. By setting up database replication and failing over the database to a second Region, you ensure automatic failover if a disruptive event occurs in one Region.

Options A and B focus on high availability within a single AWS Region but don't address automatic failover to a different Region in case of a disruptive event.

Option D uses EC2 automatic recovery, but it doesn't provide a solution for automatic failover to a different Region, which is necessary for ensuring high availability in case of a Region-wide failure.
upvoted 2 times

hardy1234567 2 years, 1 month ago

C - I am a little wonder reading same explanation, because exist a very big difference between instance cluster and database cluster.
upvoted 1 times

Uzbekistan 2 years, 2 months ago

Selected Answer: A

Option A suggests deploying two EC2 instances, each in a different Availability Zone within the same AWS Region. This ensures high availability by distributing the instances across multiple physically isolated locations. By installing the database on both EC2 instances and configuring them as a cluster, you create a highly available database setup where one instance can seamlessly take over if the other instance fails.

Additionally, setting up database replication between the instances ensures data consistency and redundancy. If one instance fails, the other instance can continue serving requests without interruption.
upvoted 4 times

derekz 2 years, 3 months ago

Selected Answer: A

A is for HA. D is for DR
upvoted 2 times

MrPCarrot 2 years, 4 months ago

Perfect Answer is A
upvoted 1 times

A company's order system sends requests from clients to Amazon EC2 instances. The EC2 instances process the orders and then store the orders in a database on Amazon RDS. Users report that they must reprocess orders when the system fails. The company wants a resilient solution that can process orders automatically if a system outage occurs.

What should a solutions architect do to meet these requirements?

- A. Move the EC2 instances into an Auto Scaling group. Create an Amazon EventBridge (Amazon CloudWatch Events) rule to target an Amazon Elastic Container Service (Amazon ECS) task.
- B. Move the EC2 instances into an Auto Scaling group behind an Application Load Balancer (ALB). Update the order system to send messages to the ALB endpoint.
- C. Move the EC2 instances into an Auto Scaling group. Configure the order system to send messages to an Amazon Simple Queue Service (Amazon SQS) queue. Configure the EC2 instances to consume messages from the queue. **Most Voted**
- D. Create an Amazon Simple Notification Service (Amazon SNS) topic. Create an AWS Lambda function, and subscribe the function to the SNS topic. Configure the order system to send messages to the SNS topic. Send a command to the EC2 instances to process the messages by using AWS Systems Manager Run Command.

Correct Answer: C

Community vote distribution

C (97%)

2%

Comments

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: C

The key reasons are:

Using an Auto Scaling group ensures the EC2 instances that process orders are highly available and scalable. With SQS, the orders are decoupled from the instances that process them via asynchronous queuing. If instances fail or go down, the orders remain in the queue until new instances can pick them up. This provides automated resilience. Any failed processing can retry by resending messages back to the queue

upvoted 16 times

awsgeek75 **Highly Voted** 1 year, 11 months ago

Selected Answer: C

A uses ECS tasks for something which makes no sense.
B does not solve the reliable processing of orders
C SQS for sending a message and processing it reliable
D is like reinventing SQS with SNS and Lambda mumbo jumbo

upvoted 5 times

satyaamm **Most Recent** 1 year, 3 months ago

Selected Answer: C

SQS queues are the most suitable here.

upvoted 1 times

jjcode 2 years ago

How does SNS capture the requests after the application fails? Those messages are ephemeral by nature and will not hold the data like SQS would. In theory one could create a subscription based service using SNS to stream the data to a service that could store the request, but why...

upvoted 2 times

pentium75 1 year, 11 months ago

That's one of the reasons why D is wrong (not to mention the "Systems Manager Run Command" nonsense).

upvoted 3 times

awsgeek75 1 year, 11 months ago

I stopped reading option D after SNS and Lambda.... it was sounding nonsense. SQS is default reliability delivery system for me.
upvoted 2 times

pavospam 2 years ago

Selected Answer: C

it's C... 4 answers wrong I have found
upvoted 1 times

Ruffyt 2 years, 1 month ago

C.
Option D suggests using Amazon SNS and AWS Lambda, which can be part of an event-driven architecture but may not be the best fit for ensuring the automatic processing of orders during system outages. It relies on an additional AWS Systems Manager Run Command step, which adds complexity and may not be as reliable as using SQS for queuing messages.
upvoted 2 times

David_Ang 2 years, 1 month ago

Selected Answer: C

"C" because they need to store the request and then be process by the system if it fails, SNS does not have that capacity. another mistake from the admin
upvoted 2 times

vijaykamal 2 years, 2 months ago

Selected Answer: C

Option D suggests using Amazon SNS and AWS Lambda, which can be part of an event-driven architecture but may not be the best fit for ensuring the automatic processing of orders during system outages. It relies on an additional AWS Systems Manager Run Command step, which adds complexity and may not be as reliable as using SQS for queuing messages.
upvoted 2 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: C

Move the EC2 instances into an Auto Scaling group. Configure the order system to send messages to an Amazon Simple Queue Service (Amazon SQS) queue. Configure the EC2 instances to consume messages from the queue.
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

C is the correct answer.

Using an Auto Scaling group with EC2 instances behind a load balancer provides high availability and scalability.

Sending the orders to an SQS queue decouples the ordering system from the processing system. The EC2 instances can poll the queue for new orders and process them even during an outage. Any failed orders will go back to the queue for reprocessing.
upvoted 2 times

cookieMr 2 years, 5 months ago

Selected Answer: C

By moving the EC2 into an ASG and configuring them to consume messages from an SQS, the system can decouple the order processing from the order system itself. This allows the system to handle failures and automatically process orders even if the order system or EC2 experience outages.

A. Using an ASG with an EventBridge rule targeting an ECS task does not provide the necessary decoupling and message queueing for automatic order processing during outages.

B. Moving the EC2 instances into an ASG behind an ALB does not address the need for message queueing and automatic processing during outages.

D. Using SNS and Lambda can provide notifications and orchestration capabilities, but it does not provide the necessary message queueing and consumption for automatic order processing during outages. Additionally, using Systems Manager Run Command to send commands for order processing adds complexity and does not provide the desired level of automation.

upvoted 4 times

pisica134 2 years, 5 months ago

D is so unnecessary this confuses people
upvoted 2 times

cookieMr 2 years, 5 months ago

Thx Allmightly for voting system! Answers provided by the site (and not by community) are 20% wrong.
upvoted 4 times

markw92 2 years, 5 months ago

The answer D is so complex and unnecessary. Why moderator is not providing an explanation of answers when there are heavy conflicts. These kind of answers put your knowledge in question which is not good going into the exam.

upvoted 1 times

pentium75 1 year, 11 months ago

The "Correct Answers" for this exam are obviously determined by picking a random letter between A and D.

upvoted 2 times

gx2222 2 years, 8 months ago

Selected Answer: C

To meet the company's requirements of having a resilient solution that can process orders automatically in case of a system outage, the solutions architect needs to implement a fault-tolerant architecture. Based on the given scenario, a potential solution is to move the EC2 instances into an Auto Scaling group and configure the order system to send messages to an Amazon Simple Queue Service (Amazon SQS) queue. The EC2 instances can then consume messages from the queue.

upvoted 3 times

k33 2 years, 8 months ago

Selected Answer: C

Answer : C

upvoted 2 times

nickolaj 2 years, 10 months ago

Selected Answer: C

C. Move the EC2 instances into an Auto Scaling group. Configure the order system to send messages to an Amazon Simple Queue Service (Amazon SQS) queue. Configure the EC2 instances to consume messages from the queue.

To meet the requirements of the company, a solutions architect should ensure that the system is resilient and can process orders automatically in the event of a system outage. To achieve this, moving the EC2 instances into an Auto Scaling group is a good first step. This will enable the system to automatically add or remove instances based on demand and availability.

upvoted 3 times

nickolaj 2 years, 10 months ago

However, it's also necessary to ensure that orders are not lost if a system outage occurs. To achieve this, the order system can be configured to send messages to an Amazon Simple Queue Service (Amazon SQS) queue. SQS is a highly available and durable messaging service that can help ensure that messages are not lost if the system fails.

Finally, the EC2 instances can be configured to consume messages from the queue, process the orders and then store them in the database on Amazon RDS. This approach ensures that orders are not lost and can be processed automatically if a system outage occurs. Therefore, option C is the correct answer.

upvoted 3 times

nickolaj 2 years, 10 months ago

Option A is incorrect because it suggests creating an Amazon EventBridge rule to target an Amazon Elastic Container Service (ECS) task. While this may be a valid solution in some cases, it is not necessary in this scenario.

Option B is incorrect because it suggests moving the EC2 instances into an Auto Scaling group behind an Application Load Balancer (ALB) and updating the order system to send messages to the ALB endpoint. While this approach can provide resilience and scalability, it does not address the issue of order processing and the need to ensure that orders are not lost if a system outage occurs.

Option D is incorrect because it suggests using Amazon Simple Notification Service (SNS) to send messages to an AWS Lambda function, which will then send a command to the EC2 instances to process the messages by using AWS Systems Manager Run Command. While this approach may work, it is more complex than necessary and does not take advantage of the durability and availability of SQS.

upvoted 3 times

LuckyAro 2 years, 10 months ago

Selected Answer: C

My question is; can orders be sent directly into an SQS queue ? How about the protocol for management of the messages from the queue ? can EC2 instances be programmed to process them like Lambda ?

upvoted 1 times

A company runs an application on a large fleet of Amazon EC2 instances. The application reads and writes entries into an Amazon DynamoDB table. The size of the DynamoDB table continuously grows, but the application needs only data from the last 30 days. The company needs a solution that minimizes cost and development effort.

Which solution meets these requirements?

- A. Use an AWS CloudFormation template to deploy the complete solution. Redeploy the CloudFormation stack every 30 days, and delete the original stack.
- B. Use an EC2 instance that runs a monitoring application from AWS Marketplace. Configure the monitoring application to use Amazon DynamoDB Streams to store the timestamp when a new item is created in the table. Use a script that runs on the EC2 instance to delete items that have a timestamp that is older than 30 days.
- C. Configure Amazon DynamoDB Streams to invoke an AWS Lambda function when a new item is created in the table. Configure the Lambda function to delete items in the table that are older than 30 days.
- D. Extend the application to add an attribute that has a value of the current timestamp plus 30 days to each new item that is created in the table. Configure DynamoDB to use the attribute as the TTL attribute. **Most Voted**

Correct Answer: D

Community vote distribution

D (95%)

5%

Comments

Gil80 **Highly Voted** 3 years, 6 months ago

Selected Answer: D

changing my answer to D after researching a bit.

The DynamoDB TTL feature allows you to define a per-item timestamp to determine when an item is no longer needed. Shortly after the date and time of the specified timestamp, DynamoDB deletes the item from your table without consuming any write throughput.

upvoted 42 times

satyaamm **Most Recent** 1 year, 3 months ago

Selected Answer: D

TTL is the most suitable here.

upvoted 1 times

1e22522 1 year, 10 months ago

Selected Answer: D

Always bet on the TTL

upvoted 2 times

Nawaff 1 year, 11 months ago

Selected Answer: C

I would say C

Because D requires extending the application to add the timestamp attribute.

Which is by itself a development effort.

upvoted 2 times

Anthony_Rodrigues 1 year, 9 months ago

It's to minimize development effort.

C is not the correct one because it needs to enable Dynamo Streams and configure them to send to Lambda, then create the Lambda code for deleting items.

Besides that, the Lambda is triggered only on new items, and since it's ephemeral, it would have to query items older than 30 days and then delete

them.

With D, you just need to make two small changes, and it's done—no extra service, no overhead, and no concern about failing to remove.

upvoted 4 times

Hkayne 2 years, 1 month ago

Selected Answer: D

D is the correct answer

upvoted 2 times

soufiyane 2 years, 2 months ago

Selected Answer: D

D is the best answer, dynamostreams is not suitable for this use cases

upvoted 2 times

Uzbekistan 2 years, 2 months ago

Option D is the most suitable solution to meet the company's requirements while minimizing cost and development effort.

TTL (Time to Live) Attribute: DynamoDB provides a feature called Time to Live (TTL), which allows you to automatically delete items from a table after a specified period. By adding a TTL attribute to each item with a value of the current timestamp plus 30 days, you can let DynamoDB automatically delete items older than 30 days. This eliminates the need for manual deletion efforts or periodic stack redeployment.

Minimal Development Effort

Cost-Effective

upvoted 3 times

scar0909 2 years, 3 months ago

Selected Answer: D

use ttl

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: D

A and B don't solve anything.

Between C and D, C requires more cost due to Lambda executions. D uses the TTL built-in feature so it won't cost extra. Also, D does not require extra development and is a matter of configuration. In old-school developer speak, don't write code if your DBA can do some work!

upvoted 3 times

awsgeek75 2 years, 4 months ago

<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/TTL.html>

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: D

DynamoDB Time to Live was designed to handle this kind of requirement where an item is no longer needed. TTL is provided at no extra cost as a means to reduce stored data volumes by retaining only the items that remain current for your workload's needs

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

D. Extend the application to add an attribute that has a value of the current timestamp plus 30 days to each new item that is created in the table. Configure DynamoDB to use the attribute as the TTL attribute.

The main reasons are:

Using DynamoDB's built-in TTL functionality is the most direct way to handle data expiration.

It avoids the complexity of triggers, streams, and lambda functions in option C.

Modifying the application code to add the TTL attribute is relatively simple and minimizes operational overhead

upvoted 3 times

cookieMr 2 years, 11 months ago

Selected Answer: D

By adding a TTL attribute to the DynamoDB table and setting it to the current timestamp plus 30 days, DynamoDB will automatically delete the items that are older than 30 days. This solution eliminates the need for manual deletion or additional infrastructure components.

A. Redeploying the CloudFormation stack every 30 days and deleting the original stack introduces unnecessary complexity and operational overhead

B. Using an EC2 instance with a monitoring application and a script to delete items older than 30 days adds additional infrastructure and maintenance efforts.

C. Configuring DynamoDB Streams to invoke a Lambda function to delete items older than 30 days adds complexity and requires additional development and operational effort compared to using the built-in TTL feature of DynamoDB.

upvoted 3 times

pisica134 2 years, 11 months ago

D: <https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/TTL.html>

upvoted 2 times

Abrar2022 3 years ago

Amazon DynamoDB Time to Live (TTL) allows you to define a per-item timestamp to determine when an item is no longer needed.

upvoted 4 times

studynoplay 3 years ago

Selected Answer: D

C is incorrect because it can take more than 15 minutes to delete the old data. Lambda won't work

upvoted 2 times

Konb 3 years, 1 month ago

Selected Answer: D

Clear case for TTL - every object gets deleted after a certain period of time

upvoted 2 times

rushi0611 3 years, 1 month ago

Selected Answer: D

Use DynamoDB TTL feature to achieve this..

upvoted 2 times

A company has a Microsoft .NET application that runs on an on-premises Windows Server. The application stores data by using an Oracle Database Standard Edition server. The company is planning a migration to AWS and wants to minimize development changes while moving the application. The AWS application environment should be highly available.

Which combination of actions should the company take to meet these requirements? (Choose two.)

- A. Refactor the application as serverless with AWS Lambda functions running .NET Core.
- B. Rehost the application in AWS Elastic Beanstalk with the .NET platform in a Multi-AZ deployment. **Most Voted**
- C. Replatform the application to run on Amazon EC2 with the Amazon Linux Amazon Machine Image (AMI).
- D. Use AWS Database Migration Service (AWS DMS) to migrate from the Oracle database to Amazon DynamoDB in a Multi-AZ deployment.
- E. Use AWS Database Migration Service (AWS DMS) to migrate from the Oracle database to Oracle on Amazon RDS in a Multi-AZ deployment. **Most Voted**

Correct Answer: BE

Community vote distribution

BE (97%)

3%

Comments

DavidNamy **Highly Voted** 3 years, 5 months ago

Selected Answer: BE

- B. Rehost the application in AWS Elastic Beanstalk with the .NET platform in a Multi-AZ deployment.
- E. Use AWS Database Migration Service (AWS DMS) to migrate from the Oracle database to Oracle on Amazon RDS in a Multi-AZ deployment.

Rehosting the application in Elastic Beanstalk with the .NET platform can minimize development changes. Multi-AZ deployment of Elastic Beanstalk will increase the availability of application, so it meets the requirement of high availability.

Using AWS Database Migration Service (DMS) to migrate the database to Amazon RDS Oracle will ensure compatibility, so the application can continue to use the same database technology, and the development team can use their existing skills. It also migrates to a managed service, which will handle the availability, so the team do not have to worry about it. Multi-AZ deployment will increase the availability of the database.

upvoted 17 times

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: BE

- B. Rehost the application in AWS Elastic Beanstalk with the .NET platform in a Multi-AZ deployment.
- E. Use AWS Database Migration Service (AWS DMS) to migrate from the Oracle database to Oracle on Amazon RDS in a Multi-AZ deployment.

To minimize development changes while moving the application to AWS and to ensure a high level of availability, the company can rehost the application in AWS Elastic Beanstalk with the .NET platform in a Multi-AZ deployment. This will allow the application to run in a highly available environment without requiring any changes to the application code.

The company can also use AWS Database Migration Service (AWS DMS) to migrate the Oracle database to Oracle on Amazon RDS in a Multi-AZ deployment. This will allow the company to maintain the existing database platform while still achieving a high level of availability.

upvoted 7 times

lofzee **Most Recent** 2 years ago

Selected Answer: BE

BE = least effort approach.. basically a lift and shift which is what the questions is asking for

upvoted 2 times

hardy1234567 2 years, 1 month ago

d - incorrect at all. Doesn't exist way for migration oracle to dinamoDB.

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: BE

E for minimizing development changes by using same Oracle engine but in highly available deployment.
C and D require platform change so it won't work as it increases development.
A is also development work of converting .Net to .Net core Lambda functions. May not even be possible.
B is simple lift and shift
BE is correct
upvoted 3 times

vijaykamal 2 years, 8 months ago

Selected Answer: BE

DynamoDB is NoSQL - E it out
Replatform requires considerable overhead - C is out
Lambda function is for running code for short duration - A is out
Answer - BE
upvoted 6 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: BE

Minimize development changes + High availability = AWS Elastic Beanstalk and Oracle on Amazon RDS in a Multi-AZ deployment
upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

B) Rehost the application in AWS Elastic Beanstalk with the .NET platform in a Multi-AZ deployment.

E) Use AWS Database Migration Service (AWS DMS) to migrate from the Oracle database to Oracle on Amazon RDS in a Multi-AZ deployment.

The reasons are:

° Rehosting in Elastic Beanstalk allows lifting and shifting the .NET application with minimal code changes. Multi-AZ deployment provides high availability.
° Using DMS to migrate the Oracle data to RDS Oracle in Multi-AZ deployment minimizes changes for the database while achieving high availability.
° Together this "lift and shift" approach minimizes refactoring needs while providing HA on AWS.
upvoted 2 times

cookieMr 2 years, 11 months ago

Selected Answer: BE

B. This allows the company to migrate the application to AWS without significant code changes while leveraging the scalability and high availability provided by EBS's Multi-AZ deployment.

E. This enables the company to migrate the Oracle database to RDS while maintaining compatibility with the existing application and leveraging the Multi-AZ deployment for high availability.

A. would require significant development changes and may not provide the same level of compatibility as rehosting or replatforming options.

C. would still require changes to the application and the underlying infrastructure, whereas rehosting with EBS minimizes the need for modification.

D. would likely require significant changes to the application code, as DynamoDB is a NoSQL database with a different data model compared to Oracle.
upvoted 4 times

markw92 2 years, 11 months ago

Answer is BE. No idea why D was chosen. That requires development work and question clearly states minimize development changes, changing db from Oracle to DynamoDB is LOT of development.
upvoted 3 times

Bmarodi 3 years ago

Selected Answer: BE

B + E are the answers that fulfil the requirements.
upvoted 2 times

cheese929 3 years, 1 month ago

Selected Answer: BE

B and E
upvoted 2 times

Nikhilcy 3 years, 1 month ago

why not C?

upvoted 2 times

AlankarJ 3 years ago

It runs on a windows server, shifting the whole this to Linux based EC2 would be extra work and of no sense

upvoted 3 times

AWSSURI 1 year, 9 months ago

Retire(simplest) < Retain < Relocate < Rehost < Repurchase < Replatform < Re-architect/Refactor (most complex)....So by this order rehosting less complex than replatform thats why we go for BE

upvoted 2 times

k33 3 years, 2 months ago

Selected Answer: BE

Answer : BE

upvoted 2 times

waiyiu9981 3 years, 5 months ago

Why A is wrong?

upvoted 1 times

gustavtd 3 years, 5 months ago

Because that needs some development,

upvoted 3 times

techhb 3 years, 5 months ago

Selected Answer: BE

B&E Option ,because D is for No-Sql

upvoted 2 times

JayBee65 3 years, 5 months ago

And requires additional development effort

upvoted 2 times

career360guru 3 years, 5 months ago

B&E Option

upvoted 2 times

A company runs a containerized application on a Kubernetes cluster in an on-premises data center. The company is using a MongoDB database for data storage. The company wants to migrate some of these environments to AWS, but no code changes or deployment method changes are possible at this time. The company needs a solution that minimizes operational overhead.

Which solution meets these requirements?

- A. Use Amazon Elastic Container Service (Amazon ECS) with Amazon EC2 worker nodes for compute and MongoDB on EC2 for data storage.
- B. Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate for compute and Amazon DynamoDB for data storage.
- C. Use Amazon Elastic Kubernetes Service (Amazon EKS) with Amazon EC2 worker nodes for compute and Amazon DynamoDB for data storage.
- D. Use Amazon Elastic Kubernetes Service (Amazon EKS) with AWS Fargate for compute and Amazon DocumentDB (with MongoDB compatibility) for data storage. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Marge_Simpson **Highly Voted** 2 years, 12 months ago

Selected Answer: D

If you see MongoDB, just go ahead and look for the answer that says DocumentDB.
upvoted 40 times

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: D

Option D is the correct solution that meets all the requirements:
° Use Amazon Elastic Kubernetes Service (Amazon EKS) with AWS Fargate for compute and Amazon DocumentDB (with MongoDB compatibility) for data storage.
The key reasons are:
° EKS allows running the Kubernetes environment on AWS without changes.
° Using Fargate removes the need to provision and manage EC2 instances.
° DocumentDB provides MongoDB compatibility so the data layer is unchanged.
upvoted 10 times

MehulKapadia **Most Recent** 1 year, 8 months ago

Selected Answer: D

Applications are already containerized. Amazon EKS is fully managed Kubernetes service.
Fargate = Less overhead of managing infrastructure.
Amazon DocumentDB is MongoDB Compatible.

Answer D
upvoted 2 times

LoXoL 1 year, 10 months ago

Selected Answer: D

no brainer says D
upvoted 2 times

james2033 2 years, 4 months ago

Selected Answer: D

Question keyword "containerized application", "Kubernetes cluster", "no changes or deployment method changes". Choose C, not D.

But "minimizes operational overhead", choose D.

upvoted 2 times

cookieMr 2 years, 5 months ago

Selected Answer: D

This solution allows the company to leverage EKS to manage the K8s cluster and Fargate to handle the compute resources without requiring manual management of EC2 worker nodes. The use of DocumentDB provides a fully managed MongoDB-compatible database service in AWS.

A. would require managing and scaling the EC2 instances manually, which increases operational overhead.

B. would require significant changes to the application code as DynamoDB is a NoSQL database with a different data model compared to MongoDB.

C. would also require code changes to adapt to DynamoDB's different data model, and managing EC2 worker nodes increases operational overhead.

upvoted 4 times

Bmarodi 2 years, 6 months ago

Selected Answer: D

The solution meets these requirements is option D.

upvoted 2 times

studynoplay 2 years, 6 months ago

Selected Answer: D

minimizes operational overhead = Serverless (Fargate)

MongoDB = DocumentDB

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Selected Answer: D

To minimize operational overhead and avoid making any code or deployment method changes, the company can use Amazon Elastic Kubernetes Service (EKS) with AWS Fargate for computing and Amazon DocumentDB (with MongoDB compatibility) for data storage. This solution allows the company to run the containerized application on EKS without having to manage the underlying infrastructure or make any changes to the application code.

AWS Fargate is a fully-managed container execution environment that allows you to run containerized applications without the need to manage the underlying EC2 instances.

Amazon DocumentDB is a fully-managed document database service that supports MongoDB workloads, allowing the company to use the same database platform as in their on-premises environment without having to make any code changes.

upvoted 5 times

techhb 2 years, 11 months ago

Selected Answer: D

Reason A & B Eliminated as its Kubernetes

why D read here <https://containersonaws.com/introduction/ec2-or-aws-fargate/>

upvoted 3 times

career360guru 2 years, 11 months ago

Selected Answer: D

Option D

upvoted 3 times

dcyberguy 3 years ago

DDDDDDD

upvoted 2 times

Gabs90 3 years ago

Selected Answer: D

<https://www.examttopics.com/discussions/amazon/view/67897-exam-aws-certified-solutions-architect-associate-saa-c02/>

upvoted 2 times

leonn 3 years ago

Selected Answer: D

D meets the requirements

upvoted 2 times

Nigma 3 years ago

Selected Answer: D

D

EKS because of Kubernetes so A and B are eliminated
not C because of MongoDB and Fargate is more expensive

upvoted 3 times

A telemarketing company is designing its customer call center functionality on AWS. The company needs a solution that provides multiple speaker recognition and generates transcript files. The company wants to query the transcript files to analyze the business patterns. The transcript files must be stored for 7 years for auditing purposes.

Which solution will meet these requirements?

- A. Use Amazon Rekognition for multiple speaker recognition. Store the transcript files in Amazon S3. Use machine learning models for transcript file analysis.
- B. Use Amazon Transcribe for multiple speaker recognition. Use Amazon Athena for transcript file analysis. **Most Voted**
- C. Use Amazon Translate for multiple speaker recognition. Store the transcript files in Amazon Redshift. Use SQL queries for transcript file analysis.
- D. Use Amazon Rekognition for multiple speaker recognition. Store the transcript files in Amazon S3. Use Amazon Textract for transcript file analysis.

Correct Answer: B

Community vote distribution

B (93%)

4%

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 11 months ago

Selected Answer: B

The correct answer is B: Use Amazon Transcribe for multiple speaker recognition. Use Amazon Athena for transcript file analysis.

Amazon Transcribe is a service that automatically transcribes spoken language into written text. It can handle multiple speakers and can generate transcript files in real-time or asynchronously. These transcript files can be stored in Amazon S3 for long-term storage.

Amazon Athena is a query service that allows you to analyze data stored in Amazon S3 using SQL. You can use it to analyze the transcript files and identify patterns in the data.

Option A is incorrect because Amazon Rekognition is a service for analyzing images and videos, not transcribing spoken language.

Option C is incorrect because Amazon Translate is a service for translating text from one language to another, not transcribing spoken language.

Option D is incorrect because Amazon Textract is a service for extracting text and data from documents and images, not transcribing spoken language.

upvoted 27 times

TheAbsoluteTruth 2 years, 8 months ago

What bothers me is the 7 years of storage.

upvoted 8 times

enzomv 2 years, 10 months ago

The correct answer is C.

<https://docs.aws.amazon.com/transcribe/latest/dg/what-is.html>

You can transcribe streaming media in real time or you can upload and transcribe media files. To see which languages are supported for each type of transcription, refer to the Supported languages and language-specific features table.

upvoted 4 times

enzomv 2 years, 10 months ago

Disregard. I meant B

upvoted 2 times

enzomv 2 years, 10 months ago

<https://aws.amazon.com/about-aws/whats-new/2022/06/amazon-transcribe-supports-automatic-language-identification-multi-lingual-audio/>
Amazon Translate is a service for multi-language identification, which identifies all languages spoken in the audio file and creates transcript using each identified language.

upvoted 2 times

enzomv 2 years, 10 months ago

Disregard. I meant Amazon Transcribe

upvoted 1 times

awsgeek75 Highly Voted 1 year, 11 months ago

Selected Answer: B

This is a poorly worded question with poorly worded options. Rekognition and Translate cannot convert speech to text so those options A, C & D are gone. B is the closes option but it does not mention S3 or retention policy of 7 years. Just a best guess on massive assumptions.

upvoted 7 times

kaleido Most Recent 12 months ago

Selected Answer: B

The closest answer here is B, but it's not the perfect answer since it doesn't address the 7 year data retention.

upvoted 1 times

bora4motion 1 year ago

Selected Answer: B

B - amazon transcribe is the right tool for call centers

upvoted 1 times

satyaamm 1 year, 3 months ago

Selected Answer: B

Transcribe for speech to text conversion and Athena for analysis.

upvoted 1 times

james_3005 1 year, 8 months ago

Option B is the most relevant one, but it doesn't mention how to retain data in 7 years...

upvoted 2 times

SinghJagdeep 1 year, 11 months ago

Selected Answer: B

check out this blog here: <https://aws.amazon.com/de/blogs/machine-learning/automating-the-analysis-of-multi-speaker-audio-files-using-amazon-transcribe-and-amazon-athena/>

upvoted 3 times

pentium75 1 year, 11 months ago

Selected Answer: B

Perfectly explained here: <https://aws.amazon.com/de/blogs/machine-learning/automating-the-analysis-of-multi-speaker-audio-files-using-amazon-transcribe-and-amazon-athena/>

upvoted 4 times

youdelin 2 years, 2 months ago

really hope I could have this kind of question during the exam, 4 different techs in the first 5 words of the answer! Just go with the correct one and ignore the rest of the text XDDD

upvoted 3 times

paniya93 2 years, 2 months ago

Selected Answer: B

<https://aws.amazon.com/blogs/machine-learning/automating-the-analysis-of-multi-speaker-audio-files-using-amazon-transcribe-and-amazon-athena/>

upvoted 2 times

vijaykamal 2 years, 2 months ago

Selected Answer: B

Amazon Rekognition is primarily designed for image and video analysis, not for transcribing audio or recognizing multiple speakers. -> Option A and D are ruled out

Amazon Translate is used for language translation -> Option C is ruled out

upvoted 3 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: B

Provide multiple speaker recognition and generate transcript files = Amazon Transcribe

Query the transcript files = Amazon Athena

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

The correct answer is B: Use Amazon Transcribe for multiple speaker recognition. Use Amazon Athena for transcript file analysis.

upvoted 3 times

Thornessen 2 years, 4 months ago

Selected Answer: B

Tricky or incomplete question..

B is the answer because Transcribe is the right service for processing voice calls.

But 7 years of storage is not covered (should add S3 storage)

And Athena querying is just SQL querying, it cannot help you much to recognize business patterns, for that I would think some text analysis service like Comprehend would be needed.

Unless... We use Transcribe not only to transcribe, but also to recognize some key words, and then create a DB/S3 record with multiple fields, e.g. if it is a telemarketing questionnaire, record answer for each question. Then SQL querying might be useful.

upvoted 3 times

pentium75 1 year, 11 months ago

"The company wants to query the transcript files" is the requirement. How they will be using the query results "to analyze the business patterns" is not our issue.

The "7 years" are not mentioned in any of the options, but Transcribe stores results in S3.

upvoted 2 times

sickcow 2 years, 5 months ago

Selected Answer: C

Transcribe and (s3) + Athena is the way to go here.

Redshift sounds like an overkill

upvoted 3 times

cookieMr 2 years, 5 months ago

Amazon Transcribe provides accurate transcription of audio recordings with multiple speakers, generating transcript files. These files can be stored in Amazon S3. To analyze the transcripts and extract insights, Amazon Athena allows SQL-based querying of the stored files.

A. Amazon Rekognition is for image and video analysis, not audio transcription.

C. Amazon Translate is for language translation, not speaker recognition or transcript analysis. Amazon Redshift may not be the best choice for storing and querying transcript files.

D. Amazon Rekognition is for image and video analysis, and Amazon Textract is for document extraction, not suitable for audio transcription or analysis. Storing the transcript files in S3 is appropriate, but the analysis requires a different service like Amazon Athena.

upvoted 2 times

Bmarodi 2 years, 6 months ago

Selected Answer: B

the solution that meets these requirements is option B.

upvoted 2 times

A company hosts its application on AWS. The company uses Amazon Cognito to manage users. When users log in to the application, the application fetches required data from Amazon DynamoDB by using a REST API that is hosted in Amazon API Gateway. The company wants an AWS managed solution that will control access to the REST API to reduce development efforts.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Configure an AWS Lambda function to be an authorizer in API Gateway to validate which user made the request.
- B. For each user, create and assign an API key that must be sent with each request. Validate the key by using an AWS Lambda function.
- C. Send the user's email address in the header with every request. Invoke an AWS Lambda function to validate that the user with that email address has proper access.
- D. Configure an Amazon Cognito user pool authorizer in API Gateway to allow Amazon Cognito to validate each request.

Most Voted

Correct Answer: D

Community vote distribution

D (98%)

2

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 11 months ago

Selected Answer: D

KEYWORD: LEAST operational overhead

To control access to the REST API and reduce development efforts, the company can use an Amazon Cognito user pool authorizer in API Gateway. This will allow Amazon Cognito to validate each request and ensure that only authenticated users can access the API. This solution has the LEAST operational overhead, as it does not require the company to develop and maintain any additional infrastructure or code.

Therefore, Option D is the correct answer.

Option D. Configure an Amazon Cognito user pool authorizer in API Gateway to allow Amazon Cognito to validate each request.

upvoted 15 times

k1kavi1 **Highly Voted** 2 years, 11 months ago

Selected Answer: D

<https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-integrate-with-cognito.html>

upvoted 5 times

MutiverseAgent 2 years, 4 months ago

up this

upvoted 1 times

kaleido **Most Recent** 12 months ago

Selected Answer: D

The answer is D. There are 2 ways to control user access to the API gateway REST API. That is through Lambda authorizers and Cognito user pools.

Use Lambda authorizers if you want complete control over the authorization process. This allows you to customize what IAM policies a user is assigned.

Use Cognito if your authorization needs are basic.

upvoted 1 times

ad2dj28 1 year, 4 months ago

Selected Answer: D

you can use an Amazon Cognito user pool to control who can access your API in Amazon API Gateway

upvoted 1 times

drich22 1 year, 6 months ago

Control access to a REST API using Amazon Cognito user pools as authorizer

<https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-integrate-with-cognito.html>

upvoted 3 times

MehulKapadia 1 year, 8 months ago

Selected Answer: D

Answer D

By integrating Amazon Cognito User Pools with API Gateway, you can secure your APIs and control access based on user authentication and authorization, allowing you to build secure and scalable web and mobile applications.

upvoted 2 times

Adi312100 1 year, 8 months ago

Selected Answer: D

Option D. Configure an Amazon Cognito user pool authorizer in API Gateway to allow Amazon Cognito to validate each request.

upvoted 2 times

awsgeek75 1 year, 11 months ago

Selected Answer: D

A is possible if the authorisation logic makes sense and does not require operational overhead.

B is too much overhead for each new user.

C is lol

D Company already has Cognito for it's users so just integrate it with the API gateway

This question and options are poorly worded an A could be a reasonable choice if more information is provided. Just keep that in mind for the exam

upvoted 5 times

osmk 1 year, 12 months ago

<https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-integrate-with-cognito.html>

upvoted 3 times

Tom123456ac 2 years, 2 months ago

The description of this question is really bad. Company is using Cognito to manage users already, but still verifying user info from dynamodb, very wired situation. But just select Cognito when you see Api gateway + cognito + authentication + least efforts

upvoted 4 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: D

use Amazon Cognito to authorize user requests.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

D. Configure an Amazon Cognito user pool authorizer in API Gateway to allow Amazon Cognito to validate each request

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

Option D is the best solution with the least operational overhead:

Configure an Amazon Cognito user pool authorizer in API Gateway to allow Amazon Cognito to validate each request.

The key reasons are:

° Cognito user pool authorizers allow seamless integration between Cognito and API Gateway for access control.

° API Gateway handles validating the access tokens from Cognito automatically without any custom code.

° This is a fully managed solution with minimal ops overhead.

upvoted 3 times

cookieMr 2 years, 5 months ago

By configuring an Amazon Cognito user pool authorizer in API Gateway, you can leverage the built-in functionality of Amazon Cognito to authenticate and authorize users. This eliminates the need for custom development or managing access keys. Amazon Cognito handles user authentication, securely manages user identities, and provides seamless integration with API Gateway for controlling access to the REST API.

A. Configuring an AWS Lambda function as an authorizer in API Gateway would require custom implementation and management of the authorization logic.

B. Creating and assigning an API key for each user would require additional management and validation logic in an AWS Lambda function.

C. Sending the user's email address in the header and validating it with an AWS Lambda function would also require custom implementation and management of the authorization logic.

Option D, using an Amazon Cognito user pool authorizer, provides a streamlined and managed solution for controlling access to the REST API with minimal operational overhead.

upvoted 3 times

Bmarodi 2 years, 6 months ago

Selected Answer: D

solution will meet these requirements with the LEAST operational overhead is option D.

upvoted 2 times

studynoplay 2 years, 6 months ago

Selected Answer: D

LEAST operational overhead = Serverless = Cognito user pool

upvoted 2 times

cheese929 2 years, 7 months ago

Selected Answer: D

D is correct.

upvoted 2 times

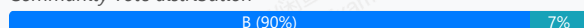
A company is developing a marketing communications service that targets mobile app users. The company needs to send confirmation messages with Short Message Service (SMS) to its users. The users must be able to reply to the SMS messages. The company must store the responses for a year for analysis.

What should a solutions architect do to meet these requirements?

- A. Create an Amazon Connect contact flow to send the SMS messages. Use AWS Lambda to process the responses.
- B. Build an Amazon Pinpoint journey. Configure Amazon Pinpoint to send events to an Amazon Kinesis data stream for analysis and archiving. **Most Voted**
- C. Use Amazon Simple Queue Service (Amazon SQS) to distribute the SMS messages. Use AWS Lambda to process the responses.
- D. Create an Amazon Simple Notification Service (Amazon SNS) FIFO topic. Subscribe an Amazon Kinesis data stream to the SNS topic for analysis and archiving.

Correct Answer: B

Community vote distribution



Comments

TariqKipkemei **Highly Voted** 2 years, 2 months ago

Selected Answer: B

Marketing communications = Amazon Pinpoint
upvoted 17 times

cookieMr **Highly Voted** 2 years, 5 months ago

Selected Answer: B

By using Pinpoint, the company can effectively send SMS messages to its mobile app users. Additionally, Pinpoint allows the configuration of journeys, which enable the tracking and management of user interactions. The events generated during the journey, including user responses to SMS, can be captured and sent to an Kinesis data stream. This data stream can then be used for analysis and archiving purposes.

A. Creating an Amazon Connect contact flow is primarily focused on customer support and engagement, and it lacks the capability to store and process SMS responses for analysis.

C. Using SQS is a message queuing service and is not specifically designed for handling SMS responses or capturing them for analysis.

D. Creating an SNS FIFO topic and subscribing a Kinesis data stream is not the most appropriate solution for capturing and storing SMS responses, a SNS is primarily used for message publishing and distribution.

In summary, option B is the best choice as it leverages Pinpoint to send SMS messages and captures user responses for analysis and archiving using an Kinesis data stream.

upvoted 11 times

pabloss **Most Recent** 2 weeks, 4 days ago

Selected Answer: B

It's B, but Pinpoint will be deprecated soon
upvoted 1 times

scar0909 1 year, 9 months ago

Selected Answer: A

<https://docs.aws.amazon.com/connect/latest/adminguide/setup-sms-messaging.html>
upvoted 1 times

FlyingHawk 1 year, 4 months ago

Amazon Connect is primarily a contact center service, not designed for sending bulk SMS messages or handling two-way SMS communication for marketing purposes. It is overkill for this use case.

upvoted 1 times

awsgeek75 1 year, 11 months ago

Selected Answer: B

<https://docs.aws.amazon.com/pinpoint/latest/userguide/welcome.html>

Amazon Pinpoint is the easiest solution.

Amazon Connect is Contact Centre as a Service so this option is not relevant to the requirement.

SQS and SNS options are overengineered or under engineered for the requirements so natural choice is "B"

upvoted 2 times

whoob 2 years, 2 months ago

base function of AWS Pinpoint

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B. AWS Pinpoint is for Marketing communications.

upvoted 4 times

Bmarodi 2 years, 6 months ago

Selected Answer: B

Option B is correct answer: link: <https://aws.amazon.com/pinpoint/>, and video under the link.

upvoted 3 times

studynoplay 2 years, 6 months ago

Selected Answer: B

Two-Way Messaging

Receive SMS messages from your customers and reply back to them in a chat-like interactive experience. With Amazon Pinpoint, you can create automatic responses when customers send you messages that contain certain keywords.

upvoted 2 times

CLOUDUMASTER 2 years, 7 months ago

Based on my research Kinesis stream is real time data ingestion, and also stores only event data and not the actual people responses, furthermore there is no requirement to have real time data streaming. That is probably why I am hesitating agree here with everyone on B and rather choose A.

upvoted 1 times

pentium75 1 year, 11 months ago

"A Kinesis data stream stores records for 24 hours by default, up to 365 days (8,760 hours)."

<https://aws.amazon.com/de/blogs/big-data/retaining-data-streams-up-to-one-year-with-amazon-kinesis-data-streams/#:~:text=A%20Kinesis%20data%20stream%20stores,parallel%20and%20at%20low%20latency.>

upvoted 2 times

jayce5 2 years, 7 months ago

Selected Answer: B

The answer is B. AWS Pinpoint is for Marketing communications.

AWS Connect is for Contact center.

upvoted 3 times

jaswantn 2 years, 7 months ago

Selected Answer: A

According to the following link I would choose Option A.

<https://docs.aws.amazon.com/connect/latest/adminguide/web-and-mobile-chat.html>

upvoted 1 times

smartegnine 2 years, 6 months ago

no no, there is no SMS, note the question stated all activities through SMS, also Amazon connect flow most likely working on web application UI, but if you see question clearly, this is receiving and sending SMS not through application UI (Web/Mobile App). So for those reason we choose B

upvoted 3 times

ProfXsamson 2 years, 10 months ago

Selected Answer: B

Amazon Pinpoint is a flexible, scalable and fully managed push notification and SMS service for mobile apps.

upvoted 4 times

Foucault 2 years, 10 months ago

It's B, see following link <https://docs.aws.amazon.com/pinpoint/latest/developerguide/event-streams.html>

upvoted 3 times

LuckyAro 2 years, 10 months ago

Selected Answer: B

[https://aws.amazon.com/pinpoint/product-details/sms/Two-Way Messaging:](https://aws.amazon.com/pinpoint/product-details/sms/Two-Way-Messaging/)

Receive SMS messages from your customers and reply back to them in a chat-like interactive experience. With Amazon Pinpoint, you can create automatic responses when customers send you messages that contain certain keywords. You can even use Amazon Lex to create conversational bots. A majority of mobile phone users read incoming SMS messages almost immediately after receiving them. If you need to be able to provide your customers with urgent or important information, SMS messaging may be the right solution for you.

You can use Amazon Pinpoint to create targeted groups of customers, and then send them campaign-based messages. You can also use Amazon Pinpoint to send direct messages, such as appointment confirmations, order updates, and one-time passwords.

upvoted 3 times

DavidNamy 2 years, 11 months ago

Selected Answer: D

D:

Amazon Simple Notification Service (SNS) is a fully managed messaging service that enables you to send and receive SMS messages in a cost-effective and highly scalable way. By creating an SNS FIFO topic, you can ensure that the SMS messages are delivered to your users in the order they were sent and that the SMS responses are processed and stored in the same order. You can also configure your SNS FIFO topic to publish SMS responses to an Amazon Kinesis data stream, which will allow you to store and analyze the responses for a year.

Amazon Pinpoint ??? NO!

is not correct solution because while Amazon Pinpoint allows you to send SMS and Email campaigns, as well as handle push notifications to a user base, it doesn't provide SMS sending feature by itself. Furthermore, it's a service mainly focused on sending and tracking marketing campaigns, not for managing two-way SMS communication and the reception of reply.

upvoted 3 times

Omok 2 years, 10 months ago

What do think about <https://docs.aws.amazon.com/pinpoint/latest/userguide/channels-sms-two-way.html?>

upvoted 2 times

Buruguduystunstugudunstuy 2 years, 11 months ago

Selected Answer: B

To send SMS messages and store the responses for a year for analysis, the company can use Amazon Pinpoint. Amazon Pinpoint is a fully-managed service that allows you to send targeted and personalized SMS messages to your users and track the results.

To meet the requirements of the company, a solutions architect can build an Amazon Pinpoint journey and configure Amazon Pinpoint to send events to an Amazon Kinesis data stream for analysis and archiving. The Kinesis data stream can be configured to store the data for a year, allowing the company to analyze the responses over time.

So, Option B is the correct answer.

Option B. Build an Amazon Pinpoint journey. Configure Amazon Pinpoint to send events to an Amazon Kinesis data stream for analysis and archiving

upvoted 5 times

A company is planning to move its data to an Amazon S3 bucket. The data must be encrypted when it is stored in the S3 bucket. Additionally, the encryption key must be automatically rotated every year.

Which solution will meet these requirements with the LEAST operational overhead?

- A. Move the data to the S3 bucket. Use server-side encryption with Amazon S3 managed encryption keys (SSE-S3). Use the built-in key rotation behavior of SSE-S3 encryption keys. **Most Voted**
- B. Create an AWS Key Management Service (AWS KMS) customer managed key. Enable automatic key rotation. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Move the data to the S3 bucket.
- C. Create an AWS Key Management Service (AWS KMS) customer managed key. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Move the data to the S3 bucket. Manually rotate the KMS key every year.
- D. Encrypt the data with customer key material before moving the data to the S3 bucket. Create an AWS Key Management Service (AWS KMS) key without key material. Import the customer key material into the KMS key. Enable automatic key rotation.

Correct Answer: A

Community vote distribution



Comments

Buruguduystunstugudunstuy **Highly Voted** 3 years, 5 months ago

Selected Answer: A

KEYWORD: LEAST operational overhead

To encrypt the data when it is stored in the S3 bucket and automatically rotate the encryption key every year with the least operational overhead, the company can use server-side encryption with Amazon S3-managed encryption keys (SSE-S3). SSE-S3 uses keys that are managed by Amazon S3, and the built-in key rotation behavior of SSE-S3 encryption keys automatically rotates the keys every year.

To meet the requirements of the company, the solutions architect can move the data to the S3 bucket and enable server-side encryption with SSE-S3. This solution requires no additional configuration or maintenance and has the least operational overhead.

Hence, the correct answer is;

Option A. Move the data to the S3 bucket. Use server-side encryption with Amazon S3-managed encryption keys (SSE-S3). Use the built-in key rotation behavior of SSE-S3 encryption keys.

upvoted 45 times

bicrasse 2 years, 6 months ago

The good answer was B before May 2022, because the rotation schedule for AWS managed keys was 3 years (SSE-S3 is based on it)...

From May 2022 the schedule rotation is 1 year, then A is now the best answer because there is NO operational task to do: S3 is by default encrypted at rest with SSE-S3 (rotation every year)... So it depends if the question has been updated since 2022

upvoted 22 times

LuckyAro 3 years, 4 months ago

The order of these events is being ignored here in my opinion. The encryption checkbox needs to be checked before data is moved into the S3 bucket or it will not be encrypted otherwise, you'll have to encrypt manually and reload into S3 bucket. If the box was checked before moving data into S3 then you are good to go !

upvoted 10 times

LuckyAro 3 years, 4 months ago

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/default-bucket-encryption.html>

upvoted 2 times

Wang87 2 years, 5 months ago

SSE DOES not rotate encryption keys, it changes master key used to lock encryption keys which creates new ciphered key and stores it.
upvoted 4 times

Smart 2 years, 10 months ago

Ignoring the new changes that the default encryption is already enabled. I agree that the encryption should be configured before moving the data into the bucket. Otherwise, the existing objects will remain unencrypted.

Correct Answer is B.

Additionally, where is the reference that SSE-S3 will rotate keys every year (which is the question's requirement).

upvoted 3 times

pentium75 2 years, 5 months ago

SSE-S3 rotates the keys when AWS wants it, not "every year" like required here.

upvoted 4 times

pentium75 2 years, 5 months ago

No, I stand corrected.

All AWS managed keys are automatically rotated every year. You cannot change this rotation schedule.

upvoted 5 times

tohegajaf 2 years, 5 months ago

https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#master_keys

upvoted 1 times

JA2018 1 year, 6 months ago

For AWS managed key (<https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#aws-managed-cmk>)

Automatic rotation (<https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html>) --> Required. Every year (approximately 365 days).

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Amazon S3 encrypts each object with a unique key. As an additional safeguard, it encrypts the key itself with a root key that it regularly rotates. However, AWS does not publicly disclose the specific frequency of these rotations

upvoted 1 times

awsgeek75 2 years, 5 months ago

I want to find a source for this yearly rotation because SSE-S3 just rotates periodically and doesn't say it follows the same policy as other managed key. I think you may be right but just need a doc link

upvoted 3 times

Maru86 2 years, 3 months ago

https://repost.aws/questions/QUES_1VN01TU-eRSO3LXergA/s3-managed-key-sse-s3-rotation-period

upvoted 2 times

JA2018 1 year, 6 months ago

Check out the AWS link mentioned in the response:

<https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#key-mgmt>

For AWS managed key (<https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#aws-managed-cmk>)

Automatic rotation (<https://docs.aws.amazon.com/kms/latest/developerguide/rotate-keys.html>) --> Required. Every year (approximately 365 days).

upvoted 1 times

Buruguduystunstugudunstuy 3 years, 5 months ago

Option B involves using a customer-managed AWS KMS key and enabling automatic key rotation, but this requires the company to manage the KMS key and monitor the key rotation process.

Option C involves using a customer-managed AWS KMS key, but this requires the company to manually rotate the key every year, which introduces additional operational overhead.

Option D involves encrypting the data with customer key material and creating a KMS key without key material, but this requires the company to manage the customer key material and import it into the KMS key, which introduces additional operational overhead.

upvoted 4 times

JayBee65 3 years, 5 months ago

But...

For A there is no reference to how often these keys are rotated, and to rotate to a new key, you need to upload it, which is operational overhead. So not only does it not necessarily meet the 'rotate keys every year' requirement, but every year it requires operational overhead.

More importantly, the question states move the objects first, and then configure encryption, but ... "There is no change to the encryption of the objects that existed in the bucket before default encryption was enabled." from <https://docs.aws.amazon.com/AmazonS3/latest/userguide/default-bucket-encryption.html>

So A is clearly wrong.

For B, whilst you have to set up KMS once, you then don't have to anything else, which i would say is LEAST operational overhead.

upvoted 29 times

ocbn3wby 3 years, 5 months ago

God bless you, man! The most articulated answers, easy to understand. Good job!

upvoted 5 times

JayBee65 3 years, 5 months ago

But wrong :)

upvoted 5 times

ocbn3wby 3 years, 4 months ago

Reviewed it the second time. Some of them are wrong, indeed.

upvoted 1 times

techhb Highly Voted 3 years, 5 months ago

Selected Answer: B

SSE-S3 - is free and uses AWS owned CMKs (CMK = Customer Master Key). The encryption key is owned and managed by AWS, and is shared among many accounts. Its rotation is automatic with time that varies as shown in the table here. The time is not explicitly defined.

SSE-KMS - has two flavors:

AWS managed CMK. This is free CMK generated only for your account. You can only view it policies and audit usage, but not manage it. Rotation is automatic - once per 1095 days (3 years),

Customer managed CMK. This uses your own key that you create and can manage. Rotation is not enabled by default. But if you enable it, it will be automatically rotated every 1 year. This variant can also use an imported key material by you. If you create such key with an imported material, there is no automated rotation. Only manual rotation.

SSE-C - customer provided key. The encryption key is fully managed by you outside of AWS. AWS will not rotate it.

upvoted 34 times

ruqui 3 years ago

AWS managed CMK rotates every 365 days (not 1095 days). Reference:

<https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#key-mgmt>

upvoted 3 times

mahesuma Most Recent 2 weeks, 2 days ago

Selected Answer: B

For annual automatic key rotation, use: SSE-KMS with AWS KMS customer managed key

upvoted 1 times

DanRuggles 1 month, 2 weeks ago

Selected Answer: B

SSE-S3 Key Rotation happens regularly, but AWS's frequency is not publicly documented, and there is no guarantee it occurs exactly once a year. KMS can be set to annually

upvoted 1 times

plucio 5 months, 1 week ago

Selected Answer: B

SSE-S3 cannot be rotated automatically once a year

upvoted 1 times

MultipleWorkerMirroredStrategy 9 months, 1 week ago

Selected Answer: B

AWS documentation about S3 SSE doesn't really commit to the yearly rotation, so to be it's B

upvoted 1 times

Clouddon 11 months, 1 week ago

Selected Answer: B

B. Create an AWS Key Management Service (AWS KMS) customer managed key. Enable automatic key rotation. Set the S3 bucket's default encryption behavior to use the customer managed KMS key. Move the data to the S3 bucket.

upvoted 1 times

In2718 11 months, 2 weeks ago

Selected Answer: B

AWS Key Management Service (AWS KMS) customer managed keys are better to use than Amazon S3 managed encryption keys (SSE-S3). AWS training states to first use KMS and not SSE-S3). The "better" correct answer is B.

upvoted 1 times

Charlesvg 11 months, 4 weeks ago

Selected Answer: B

AWS doesn't have a published frequency to rotate the keys used in SSE-S3

https://repost.aws/questions/QUES_1VN01TU-eRSO3LXergA/s3-managed-key-sse-s3-rotation-period

upvoted 1 times

chest_jd 1 year, 2 months ago

Selected Answer: B

Why it is not B - we can not rely on implicit period of rotation for S3 kry.

upvoted 1 times

chest_jd 1 year, 2 months ago

Sorry - it is not A - we can not rely on implicit period of rotation for S3 key.

upvoted 1 times

AwsAbhiKumar 1 year, 2 months ago

Selected Answer: B

Amazon S3 managed encryption keys (SSE-S3), Amazon S3 handles the key rotation automatically and regularly. Key points regarding the rotation schedule:

SSE-S3 uses a unique encryption key for each object.

These individual object keys are themselves encrypted with a root key.

Amazon S3 regularly rotates this root key as an additional security measure.

The exact rotation schedule for the root key is not publicly disclosed for security reasons.

Hence Answer is B

upvoted 1 times

SirDNS 1 year, 2 months ago

Selected Answer: B

The answer should be B in this case cause we exactly know that AWS Managed (SSE-KMS) is automatically rotated annually. AWS does not disclose the rotation schedule of SSE-S3 keys

upvoted 1 times

yangbo 1 year, 3 months ago

Selected Answer: B

SSE-S3 is not rotate keys every year

upvoted 2 times

Dharmarajan 1 year, 4 months ago

Selected Answer: A

It can be done with A & B, but with A, SSE-S3 there are no charges.

upvoted 1 times

FlyingHawk 1 year, 4 months ago

Selected Answer: B

AWS does not specify the exact rotation frequency for SSE-S3 root keys, AWS Managed Keys are rotated annually, and Customer Managed Keys offer customizable rotation options.

upvoted 1 times

suresh_01 1 year, 4 months ago

Selected Answer: A

Amazon S3 now applies server-side encryption with Amazon S3 managed keys (SSE-S3) as the base level of encryption for every bucket in Amazon S3. Starting January 5, 2023, all new object uploads to Amazon S3 are automatically encrypted at no additional cost and with no impact on performance. The automatic encryption status for S3 bucket default encryption configuration and for new object uploads is available in AWS CloudTrail logs, S3 Inventory, S3 Storage Lens, the Amazon S3 console, and as an additional Amazon S3 API response header in the AWS Command Line Interface and AWS SDKs.

upvoted 1 times

4729e6c 1 year, 4 months ago

Selected Answer: B

A is incorrect because SSE-S3 rotates keys when AWS thinks is right, not when customer wants ("every year")

B is correct because AWS KMS customer managed keys allow automatic key rotation every year, meeting the requirement while minimizing operational overhead.

upvoted 1 times

The customers of a finance company request appointments with financial advisors by sending text messages. A web application that runs on Amazon EC2 instances accepts the appointment requests. The text messages are published to an Amazon Simple Queue Service (Amazon SQS) queue through the web application. Another application that runs on EC2 instances then sends meeting invitations and meeting confirmation email messages to the customers. After successful scheduling, this application stores the meeting information in an Amazon DynamoDB database.

As the company expands, customers report that their meeting invitations are taking longer to arrive.

What should a solutions architect recommend to resolve this issue?

- A. Add a DynamoDB Accelerator (DAX) cluster in front of the DynamoDB database.
- B. Add an Amazon API Gateway API in front of the web application that accepts the appointment requests.
- C. Add an Amazon CloudFront distribution. Set the origin as the web application that accepts the appointment requests.
- D. Add an Auto Scaling group for the application that sends meeting invitations. Configure the Auto Scaling group to scale based on the depth of the SQS queue. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

Buruguduystunstugudunstuy **Highly Voted** 2 years, 11 months ago

Selected Answer: D

Option D. Add an Auto Scaling group for the application that sends meeting invitations. Configure the Auto Scaling group to scale based on the depth of the SQS queue.

To resolve the issue of longer delivery times for meeting invitations, the solutions architect can recommend adding an Auto Scaling group for the application that sends meeting invitations and configuring the Auto Scaling group to scale based on the depth of the SQS queue. This will allow the application to scale up as the number of appointment requests increases, improving the performance and delivery times of the meeting invitations.

upvoted 17 times

cookieMr **Highly Voted** 2 years, 5 months ago

Selected Answer: D

By adding an ASG for the application that sends meeting invitations and configuring it to scale based on the depth of the SQS, the system can automatically adjust its capacity based on the number of pending messages in the queue. This ensures that the application can handle increased message load and process the meeting invitations more efficiently, reducing the delay experienced by customers.

A. Adding a DynamoDB Accelerator (DAX) cluster in front of the DynamoDB database would improve read performance for DynamoDB, but it does not directly address the issue of delayed meeting invitations.

B. Adding an API Gateway API in front of the web application that accepts the appointment requests may help with request handling and management, but it does not directly address the issue of delayed meeting invitations.

C. Adding an CloudFront distribution with the web application as the origin would improve content delivery and caching, but it does not directly address the issue of delayed meeting invitations.

upvoted 11 times

67a3f49 **Most Recent** 1 year, 9 months ago

First question with consistent answer :)

upvoted 5 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: D

Add an Auto Scaling group for the application that sends meeting invitations. Configure the Auto Scaling group to scale based on the depth of the SQS queue.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

Add an Auto Scaling group for the application that sends meeting invitations. Configure the Auto Scaling group to scale based on the depth of the SQS queue.

upvoted 2 times

career360guru 2 years, 11 months ago

Selected Answer: D

Option D is the right Answer,

upvoted 3 times

k1kavi1 2 years, 12 months ago

Selected Answer: D

Agreed

upvoted 2 times

jambajuice 3 years ago

Selected Answer: D

ANswer d

upvoted 2 times

leonnnn 3 years ago

Selected Answer: D

D meets the requirements

upvoted 2 times

Nigma 3 years ago

Selected Answer: D

Answer : D

upvoted 2 times